

BASE24 Release 6.0 Version 10

Implementation Guide



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Preface

Document Control

A description of the document revisions is shown below. Any changes of text from one document version to another are indicated with the symbol “|” in the outside margin.

BASE24 Release 6.0 Version 10 (D. Barton, S. Eschliman, R. McCall, C. Meyers, C. Rink, E. Rock, R. Schieuer, F. Slevin, L. Watson, J. Wishart)
– Dec-2009

Notification Percent for PTD Extended Memory Tables Fix, Salesforce Case #1088099 (C. Sader). Pages 6-37 and 6-38.
– Sep-2010

Diebold ENTIBOLD Entry Program Reorganization, (L. Watson). Clarification regarding source and object files/subvols provided to customers. March 2011.

Replaced all references to AT60IBLD with AT20IBLD, (L. Watson). April 2011.

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1: Introduction

The ***BASE24 Release 6.0 Version 10 Implementation Guide*** is the primary reference document for information needed to upgrade existing BASE24 Release 6.0 Version 9 systems to BASE24 Release 6.0 Version 10.

This section provides an overview of the planning requirements for BASE24 users migrating to BASE24 Release 6.0 Version 10. It also provides guidance for using this manual.

Also included in this section is a listing of other documents that a BASE24 user should obtain for implementation planning, the general software requirements for BASE24 Release 6.0 Version 10, and a glossary of terms used throughout the guide.

Implementation Planning

BASE24 users must develop their own project plans for migrating their BASE24 systems to Release 6.0 Version 10. Project plans should address such issues as staffing and training requirements, host external message changes, device programming enhancements, hardware and software requirements, and the migration method to be used.

ACI has provided BASE24 users with the **BASE24 Release 6.0 Version 10 Implementation Guide** to assist in preparing project plans. This guide includes a list of services available to the BASE24 user and information that existing BASE24 users will need to upgrade to Release 6.0 Version 10.

BASE24 Release 6.0 products include the following:

- ☐ BASE24-atm
- ☐ BASE24-pos
- ☐ BASE24-teller
- ☐ BASE24-remote banking
- ☐ BASE24 Interchange Interfaces
- ☐ BASE24 Value-Added Products:
 - o BASE24-card
 - o BASE24-customer service
 - o BASE24-infoBase
 - o BASE24-inventory

Who should use this guide?

There are two audiences for this guide: those who plan and manage the migration process and those who implement the migration to BASE24 Release 6.0 Version 10. In either case, the reader should be experienced with BASE24 software, HP NSK products, and the BASE24 user's current configuration. This guide is not directed at new BASE24 users who will be installing a BASE24 application for the first time. New users should refer to the formal BASE24 publications listed under the Publications heading in Section 3.

To migrate to BASE24 Release 6.0 Version 10, users must be on BASE24 Release 6.0 Version 9. Refer to the ***BASE24 Release 6.0 Version 9 Implementation Guide*** to assist in planning the tasks necessary to get to BASE24 Release 6.0 Version 10.

Using this guide, you can do the following:

- ☐ Determine what ACI services are available to assist in the migration to BASE24 Release 6.0 Version 10.
- ☐ Develop a project plan to migrate to BASE24 Release 6.0 Version 10.
- ☐ Determine staffing and training needs.
- ☐ Gain a detailed understanding of the BASE24 Release 6.0 Version 10 enhancements.

Software Requirements

HP NSK Software Requirements

ACI recommends that the BASE24 user be on a current HP NSK-supported operating system to obtain maximum performance and support. However, BASE24 Release 6.0 is currently compatible with the HP NSK Guardian D45 Operating System. Customers choosing to implement Format 2 File support are required to be on a minimum Operating System level of D46. In addition, BASE24 Release 6.0 requires TMF for several files.

XPNET

ACI requires that the BASE24 user be on an XPNET 3.0 or 3.1 system.

FIX

FIX 2.28 (or newer) is required to apply any new BASE24 fix files produced after 19 October 2006. FIX 2.28, like the previous version of FIX, will allow application of fix files to a specified file name in a concatenated list of fix files. This option can be combined with the FIRSTSCN and LASTSCN options to apply only fixes to the specified file name in a specified SCN range. As of 19 October 2006, FIX 2.28 (or the most current version of FIX) was automatically shipped with each software order. Please replace your existing FIX object with the most current version.

Conversion Planning

BASE24 users must develop their own project plans for upgrading their systems to BASE24 Release 6.0 Version 10. Project plans should address such issues as staffing and training requirements, host external message changes, device programming enhancements, hardware and software requirements, and the migration method to be used.

In general, all changes for BASE24 Release 6.0 Version 10 can be obtained by requesting the latest fixes from ACI and then retrofitting all fixes to the user's BASE24 Release 6.0 Version 9 modules.

Conversion Program Guidelines

Users of previous releases of BASE24 and previous versions of BASE24 Release 6.0 must run file conversion programs in order to migrate to the latest version of Release 6.0. Information about specific conversion programs can be found in subsequent sections of this document under the applicable enhancements. Following are general conversion program guidelines.

The *UC04 subvolumes (e.g., AT60UC04, BA60UC04, PS60UC04) contain the conversion programs for users migrating from BASE24 Release 5.3 to the latest version of BASE24 Release 6.0. The conversion programs on these subvolumes are kept up-to-date with each new version of BASE24 Release 6.0.

The *UC05 subvolumes (e.g., AT60UC05, BA60UC05, PS60UC05) contain the conversion programs for users migrating from Release 6.0 Version 1 to Release 6.0 Version 2. Also residing on these subvolumes are the Transaction Security Services (TSS) conversion programs, which are used to move key data to the new TSS database. The TSS conversion programs are required only if the user is implementing TSS.

The *UC06 subvolumes (e.g., AT60UC06, BA60UC06, PS60UC06) contain the conversion program for users migrating from Release 6.0 Version 2 to Release 6.0 Version 3. No changes are made to these programs other than necessary fixes.

The *UC07 subvolumes (e.g., AT60UC07, BA60UC07, PS60UC07) contain the conversion program for users migrating from Release 6.0 Version 3 to Release 6.0 Version 4. No changes are made to these programs other than necessary fixes.

The *UC08 subvolumes (e.g., AT60UC08, BA60UC08, PS60UC08) contain the conversion program for users migrating from Release 6.0 Version 4 to Release 6.0 Version 5. No changes are made to these programs other than necessary fixes.

The *UC09 subvolumes (e.g., AT60UC09, BA60UC09, PS60UC09) contain the conversion program for users migrating from Release 6.0 Version 5 to Release 6.0 Version 6. No changes are made to these programs other than necessary fixes.

The *UC0A subvolumes (e.g., AT60UC0A, BA60UC0A, PS60UC0A) contain the conversion program for users migrating from Release 6.0 Version 6 to Release 6.0 Version 7. No changes are made to these programs other than necessary fixes.

The *UC0B subvolumes (e.g., AT60UC0B, BA60UC0B, PS60UC0B) contain the conversion program for users migrating from Release 6.0 Version 7 to Release 6.0 Version 8. No changes are made to these programs other than necessary fixes.

The *UC0C subvolumes (e.g., AT60UC0C, BA60UC0C, PS60UC0C) contain the conversion program for users migrating from Release 6.0 Version 8 to Release 6.0 Version 9. No changes are made to these programs other than necessary fixes.

The *UC0D subvolumes (e.g., AT60UC0D, BA60UC0D, PS60UC0D) contain the conversion program for users migrating from Release 6.0 Version 9 to Release 6.0 Version 10. No changes are made to these programs other than necessary fixes.

If a Release 5.3 user wants to convert to the latest Version of Release 6.0, the conversion programs on *UC04 should be run. If the user is implementing TSS, the TSS-specific conversion programs on *UC05 should also be run.

If a Release 6.0 Version 1 user wants to convert to Release 6.0 Version 2, the conversion programs on *UC05 should be run.

If a Release 6.0 Version 2 user wants to convert to Release 6.0 Version 3, the conversion programs on *UC06 should be run.

If a Release 6.0 Version 3 user wants to convert to Release 6.0 Version 4, the conversion programs on *UC07 should be run.

If a Release 6.0 Version 4 user wants to convert to Release 6.0 Version 5, the conversion programs on *UC08 should be run.

If a Release 6.0 Version 5 user wants to convert to Release 6.0 Version 6, the conversion programs on *UC09 should be run.

If a Release 6.0 Version 6 user wants to convert to Release 6.0 Version 7, the conversion programs on *UC0A should be run.

If a Release 6.0 Version 7 user wants to convert to Release 6.0 Version 8, the conversion programs on *UC0B should be run.

If a Release 6.0 Version 8 user wants to convert to Release 6.0 Version 9, the conversion programs on *UC0C should be run.

If a Release 6.0 Version 9 user wants to convert to Release 6.0 Version 10, the conversion programs on *UC0D should be run.

If a Release 6.0 user wants to convert from Version 1 to Version 10, multiple conversion programs must be run starting with *UC05. If the user is implementing TSS, the TSS-specific conversion programs on *UC05 should also be run. Following this, the user needs to run the conversion programs on *UC06, *UC07, *UC08, *UC09, *UC0A, *UC0B and on *UC0C. Finally, the user should run the conversion programs on *UC0D.

Other Documents

In addition to this guide, the reader can obtain other BASE24 and NET24-XPNET publications that cover operational and product-specific information required to fully understand and implement the BASE24 Release 6.0 Version 10 software. Such manuals include the formal publications mentioned under the Publications heading in Section 3, the event message documentation located on the BA60LOGM and OK60LOGM subvolumes, and the following documents:

BASE24 6.0 Installation Guide (Informal) - BA-AE000-60

A primary reference document for information about the installation of a BASE24 Release 6.0 system from scratch. This document is included as a Microsoft Word document and is shipped on the BASE24 6.0 CD-ROM.

The BASE24 6.0 Installation guide is also provided on the software tape with the code. The document is located on the BA60UD04 subvolume as a file called BASE24. On this subvolume there is also a readme file called AAREAD04, which details how to move the install document to your hard drive and open it with Microsoft Word.

BASE24 Release 6.0 Interchange Interface Implementation Guide (Informal) - SW-DS0000-60

Describes the function and purpose of ACI's Switch 6.0 interfaces. It explains how customers can implement Switch 6.0 interfaces in their existing networks as well as in a BASE24 6.0 network. It also explains how customers can implement their Fortify.1/Fortify.3 Interfaces in a 6.0 network. This document is intended to assist customers in configuring and controlling their Switch 6.0 Interchange Interfaces in all BASE24 release networks and their Fortify.1/Fortify.3 Interchange Interfaces in a BASE24 Release 6.0 network. This document is included as a Microsoft Word document and is shipped with the Release 6.0 Interchange Interface software. This document is named NSTLSW60 and is located on the SW60UD04 subvolume.

Note: This guide is to be used with BASE24 releases 4.0, 5.0, 5.1.2, 5.3, and 6.0.

Release 6.0 Data File Impacts Document (Informal)

A primary reference document for information about the data file impacts and the methods that can be used to upgrade to BASE24 Release 6.0 Version 10. This document is included as a HP NSK TGAL document and is shipped with the BASE24 Release 6.0 Version 10 software. This document is named AAREAD0D and is located on the BA60UC0D subvolume.

HELP24 and HELP24 InfoLink

HELP24 and HELP24 InfoLink are excellent sources of assistance and information. The following access options appear on the support page of the ACI Worldwide Inc. Internet website (<http://www.aciworldwide.com/support>).

- ☐ A current listing of HELP24 phones worldwide.
- ☐ An option to send an e-mail message to HELP24.
- ☐ An option to sign up for HELP24 InfoLink.
- ☐ An option to access HELP24 InfoLink once you have received a user ID and password.

Once you can log on to HELP24 InfoLink, you can do the following:

- ☐ Access the HELP24 database to open a new case.
- ☐ Access installation and fix release procedures online.
- ☐ Access all BASE24, BASE24-eps, and NET24-XPNET documentation online.
- ☐ Search the site for information about BASE24.

The **HELP24 eSupport Users Guide** contains procedures for opening a new case, accessing the database, and accessing documentation on-line. Once you log on to HELP24, the path to the **HELP24 eSupport Users Guide** is Support Center> InfoLink > HELP24 eSupport Users Guide. Once you log on to HELP24, the path to the installation procedures is Support Center> InfoLink > Product Info >BASE24> BASE24 6.0 Install Info and Guides. The menu displayed gives you three options:

- ☐ Pre-Install Checklists- for BASE24 6.0 and ITS
- ☐ Install Guides—Procedures for installing a software release or Version for the first time.
- ☐ Retail Install Guides—Procedures for installing Retail 3.1 or Check Auth 3.2.

Once you log on to HELP24, the path to the Fix Information is Support Center> InfoLink > Fix Information > BASE24

Definition of Terminology

The following list has been defined to assist the reader in understanding the terms referred to within this document.

6.0 Interchange Interfaces

The BASE24 Interchange Interfaces developed in conjunction with BASE24 Release 6.0. These replace the Fortify.3 Interchange Interfaces and can be used with releases 4.0, 5.0, 5.1, 5.1.2, 5.3 and 6.0 of BASE24.

Accountholder

A person for whom a financial account has been established for some purpose. Accountholders may or may not use cards to initiate transactions on their accounts.

BASE24 User

An entity that has licensed BASE24 software products.

Cardholder

A person who has been issued a plastic card for the purpose of initiating transactions.

Consumer

A person who utilizes services offered by a BASE24 user who may or may not require a plastic card.

Conversion Program

A utility within existing code that reads the records from an input file, adds new fields and initializes appropriate fields and then writes the record to an output file. The input file (existing file) and output file (converted/initialized file) are different files. This method (versus the initialization program and online conversion) is required when the record structure changes (e.g., increasing the record length, adding/deleting fields from segments, etc.). This method can also be used to initialize existing user fields in a record.

This method requires system down time to convert the file (also requires an operator to manually rename the file that has been converted), move in the new objects of modules that access the file (e.g., device handler, auth, etc.) and then restart the objects. This method impacts all BASE24 users who use the file being converted even if they are not taking advantage of the enhancement for which the modifications are being made.

Extended Memory Table file (EMT)

A data file produced by the EMT Build utility program. Once the EMT has been built with the EMT Build utility, processes that use the EMT must be warmbooted (or started if not running).

Extended Memory Table Build utility (EMT Build)

The EMT Build utility is a stand-alone program used to build tables in extended memory from information contained in disk files. This utility can build hash tables, radix tables, or custom designed tables. Refer to the ***BASE24-atm Transaction Processing Manual***, ***BASE24-pos Transaction Processing Manual***, or the ***BASE24 ITS Kernel Configuration Manual*** for additional information about this utility.

Online (Inline) Conversion

A software routine within existing code that reads and updates each record in a file to initialize user fields that have been redefined as new fields for use by an enhancement. The file is converted in place (i.e., a new file is not created). The record is read, updated and then rewritten to the same file. This method (versus the conversion program and online conversion) cannot be used when the record structure changes (e.g., increasing the record length, adding/deleting fields from segments, etc.).

The online conversion code will be executed each time a record is accessed (versus the conversion and initialization programs that are run once). When a record is read, the process will determine if the value of a new field is valid. If the value is not valid, the field will be initialized to the default value. The database is gradually converted as records are read and updated.

This method requires an operator to stop a process, roll in the new object and start processes that use the files that are impacted. This method does not require any additional steps to be performed by an operator (e.g., running conversion/initialization programs, renaming files).

Operator

A person who carries out network operations functions, such as starting and stopping the system, entering ONCF commands, controlling logging, dealing with event messages, performing refreshes and extracts. The term *operator* also can refer to persons using BASE24 screens for such purposes as file maintenance, perusal, etc.

2: Functionality

This section contains an overview of the BASE24 Release 6.0 Version 10 enhancements.

Functionality has been organized by product, and within each product, according to whether the enhancement is a general enhancement or an optional enhancement.

- ☐ General enhancements are those enhancements that are inherent in the BASE24 Release 6.0 Version 10 offerings. These enhancements affect all BASE24 users of a specific product or module. BASE24 users must review the features that are part of these enhancements to ensure an understanding of the new processing and ensure proper configuration within the BASE24 environment.
- ☐ Optional enhancements are those enhancements that require a new separate licensable module. BASE24 users can implement these enhancements based on their own BASE24 processing and marketing requirements.

Enhancements are described in this section by product as follows:

- ☐ BASE24 Base (applicable to all products)
- ☐ BASE24-atm
- ☐ BASE24-pos
- ☐ BASE24-teller
- ☐ BASE24-remote banking
- ☐ BASE24 Interchange Interfaces
- ☐ BASE24 Value Added Products

BASE24 users should familiarize themselves with the BASE24 Base section (Section 4) as well as the applicable product-specific sections in order to obtain an understanding of the overall BASE24 Release 6.0 Version 10 offering.

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BASE24 Base Enhancements

The BASE24 Base product enhancements are divided into the following groups:

- ☐ General Enhancements
- ☐ Optional Enhancements

For more information about the enhancements in the BASE24 Base product, refer to Section 4.

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General Enhancements

The following enhancements are common to all products or provide the base for product-specific enhancements. As these enhancements are inherent in BASE24 Release 6.0 Version 10, BASE24 users must review them to ensure an understanding of the new processing and proper configuration within the BASE24 environment.

General Changes

- ☐ Changes for all BASE24 Release 6.0 Version 10 enhancements affecting the DDL files and the BASE24 AFT Subsystem files (i.e., screens, requesters, servers) have been merged and are cataloged under one SCN for each affected file. Applying changes for individual enhancements to the screen files does not allow BASE24 users to identify which changes go with a particular enhancement, as screen files do not have sequence numbers at the end of each line. Since the screen file changes usually involve new fields being added, all associated requester, server and DDL changes are required. Even though requester, server and DDL files do contain sequence numbers, all SCNs would need to be applied in order to accept the changes to the screen files. Therefore, all requester, server and DDL file changes, for all enhancements, were done under one SCN per file as well.
- ☐ The Card Prefix File (CPF) has been modified to increase the length of the ATM, POS and EMV segments for the Increase Size of Segments in the CPF enhancement, and to include additional fields on screen 8 for the Network 2008 Mandates for POS Acquiring enhancement. The Card Prefix File screen, requester and server have been modified to process the new fields. For further information, refer to the sections on the Increase Size of Segments in the CPF and the Network 2008 Mandates for POS Acquiring enhancements.
- ☐ SECUTILS Data Stack Relief
Many SECUTILS procedures allocate local message buffer arrays that are very large. This change modified several message buffers to relieve the data stack overflows occurring within the SECUTILS library of procedures.
- ☐ XPNET Output Buffer Fix
This change to the HISO and BIC ISO interfaces allowed the XPNET output buffer size to be larger and help with performance.

Increase Size of Segments in the CPF

The Increase Size of Segments in the CPF enhancement expands the ATM, EMV, and POS segments in the Card Prefix File (CPF). The BASE24 Release 6.0 Version 10 project to support Network 2008 Mandates for POS Acquiring requires the

expansion of the POS segment. Since both the ATM and EMV segments are also near capacity, this project expands all 3 segments at the same time.

By increasing the size of these segments, the ATM, EMV and POS segments of the CPF will also have enough space to support additional CSMs.

The existing segment lengths (in bytes) in the CPF are shown in the table below.

Base	ATM	POS	EMV
340	98	164	86

New segment lengths (in bytes) in the CPF are shown in the table below.

Base	ATM	POS	EMV
340	110	220	116

Prerequisites: None

Inhibit CVD Validation for EMV Transactions

The enhancement resolves an issue associated with iCVV processing by allowing customers to configure BASE24 to inhibit CVD validation on EMV transactions where the expiration date of the card is prior to the card verification date on the EMV segment of the appropriate Card Prefix File (CPF) record.

Prerequisites: EMV Extensions

Optional General Enhancements

There are no BASE24 Base Release 6.0 Version 10 enhancements that are optional.

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BASE24-atm Enhancements

The BASE24-atm product enhancements are divided into the following groups:

- ☐ General Enhancements
- ☐ Optional Enhancements

For more information about the enhancements in the BASE24-atm product, refer to Section 5.

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General Enhancements

The following BASE24-atm enhancements are inherent in the BASE24 Release 6.0 Version 10 offerings. All BASE24-atm users must review these enhancements in order to ensure an understanding of the new processing and proper configuration within the BASE24 environment.

General Changes

- ☐ Changes for all BASE24 Release 6.0 Version 10 enhancements affecting the DDL files and the BASE24 AFT Subsystem files (i.e., screens, requesters, servers) have been merged and are cataloged under one SCN for each affected file. Applying changes for individual enhancements to the screen files does not allow BASE24 users to identify which changes go with a particular enhancement, as screen files do not have sequence numbers at the end of each line. Since the screen file changes usually involve new fields being added, all associated requester, server and DDL changes are required. Even though requester, server and DDL files do contain sequence numbers, all SCNs would need to be applied in order to accept the changes to the screen files. Therefore, all requester, server and DDL file changes, for all enhancements, were done under one SCN per file as well.
- ☐ The ATM Terminal Data File (ATD) has been modified to support new additional offset/length areas to store data required for the Shared DH 912 BNA Support enhancement and the Shared DH 912 CPM Support enhancement. The ATD has also been modified to support an additional static field in the Diebold device-dependent data for the Shared DH 912 BNA Support enhancement and the Shared DH 912 CPM Support enhancement. These fields are displayed on ATD screen 10 for Diebold terminals, so there are also modifications to the Diebold specific screens. For further information, see Shared DH 912 BNA Support and Shared DH 912 CPM Support later in Section 5.
- ☐ The Terminal Data File (TDF) has been modified to include two additional fields in the Triton device-dependent data for the Triton MACing On/Off Support enhancement. These fields are not displayed on the TDF screens, so there are no modifications to the Triton specific screens. For further information, see Triton MACing On/Off Support later in Section 5.

ATM Bulk Check Infrastructure Support

The major ATM vendors, such as Diebold, NCR and Wincor Nixdorf, are introducing deposit devices in their ATMs, respectively the IDMBd, the SCPM and the Chequeln devices, that are capable of supporting Bulk Check deposit transactions. The Bulk Check deposit transactions allow the deposit of multiple checks at once without an envelope at the ATM, and allow the consumers at the ATM to view a scanned image

of their checks, and also verify the ATM reading of the courtesy amount. Once the checks are reviewed and approved for deposit by the ATM, the device sends the information about each check (MICR, amount etc.) to BASE24 in a single transaction for authorization.

There are 3 elements of support to be considered in BASE24-atm:

1. Acquiring bulk check transactions via the device handler.
2. Authorizing the bulk check transactions with BASE24-atm authorization components
3. Passing the bulk check data through to a back-end host.

This ATM Bulk Check Infrastructure Support enhancement focuses on the authorization function and on passing the bulk check data to a back-end-host.

Since acquiring the transaction is specific to each vendor environment and would therefore affect multiple device handlers, this function is not supported as standard BASE24-atm product. The individual device handlers catering for the NCD NDC, Diebold 912, Wincor NDC and Wincor DDC/912 message standards will need to be customized to support the acquiring function. That said, the individual customized device handlers will be able to leverage the authorization function provided as part of this Bulk Check Infrastructure Support enhancement.

1. Acquiring Bulk Check transactions

The expectations in terms of customizations to acquire the bulk check transactions are that, depending on the vendor software supported, the relevant BASE24 Device Handler should store the MICR data in a new Bulk Check MICR token, and the amount data in a new Bulk Check Amount token. In addition, a new Bulk Check Self Service Banking Check (SSBC) token should be used to store SSBC data for all the checks that are part of this single transaction.

2. Authorizing Bulk Check transactions with BASE24-atm authorization and
3. Passing the bulk check data through to a back-end host.

The enhancement, provided as part of BASE24 Release 6.0 Version 10, ensures the processing of this Bulk Check transaction by BASE24-atm Authorization. The processing depends on whether the Self Service Banking Enhanced ATM module or the Self Service Banking Enhanced Check Processing module is bound in. If the Self Service Banking Enhanced Check Processing module is present, the MICR data and amount from each check are interrogated in order to process the transaction successfully. Deposit credit is calculated on a check by check basis based on the check type (pre-approved, registered or personal). BASE24-atm Authorization adds a new token, the Bulk Check disposition token with an indicator for each check that identifies whether that particular check is to be deposited (approved) or returned to the customer (denied). The Amount field in the Standard Internal Message (STM) includes the sum of the amounts from all the checks to be deposited (approved) only.

Bulk Check transactions are logged to the Transaction Log File (TLF) as a single transaction, and passed to the Host via the HISO interface as a single transaction. The HISO interface was enhanced to be able to process this transaction with large amounts of MICR and amount data. The four new tokens are not placed in the external message in the current token buffer (Secondary field 126), as they are too large. Instead, other secondary fields in the external message that are not currently used for ATM transactions were redefined to carry this data. These new tokens are only present in the STM if the customer is configured to use the Self Service Banking enhanced check application. Bulk Check deposit transactions can be processed without using the Self Service Banking enhanced check application but in that case the authorization is based on the total deposit amount only. Host changes are therefore only required if the Self Service Banking enhanced check application is being used. In addition, a new value is allowed in the avail-dep-typ field and dep-typ of the STM. A value of 5 indicates a bulk check depositor. This allows customers to be able to identify transactions as performed at a bulk check device.

Prerequisites: AT-AE200 Self-Service Banking Enhanced ATM Application and AT-AE210 Self-Service Banking Enhanced Check Processing

Diebold Agilis 2.3 Upgrade

Diebold released the Agilis 91x terminal programming manual Version 2.3 in June 2008, and the corresponding Status Reference Manual in April 2008. Some of the changes introduced by the specifications in support of new ATM features or hardware fixes affect the communication between the ATM and BASE24-atm. As a result, the Diebold 10xx/478x Device Handler was enhanced to support the Agilis Version 2.3 specifications. Specifically the changes ensured:

1. Maintenance to functionality already supported by the Diebold 10xx/478x Device Handler, including:
 - ☐ Support for the Diebold ESC-ZV Screen Code to Allow Persistent Browser Windows
 - ☐ Support for Envelope Rejects Count
 - ☐ Support for New Coin Dispenser Status Messages
 - ☐ Support for New Alarm Status Messages
2. Support of anti-skimming device statuses for motorized card readers. This enables power on/off actions to be automated if a skimming device is detected on or removed from the ATM. This is an essential fraud prevention measure.
 - ☐ Support for Skimmer Code "CR01" for Message Type "5"

As mentioned in the ATM Bulk Check Infrastructure Support enhancement description above, the Diebold IDMbd hardware module introduced with Agilis 2.3, that enables the processing of bulk check deposits, will not be supported in the Diebold 10xx/478x Self Service Banking device handler extension. Any changes

related to acquiring bulk check deposit transactions from the IDMBd hardware module would need to be addressed as a CSM.

Support of the cash recycling hardware module (BCRM), voice guidance and the after-hour deposit module is not included in this enhancement and would also need to be addressed as CSMs.

Prerequisites: AT-DA150 Diebold 10XX/478X Device Handler

Diebold Agilis Enhanced Note Acceptor Upgrade

With Agilis Version 2.4, Diebold added support for an Enhanced Note Acceptor (ENA) module which is a replacement of the former Bunch Note Acceptor (BNA) on Diebold ATM devices. Similarly, the ENA provides the capability to deposit stacks of notes without an envelope at the ATM. BASE24 was modified to support this Diebold hardware module.

Any other changes related to Agilis Version 2.4 are not supported as part of this enhancement and would need to be addressed as CSMs.

Prerequisites: AT-DA150 Diebold 10XX/478X Device Handler

Diebold ENTIBOLD Entry Program Reorganization

This project was to reorganize the Diebold Entry Program to cater for the size of the program. BASE24 Release 6.0 Version 10 has introduced two new Entibold programs, one to support Standard Diebold functionality and the other to support Wincor functionality. The third, previously existing Diebold Entibold program which supports both the standard Diebold and the Wincor functionality remains but was not enhanced to support new functionality.

Prerequisites: AT-DA150 Diebold 10XX/478X Device Handler

Inhibit CVD Validation for EMV Transactions

This enhancement resolves an issue associated with iCVV processing by allowing customers to configure BASE24 to inhibit CVD validation on EMV transactions where the expiration date of the card is prior to the card verification date on the EMV segment of the appropriate Card Prefix File (CPF) record.

Prerequisites: AT-CC998 ATM Application EMV Extension

Issuer Fees for Non-Financial Transactions

In its 09.2 mandate, MasterCard provided support of a direct charging fee for domestic ATM transactions in the United States. As of 12 November 2009, the direct charging fee for domestic ATM transactions applied to Australian issuers as well.

Australian MasterCard members can apply the direct charging fee to balance inquiry, PIN change and PIN unblock transactions.

The direct charging fee is processed like a surcharge. BASE24-atm issuer surcharge functionality only supports financial transactions. This enhancement adds the ability to apply a surcharge to non-financial transactions.

Prerequisites: None

No Default Account on Withdrawal and Balance Inquiry

This change adds new functionality to both the Diebold and NCR device handlers. It allows a withdrawal and a balance inquiry to be processed by BASE24-atm without an account type present.

Prerequisites: None

Shared DH Maintenance

This group of BASE24-atm enhancements addresses issues raised by Wincor ATM customers requiring Wincor AID Table changes.

1. The first item is applicable to all customers running in a mixed NCR and Wincor NDC environment. This project increased the Data Objects List Size on the DCT Wincor AID Table Screen on BASE24 in order to match the size used on the NCR DCT Application ID Table and ensures customers can use the same EMV Data Object List when processing transactions acquired by either NCR or Wincor devices.
2. The second item consists of enhancing the Wincor EMV AID table for both the Wincor NDC and DDC/912 environments to provide customers the ability to delete AID records. This provides operational efficiency for customers dealing with a number of AID records that are no longer in use.

Prerequisites: AT-CC996 Shared NDC+ EMV Extension or AT-CC992 Shared Diebold 912 Device Handler EMV Extension

Shared DH Wincor EMV v1.2 Support

To align with EMVCO's requirements, on November 26, 2007, Wincor released Version 1.2 of its EMV hardware specifications. Specifically, the changes introduced were:

1. The addition of the Confirmed Candidates Checking Flag to the Wincor EMV Chip Card State 'e', Operation '000' (Contact Chip Card)
2. If a Data Object List for Solicited Status Message extension or Solicited EMV Status Message is specified in the Application Identifier table, then default tags are no longer used

3. In EMV Fault Status Messages, a Field Separator (FS) is sent instead of a Group Separator (GS)
4. Wincor EMV Chip Card State 'e', Operation '003' (Initialize EMV Transaction Data) is extended to allow setting of amount/other and optionally not to set the transaction type and currency
5. Added support for new EMV Data tags
6. Added support for additional values for the AID Options field in the Application Identifier Table

BASE24-atm customers acquiring EMV transactions with ATMs running Wincor Procash in a NDC or DDC/912 environment have encountered some issues related to these Wincor changes. This enhancement is intended to resolve these issues. Specifically, enhancements were made to the Shared NDC and Shared Diebold DDC/912 Device Handler EMV Extensions to support the changes introduced by Wincor in their EMV specifications Version 1.2.

Prerequisites: AT-CC996 Shared NDC+ EMV Extension, or AT-CC992 Shared Diebold 912 Device Handler EMV Extension

TSS New AKDS Message Format for the NCR and Diebold

This enhancement provides the ability to configure more than one set of keys within a single BASE24 system for each type of device. It is required for situations when different Host public keys are used for different regions or a fleet of ATMs is working from different Host public keys. This enhancement requires a minimum of BASE24-eps TSS Version 05.4.

Prerequisites: Shared NDC+ Device Handler AKDS Extension or Shared Diebold 912 Device Handler AKDS Extension

Triton MACing On/Off Support

The Triton Device Handler supports the processing of transactions with or without MACing. However, prior to this enhancement, once MACing had been enabled, there was no way to disable MACing via BASE24 or at the ATM. This enhancement provides the ability to send commands to the ATM to enable or disable MACing.

Prerequisites: AT-DA280 Triton Device Handler

Wincor SlimCash ATMs in NDC Environment

Wincor has released software products to allow operation of their SlimCash ATMs. A difference was identified between the SlimCash/NDC device and other already supported Wincor/NDC ATMs in BASE24-atm. Specifically, a case was submitted by a BASE24 customer stating there was a difference in the total number of bills the SlimCash ATM can dispense which created dispense errors in an NDC environment.

This difference was addressed to ensure Wincor SlimCash/NDC ATMs can be used by BASE24 customers using the NCR 5xxx Device Handler.

Prerequisites: AT-DA031 NCR 5xxx Device Handler

Optional Enhancements

The following BASE24-atm Release 6.0 Version 10 enhancements are optional. BASE24-atm users should review these changes to ensure an understanding of the new processing and proper configuration within the BASE24 environment before deciding to implement any of the changes.

Shared DH 912 BNA Support

This enhancement is based on Wincor's Nixdorf NDC_Diebold D91x Message Format Extension for CashIn V1.44 and provides support for the Wincor CashIn bunch note acceptor (BNA) for terminals controlled by the BASE24-atm Diebold 10XX/478X device handler. This functionality also enables BASE24 licensees who have terminals with BNAs provided by other vendors who have chosen to emulate Wincor's Diebold 912 standard.

The BNA provides consumers the ability to deposit stacks of notes into the device, as a single deposit, without an envelope. The notes can be of different denominations or currencies. The BNA unit utilizes sophisticated note recognition technology which counts and verifies bundles of notes, allowing consumers to deposit large amount of cash very quickly at their convenience at the ATM. BASE24 verifies the total deposit value based on the details of the currency, denomination and count of notes deposited. Unlike traditional envelope deposits, the full BNA deposit can be authorized on-line and credited for the total deposit value. BASE24 formats the transaction receipt to show the total deposit amount and the new current and ledger balance reflecting the deposit amount, giving the customer an immediate proof of deposit into their account. Deposited bills are transported into storage bins contained in the security enclosure.

Prerequisites: AT-DA150 Diebold 10xx/478x Device Handler, and AT-AE200 Self-Service Banking Enhanced ATM Application

The above enhancement is a separate licensable module: AT-DA156 Shared 912 Bunch Note Acceptor Device Handler Extension. Please contact your Customer Manager for further information.

Shared DH 912 CPM Support

This enhancement is based on Wincor Nixdorf NCD/Diebold D91x Message Format Extension for ChequeIn release 1.34 and provides support for the Wincor ChequeIn check processing module (CPM) for terminals controlled by the BASE24-atm Diebold 10XX/478X device handler. This functionality also enables BASE24 licensees who have terminals with CPMs provided by other vendors who have chosen to emulate Wincor's Diebold 912 standard.

A CPM may allow a bundle of up to 50 checks to be deposited at the ATM without an envelope. The device scans the checks, retrieves each check's MICR data, attempts to read the amount (via the optional OCR functionality), stores the images, or rejects any checks that are invalid, unrecognizable (i.e., torn, dirty, etc.), or improperly inserted, all before it sends the relevant transaction requests to the processing Host (BASE24).

At present, BASE24 supports single check and multiple check deposit transactions where each check results in a single transaction sent and processed individually by BASE24. In each case, the MICR data captured can be used for improved authorization, leveraging the feature available BASE24's Self Service Banking Enhanced Check Processing module, where the check data can be screened based on the type of check (registered, personal, etc.) The amount credited to an account can be defined based on the perceived risk for each check.

Prerequisites: AT-DA150 Diebold 10xx/478x Device Handler, AT-AE200 Self-Service Banking Enhanced ATM Application, and AT-AE210 Self-Service Banking Enhanced Check Processing

The above enhancement is a separate licensable module: *AT-DA155 Shared 912 Check Processing Module Device Handler Extension*. Please contact your Customer Manager for further information.

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BASE24-pos Enhancements

The BASE24-pos product enhancements are divided into the following groups:

- ☐ General Enhancements
- ☐ Optional Enhancements

For more information about the enhancements in the BASE24-pos product, refer to Section 6.

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General Enhancements

The following BASE24-pos enhancements are inherent in the BASE24 Release 6.0 Version 10 offerings. All BASE24-pos users must review these enhancements in order to ensure an understanding of the new processing and proper configuration within the BASE24 environment.

- ☐ Changes for all BASE24 Release 6.0 Version 10 enhancements affecting the DDL files and the BASE24 AFT Subsystem files (i.e., screens, requesters, servers) have been merged and are cataloged under one SCN for each affected file. Applying changes for individual enhancements to the screen files does not allow BASE24 users to identify which changes go with a particular enhancement, as screen files do not have sequence numbers at the end of each line. Since the screen file changes usually involve new fields being added, all associated requester, server and DDL changes are required. Even though requester, server and DDL files do contain sequence numbers, all SCNs would need to be applied in order to accept the changes to the screen files. Therefore, all requester, server and DDL file changes, for all enhancements, were done under one SCN per file as well.
- ☐ The POS Retailer Definition File (PRDF) has been modified to include an additional field on screen 4 for the Network 2008 Mandates for POS Acquiring enhancement. The POS Retailer Definition File screen, requester and server have been modified to process these new fields. For further information, see Network 2008 Mandates for POS Acquiring, where the impacts of this enhancement are described.

Address Verification (AVS) Support with No ZIP Code

The POS Address Verification module was enhanced to support conditions where address data is received without a ZIP code. As a means to improve customer service, while continuing to fight fraudulent transactions, BASE24 was enhanced to perform the AVS processing and return the appropriate address verification result code when no ZIP code is received from the acquirer, instead of declining these transactions.

Prerequisites: PS-AE110 POS Address Verification module

Contactless from Unattended PIN Required Terminals

This enhancement added functionality to the Router Authorization (RTAU) component for POS issuer transactions to support contactless transactions originating at terminals that are unattended and require a PIN.

Prerequisites: None.

Inhibit CVD Validation for EMV Transactions

The enhancement resolves an issue associated with iCVV processing by allowing customers to configure BASE24 to inhibit CVD validation on EMV transactions where the expiration date of the card is prior to the EMV CV DATE on the appropriate CPF record.

Prerequisites: AT-CC986 POS Application EMV Extension

Network 2008 Mandates for POS Acquiring

This enhancement ensures BASE24 compliance with MasterCard MDS 9.1 mandates. It addresses support for partial approvals, partial approval reversals and account balance responses, particularly where MasterCard's requirements are to provide the capability to authorize these transactions for an amount greater than the original request amount. This enhancement will primarily benefit petroleum and hotel payments that request pre-approval for a low amount, and are authorized for an amount greater based on a customer's available open-to-buy. SPDH was enhanced to support partial approvals associated with Purchase with Cash Back transactions.

Prerequisites: None

PTD Terminal Table Full Warning

This enhancement improves message processing and ensures that an event message is logged when the POS terminal table becomes full and terminal records cannot be read during online transaction processing. The enhancement allows both the Standard POS Device Handler and HyperCom POS Device Handler to monitor the condition of the table and produce an event message to alert operations when the terminal table is nearing full capacity. This will improve operability of the system.

Prerequisites: PS-DP140 Standard POS Device Handler or PS-DP390 HyperCom POS Device Handler

Optional Enhancements

The following BASE24-pos Release 6.0 Version 10 enhancements are optional. BASE24-pos users should review these changes to ensure an understanding of the new processing and proper configuration within the BASE24 environment before deciding to implement any of the changes.

There are no BASE24-pos Optional Enhancements.

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BASE24 Interchange Interface Enhancements

All BASE24 Interchange Interface product enhancements are divided into the following groups:

- ☐ General Enhancements
- ☐ Optional Enhancements

For more information about the enhancements in the BASE24 Interchange Interfaces, refer to Section 9.

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General Enhancements

There are no general enhancements to Interchange Interfaces.

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Optional Enhancements

There are no optional enhancements to Interchange Interfaces.

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BASE24 Value Added Products

All BASE24 value added product enhancements are divided into the following groups:

- ☐ General Enhancements
- ☐ Optional Enhancements

For more information about the enhancements in the BASE24 Value Added Products, refer to Section 10.

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General Enhancements

There are no general enhancements affecting the following BASE24 value-added product users:

- ☐ BASE24-card
- ☐ BASE24-customer service
- ☐ BASE24-infoBase
- ☐ BASE24-inventory Voucher Manager

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Optional Enhancements

There are no optional enhancements affecting the following BASE24 value-added product users:

- ☐ BASE24-card
- ☐ BASE24-customer service
- ☐ BASE24-infoBase
- ☐ BASE24-inventory Voucher Manager

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3: Support Services

This section summarizes the various support services available from ACI. These support services are described as follows:

- ☐ Product consulting and project management
- ☐ Publications
- ☐ Education
- ☐ HELP24
- ☐ ASTech services

Other services, besides the traditional Installation service, such as go-live support, onsite conversion assistance, and software methodology and utility training are available or are being planned; please contact your ACI Customer Manager for more details of the services ACI can offer.

Product Consulting and Project Management

No formal service programs have been created by ACI for upgrading customers to BASE24 Release 6.0 Version 10. Customers need to review this entire document and any referenced manuals, to determine what they will need to do to upgrade their BASE24 system.

After reviewing, customers should contact their ACI Customer Manager to discuss any services, such as project management, custom code upgrades, onsite staff, etc., required for upgrading, after reviewing all materials.

Project Planning Guide

The following Project Planning Guide is designed to assist in the successful upgrade of a BASE24 system to Release 6.0 Version 10. Information contained in this guide provides insight into required planning and control of the entire project.

Following this project guide helps ensure the realization of the goals and business objectives of the BASE24 user.

1. IDENTIFY PROJECT TEAM

One of the initial tasks for the BASE24 upgrade is for the user to assign a project team who will be responsible for the success of the project. A project manager and team members will need to be assigned and specific responsibilities established.

2. DETERMINE HARDWARE REQUIREMENTS

An ACI sales representative can assist in determining whether additional hardware will be required for the BASE24 upgrade. It is the BASE24 user's responsibility to order the hardware and establish the delivery date with HP NSK. HP NSK will perform diagnostic testing on the hardware when it is installed at the site. Should hardware encryption be required, it will be necessary to determine the hardware requirements. Additional hardware should be ordered for installation before testing can occur.

If applicable, the BASE24 user should order any new device firmware/software required to implement the enhancements they have chosen to support. The delivery date for the device firmware/software should be agreed upon with the vendor, as this time frame will affect the entire project.

3. IDENTIFY COMMUNICATION REQUIREMENTS

If the BASE24 user has identified additional protocol requirements, the communications lines and modems should be ordered. The customer must coordinate this effort with the local telephone company to ensure that the lines are configured in the least costly manner. The lines and modems must be installed and tested prior to the scheduled software installation date.

4. UPDATE HP NSK SOFTWARE REQUIREMENTS

ACI recommends that the BASE24 user be on a current HP NSK supported operating system to obtain maximum performance and support. BASE24 Release 6.0 requires a minimum of HP NSK Guardian D45 Operating System. Customers choosing to implement Format 2 File support are required to be on a minimum Operating System level of D46. In addition, BASE24 Release 6.0 requires TMF for several files. Customers choosing to implement BASE24-eps Transaction Security Services (known as TSS) with BASE24 Release 6.0 must review the BASE24-eps Hardware and Software Requirements for the BASE24-eps Release they will deploy.

5. UPGRADE TO XPNET 3.0 or 3.1

Customers should note that XPNET 3.0 is the minimum requirement for BASE24 Release 6.0.

6. PROJECT SCOPE MEETING

ACI does not believe this step is necessary for customers upgrading to BASE24 Release 6.0 Version 10.

7. MIGRATION PLANNING

Users will need to use this document, and any references within it, for detailed analysis when creating their migration plan.

8. DEVELOP PROJECT SCOPE DOCUMENT

ACI does not believe this step is necessary for customers upgrading to BASE24 Release 6.0 Version 10.

9. DEFINE REQUIREMENTS DEFINITION

The customer will be required to review existing CSMs to determine which ones will require retrofitting to the new version. With ACI's Fix Release methodology, BASE24 users should evaluate their CSMs to determine if they are handled as standard product in BASE24 Release 6.0 Version 10 or if they are specific to the customer's environment and will need to be upgraded to the new version.

10. DEVELOP PROJECT PLAN

The customer's project manager is responsible for developing an internal project plan. A review of the ACI project guidelines will help this person identify job responsibilities and establish time frames. A tracking mechanism should be included in the plan to ensure that the project is proceeding on schedule.

11. IDENTIFY CUSTOMER SPECIFIC MODIFICATION DEVELOPMENT

If the customer elects to have ACI develop modifications identified as required for the new version, a request will be submitted to ACI Software Development for a cost estimate. The functional descriptions and estimates will be sent to the customer for review and approval. Resources will be assigned and the final delivery dates established upon receipt of the signed contract.

Should the customer decide to develop any modifications in-house, resources will need to be assigned and scheduled. Once delivery dates are established, impacts to system testing can be determined.

12. DEFINE CERTIFICATION REQUIREMENTS

Customers currently on Fortify.3 will need to verify with each of the interchanges they connect to, to determine if they need to re-certify with the Interchanges with BASE24 Release 6.0 Version 10 code.

If re-certification is required, preliminary test times should be determined and a copy of the interchange test script secured.

The customer needs to set up test time with the interchange well in advance in order to insure time frames are met.

13. ESTABLISH TEST FACILITY

It is recommended that an area be designated specifically for testing the BASE24 Release 6.0 Version 10 software. The following items are recommended for the test area:

- ☐ 3 PCs
- ☐ Test devices
- ☐ Hard copy printer
- ☐ Test cards

14. INSTALL PRODUCTION HARDWARE

Installation of any additional computer hardware or peripherals is generally performed by the local hardware service organization or customer staff.

15. ATTEND BASE24 EDUCATION CLASSES

ACI offers a full suite of instructor led courses for BASE24. An enhancement course is also available upon request. The course descriptions and outlines can be viewed on the ACI Worldwide web site at www.aciworldwide.com.

16. DEVELOP INTERNAL PROCEDURES

There are no new internal procedures that ACI is aware of for BASE24 Release 6.0 Version 10 enhancements. User/Operational impacts are noted in the detailed sections for each enhancement described within this document.

17. DEVELOP SYSTEM TEST PLAN

The customer is responsible for developing a system test plan. Included in the test plan should be a description of each step specifying the test card to be used, the account to be used, the transaction to be executed, and the expected results. A procedure for retaining the results of the transaction processing should also be specified. Controlled testing and result verification procedures are key elements for successful system testing.

The execution of the system test plan will occur over a period of weeks to fully test all aspects of the system. BASE24 product functionality should be the first application tested. Any system modifications should then be thoroughly tested. The interface between the host and BASE24 should also be tested. This includes testing on-line authorization and the refresh and extract programs.

18. DEVELOP HOST SOFTWARE

During the course of a business day, BASE24 logs all completed and declined transactions to a transaction log file. After network settlement, the transaction log file can be extracted and used as input into the customer's nightly processing cycle on the host. If the customer would like to take advantage of the new features in BASE24 Release 6.0 Version 10, a host program will need to be developed to interpret the new token data that can be extracted with BASE24 Release 6.0 Version 10.

After the nightly processing cycle on the host or periodically during the business day the host may need to update card and account files used by BASE24 for pre-authorization or stand-in processing. If the customer would like to take advantage of the new features in BASE24 Release 6.0 Version 10 in these files, a host program will need to be developed to initialize the new data fields in BASE24 files.

19. INSTALL BASE24

BASE24 Release 6.0 Version 10 software will need to be installed at the customer site. Licensed BASE24 Release 6.0 Version 10 software modules will be provided to the customer upon request for installation on their system.

20. CONVERT DATA FILES

ACI provides database conversion programs for selected BASE24 files. These files are discussed in the detail sections for each enhancement described within this document.

21. CERTIFY INTERCHANGE

This task involves implementing the interchange test script and receiving notice of certification from the interchange. Current BASE24 users should not have to re-certify their Interchange Interfaces, unless they are implementing new functionality not previously tested or as directed by the interchange.

22. PERFORM SYSTEM TESTING

Full business day cycle testing of the system should be performed as determined by the customer testing plan and procedures.

23. DEVELOP CONVERSION PLAN

The customer should develop an implementation plan for converting existing devices and interchange links to BASE24 Release 6.0 Version 10. The installation of new devices and interchanges should also be included in this plan. A method of informing users of any changes to existing procedures should also be developed. The conversion schedule should be routed to all departments.

24. GO-LIVE

At the customer's request, an operational support analyst will visit the site one day prior to the "go-live" date to review the current software status and the BASE24 facilities. This individual will remain onsite to assist the customer with production cutover. The migration plan developed by the customer will be utilized in production cutover. On-site go-live support is a billable service.

25. COMPLETE POST IMPLEMENTATION REVIEW

In general, the purpose of this meeting is to ensure overall satisfaction of the customer's project.

26. TURN OVER TO CUSTOMER MANAGEMENT

If an ACI Project Manager has handled the BASE24 upgrade, a formal turnover meeting is held between the ACI Project Manager and the ACI Customer Manager to ensure a smooth transition in turnover of responsibilities for on-going customer support.

Publications

The BASE24 documentation set has been updated to include the BASE24 Release 6.0 Version 10 enhancements. The updated BASE24 documentation is available in three formats.

- ☐ **HELP24 eSupport InfoLink.** All files for the manuals in the BASE24 Release 6.0 documentation set are located on the HELP24 eSupport InfoLink database. You have the option of copying the files for individual manuals from the HELP24 eSupport InfoLink database to your PC.

In addition, the HELP24 eSupport InfoLink database contains documentation that is added or revised between ZIP files or CD-ROMs.

- ☐ **Software Management Electronic Distribution.** ZIP or ZIP Executable files that contain the same contents as the CD-ROM are available through the ACI electronic distribution system.
- ☐ **CD-ROM.** The CD-ROM contains all PDF files for the manuals in the documentation set.

The BASE24 Release 6.0 documentation set contains the most current version of each document. If a BASE24 manual has not been updated for BASE24 Release 6.0, the manual for the prior release appears on the BASE24 Release 6.0 menu with a note identifying the prior release.

Documentation for XPNET 3.0 and XPNET 3.1 is available in a separate set titled NET24-XPNET 3.1 that is available in the same formats as the BASE24 documentation.

BASE24 customers can download files from the HELP24 eSupport InfoLink database or order the documentation ZIP files or CD-ROM from their ACI Customer Manager.

Education

BASE24 Technical Education offers a complete set of one- to four-day modules covering the components of the BASE24 system. BASE24 courses are structured into a modular format that eliminates repetitious material, provides information specific to certain job functions, offers hands-on lab experience, and enhances retention of information taught.

A BASE24 Enhancement course is also available upon request for existing customers upgrading to BASE24 Release 6.0 Version 10.

The modular format enables BASE24 users to choose modules applicable to their employees' job responsibilities. This provides consolidated training time and offers opportunities for additional employees to receive education.

The course descriptions, outlines and current education schedule can be viewed on the ACI Worldwide web site at www.aciworldwide.com.

HELP24

Please use ACI Worldwide's eSupport site to report problems or ask questions about BASE24 Release 6.0 Version 10. eSupport is an Internet Web based application that interfaces with the HELP24 database. Using eSupport, you can search our KnowledgeBase Solutions for information and resolutions to problems you are experiencing. You can also directly open cases. An eSupport option, InfoLink, is your link to a wealth of information that can help you in your day to day business. It allows you on-line access to ACI Publications, Fix History Information, HELP24 Hotlines and much more. Contact your local HELP24 office for an eSupport logon.

As a result of reviewing new troubleshooting methods, HELP24 has acquired a tool called WebEx. This is a state of the art web-based technology that allows us to troubleshoot issues in a secure, online environment. Compared to the dial-up environment we have used in the past, WebEx provides convenient access, a more stable connection and most importantly, it provides a collaborative environment which also allows you to actively participate in the session by viewing and asking questions about everything that goes on.

To address audit security requirements, WebEx provides the option to record the session so all steps can be documented in a file that can be accessed after the fact if necessary. WebEx has a file transfer option that provides the ability to transfer files from your PC to ours, or vice versa. This feature can be used in the event the file(s) are too large to email.

For more information, please refer to the WebEx User's Guide located under Product Info on Infolink. Contact your local HELP24 office to schedule your WebEx demonstration.

ASTech Services

ACI offers customer support services through our ASTech Division which allows the selection of a la carte support services to meet individual customer requirements. ASTech, ACI Support and Technical Services, is a service support team which provides the customer with a variety of on-site and remote technical support. ASTech provides customers with supplemental staffing, system performance analysis, review and/or upgrading of previous release customer-specific software development, and detailed operational technical training.

The following are example descriptions of the various services ASTech could undertake to assist you during the BASE24 upgrade. The number of staff weeks identified to accomplish these tasks is estimated and will vary depending on the amount of customer specific code to be developed and delivered by ACI.

Your ACI Customer Manager can provide you with details of services available in your region that have been tailored to the special needs of different areas of the world.

Customer Software Analysis

Effort: Two to four staff weeks

ASTech personnel will perform a complete review of the customer's BASE24 software system and determine what current modifications (CSM code) exist at the customer location. ASTech personnel will perform a detailed comparison of existing customer specific software modifications to BASE24 Release 6.0 Version 10 functions. Upon completion of the review, the customer will be provided with a document that identifies all modules that contain code unique to their BASE24 system (file names, custom functions, custom function applicability, etc.).

Conversion Planning

Effort: One and one-half to three staff weeks depending on the size and current system configuration.

This service option provides the customer with comprehensive support during the system migration to the new version. ASTech assigns the customer a Technical Project Lead, or Tech Lead from ACI. The Tech Lead will be responsible for ensuring the success of the BASE24 upgrade project.

The Tech Lead will:

- ☐ Review the migration options and develop a detailed migration plan for the conversions project
- ☐ Review the Project Scope Document on behalf of the customer (if applicable)
- ☐ Review the customer's internal procedures to accommodate BASE24 Release 6.0 Version 10 functions.

Upgrade Custom Code

Effort: The amount of time necessary for this task varies depending on the number of modules identified as having modifications unique to the customer. Systems with a great number of customizations may take up to sixteen staff weeks. Actual estimates would be provided upon completion of the Customer Software Analysis.

Prerequisite: The Customer Software Analysis to identify all code unique to the customer.

With this service, ACI will upgrade all custom code to BASE24 Release 6.0 Version 10. This includes unit testing of the code. Customizations that are no longer necessary will be removed. Customizations that must be carried forward can either be upgraded in their current form or be reengineered to take advantage of new functionality in BASE24 Release 6.0 Version 10. Reengineering customizations will add time to the estimate.

System Test Planning

Effort: One to four staff weeks depending on the results of the Customer Software Analysis and the number of BASE24 products to be tested. Some of this time will be on-site to perform the System Test Plan.

Prerequisite: The Customer Software Analysis to identify all code unique to the customer.

ASTech personnel will develop and execute a BASE24 System test plan designed for the customer's specific needs. This service also includes the performance and documentation of the system acceptance testing.

Switch and Host Certification Support

Effort: Two to five staff weeks depending on the results of the Customer Software Analysis and the number of BASE24 products and interfaces to be certified. All of this time will be on-site time.

Prerequisite: The Customer Software Analysis to identify all code unique to the customer.

This service option allows ASTech personnel to provide complete support of the various switch and/or host communication certification testing. This includes reviewing and refining the test scripts, performance of certification testing functions, evaluation of the testing results, and documentation and presentation of the results obtained during the certification testing.

Go-Live Support

Effort: One week (on-site).

ASTech personnel will be at the customer site assisting in all conversion tasks and being present for go-live support. This includes remaining on-site for a couple of days to ensure system stability.

Post Go-Live Support

Effort: Varies.

After a successful go-live, a Tech Lead will provide on-going support which may be required by the customer. Tech Lead support can be provided at the customer location and/or remotely through ACI.

4: BASE24 Base

Section 4 discusses the changes and enhancements made to the BASE24 Base modules in Release 6.0 Version 10, along with the resulting conversion and implementation requirements that existing BASE24 users must consider when upgrading from Release 6.0 Version 9.

Changes and enhancements to the BASE24 Base modules impact all users. Consequently, all users should review the material in this section to determine how it applies to them.

This section is divided into two parts: General Enhancements and Optional Enhancements.

General enhancements describe the Base modules changes that affect all BASE24 users using a particular product or module, along with the associated conversion and implementation requirements that all BASE24 users must address.

Optional enhancements describe the Base modules changes that affect only those BASE24 users that choose to implement a new separate licensable module, along with the associated conversion and implementation requirements that BASE24 users must address to implement the enhancements.

Section 4 must be used in conjunction with Sections 5 through 10 to identify the changes that BASE24 users need to consider when upgrading to Release 6.0 Version 10.

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Standard Information Format

A standard format is used in Sections 4 through 10 to highlight and describe the various changes, conversion requirements, and implementation requirements BASE24 users need to consider when upgrading to Release 6.0 Version 10. This standard format is defined below.

In situations where the following information does not apply, the information is labeled as “Not applicable”

Software Impacts

Describes the software modules impacted for a given change or enhancement and the nature of the impacts. This information includes a high level list of the modules impacted, how the modules are impacted (i.e., inline changes, hooks, dummy procs, extensions) and which customers are affected (i.e., all, only those implementing the enhancement).

CUSTMACS Changes

Describes CUSTMACS changes required to implement a given change or enhancement. This information includes Make defines that have been added, deleted, or modified.

LCONF Assign Changes

Describes LCONF assign changes required to implement a given change or enhancement. This information includes assigns that have been added, deleted, or modified, along with an explanation of how these assigns can or must be set.

LCONF Param Changes

Describes LCONF param changes required to implement a given change or enhancement. This information includes params that have been added, deleted, or modified, along with an explanation of how these params can or must be set.

Database Configuration

Describes data file changes required to implement a given change or enhancement. This information includes file fields that have been added, deleted, or modified, along with an explanation of how these fields can or must be set. It also identifies file records and record segments that must be added, deleted, or modified.

New Data Files

Identifies new data files that must be created in order to implement a given change or enhancement.

Template Changes

Identifies file templates that must be created or modified in order to implement a given change or enhancement. It also identifies COBNAMES and Device Handler Names (xxxxNAMS) file modifications that must be made.

Device Configuration

Identifies device configuration changes that are required in order to implement a given change or enhancement.

Mega Table (MEGATBL) Changes

Identifies Mega Table changes that are required in order to implement a given change or enhancement.

PI-Table Changes

Identifies Product Indicator (PI) Table changes that are required in order to implement a given change or enhancement.

SEG-Table Changes

Identifies Segment Indicator (SEG) Table changes that are required in order to implement a given change or enhancement.

Security Table (SECTBL) Changes

Identifies Security Table changes that are required in order to implement a given change or enhancement.

PATHCONF Changes

Identifies PATHCONF changes that are required in order to implement a given change or enhancement.

CRT Access Security Changes

Identifies CRT Access security changes that are required in order to implement a given change or enhancement. This includes changes that need to be made to BASE24 operator security records.

Routing Changes

Identifies routing and configuration changes that are required in order to implement a given change or enhancement. This includes required changes to files such as SPROUTE and the BASE24-pos Authorization Selection Table (AST).

Network Configuration

Identifies configuration changes that are required in order to implement a given change or enhancement. This includes NEF changes.

File Conversions

Identifies files that need to be converted in order to implement a given change or enhancement. This implies that a given file could be impacted by multiple changes and/or enhancements. BASE24 users need to consider all of the impacts to any one specific file before the actual conversion of a file.

Host/Co-Network Impacts

Identifies impacts to a user's host or co-network as a result of a given change or enhancement. This includes changes to the external messages used to communicate with hosts and co-networks as well as changes to refresh and extract tape formats.

User/Operational Impacts

Identifies potential changes to a user's internal BASE24 operating procedures. This includes changes to operations and file maintenance screens, operator-initiated commands, and reports.

Vendor Impacts

Identifies any issues that need to be addressed with vendors in order to implement a given change or enhancement. This includes hardware, firmware, and software changes that vendors need to make in order for their products to function with the given change or enhancement.

Reference Documents

Lists any outside sources used in the design of a given change or enhancement. BASE24 users can refer to these documents for additional information.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24 users must address in order to upgrade to Release 6.0 Version 10.

- ☐ General Changes
- ☐ Increase Size of Segments in the CPF
- ☐ Inhibit CVD Validation for EMV Transactions

These areas are summarized on the following pages.

Section 2 contains a functional overview of the general enhancements described in section 4.

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General Changes

Changes for all BASE24 Release 6.0 Version 10 enhancements affecting the DDL files and the BASE24 AFT Subsystem files (i.e., screens, requesters, servers) have been merged and cataloged under one SCN for each affected file. Applying changes for individual enhancements to the screen files does not allow BASE24 users to identify which changes go with a particular enhancement, as screen files do not have sequence numbers at the end of each line. Since the screen file changes usually involve new fields being added, all associated requester, server and DDL changes are required. Even though requester, server and DDL files do contain sequence numbers, all SCNs would need to be applied in order to accept the changes to the screen files. Therefore, all requester, server and DDL file changes, for all enhancements, were done under one SCN per file as well.

The Card Prefix File (CPF) has been modified to increase the length of the ATM, POS and EMV segments for the Increase Size of Segments in the CPF enhancement, and to include additional fields on screen 8 for the Network 2008 Mandates for POS Acquiring enhancement. The Card Prefix File screen, requester and server have been modified to process the new fields. For further information, refer to the sections on the Increase Size of Segments in the CPF and the Network 2008 Mandates for POS Acquiring enhancements.

The country codes and currency codes used by BASE24 have been modified where appropriate to ensure that they match the values defined by MasterCard and Visa.

SECUTILS Data Stack Relief

The BASE24 Security Utilities (SECUTILS) have been modified to reduce the amount of data they place on a process' data stack. A targeted amount of local buffer data has been relocated to a flat extended memory segment, reducing the size of the data stack by approximately 4500 bytes, or 7% of the total 32K-word stack space available to a process. This will help prevent data stack overflows from occurring within the SECUTILS.

A complete list of the files for this enhancement can be found in the SCN file BA60UD0D.SCNSECU.

XPNET Output Buffer Fix

When response messages sent via XPNET are larger than the corresponding request messages, high transaction volume situations can result in "Reverse Channel Writes". Reverse channel writes occur when it is not possible to fit all of the blocked response messages into a single reply buffer that corresponds to the blocked request messages in a request buffer. Reverse channel writes ensure that the remaining replies are sent, but are WAITED I/O and can be extremely detrimental to system performance. Note that the issue does not arise if request messages are larger than corresponding response messages.

The solution is to set the reply buffer size specified in the call to NETINIT sufficiently large so that all responses will fit into the reply buffer. Typically, calls to NETINIT from BASE24 interface processes set the maximum output block size to 4096 bytes (arising from use of a SKEL literal in the call to NETINIT). The ISO Host Interface and ISO BIC Interface have been enhanced to change the maximum output block size to 16384 bytes. Additionally, the value of the MAXPROMSG setting in the node must be equal to or greater than 16384.

Software Impacts

All BASE24 customers:

- ☐ Inline changes to BA60AFT.RQCPFS, SCRNCPPF, SVCPPFS
- ☐ Inline changes to BA60DDL.DDLBCNST, DDLFCPPF
- ☐ Inline changes to BA60DEFS.DDLGDEFS
- ☐ Inline changes to BA60SRC.BAUTILS, SECUTILM, SECUTILS
- ☐ Inline changes to OK60LIB.OKLIBTS

All BASE24 Banknet Interface customers:

- ☐ Inline changes to SW60BNET.BNETS

All BASE24 ISO BIC Interface customers:

- ☐ Inline changes to SW60BABI.BABICIM, BABICIS

All BASE24 ISO Host Interface customers:

- ☐ Inline changes to BA60HISO.BAHISOS

The following make files were changed to add a dependency for BA60DEFS.DDLGDEFS:

- ☐ Inline changes to BA60ASRC.SECAKDM – BASE24 AKDS Security Utilities
- ☐ Inline changes to BA60HISF.BAHISFM – BASE24 ANSI Host Interface
- ☐ Inline changes to BA60IEMV.CAMUTILM – BASE24 CAM Utilities
- ☐ Inline changes to BA60QDCV.DCVUTILM – BASE24 Dynamic Card Verification Utilities
- ☐ Inline changes to AT601000.C1000M – BASE24-atm Diebold Device Handler
- ☐ Inline changes to AT60ATDL.TIDLAKDM – BASE24-atm Tidel AKDS Extension
- ☐ Inline changes to AT60ATRT.TRITAKDM – BASE24-atm Triton AKDS Extension
- ☐ Inline changes to AT60HISF.ATHISFM – BASE24-atm ANSI Host Interface
- ☐ Inline changes to AT60IW10.C10WEMVM – BASE24-atm Diebold Wincor EMV
- ☐ Inline changes to AT60N50.N50M – BASE24-atm NCR Device Handler
- ☐ Inline changes to AT60TIDL.TIDELM – BASE24-atm Tidel Device Handler
- ☐ Inline changes to AT60TRIT.TRITONM – BASE24-atm Triton Device Handler
- ☐ Inline changes to AT60YATH.AUTHTKIM – BASE24-atm Authorization TSS Key Index Utilities

- ☐ Inline changes to AT60YHIS.ATHISTKM – BASE24-atm HISO TKI Utilities
- ☐ Inline changes to PS60HISF.PSHISFM – BASE24-pos ANSI Host Interface
- ☐ Inline changes to PS60YHIS.PSHISTKM – BASE24-pos HISO TKI Utilities
- ☐ Inline changes to SW60ATBI.ATBICIM – BASE24-atm BIC ISO
- ☐ Inline changes to SW60CSSI.CSSIMM – BASE24 Cash Station ISO
- ☐ Inline changes to SW60PSBI.PSBICIM – BASE24-pos BIC ISO
- ☐ Inline changes to TB60HISO.TBHISOM – BASE24-telebanking HISO

Note: The list of files impacted by the merged changes is also included in the **Software Impacts** section of the applicable enhancement(s).

A complete list of the merged files can be found in the SCN file
BA60UD0D.SCNMERGE.

CUSTMACS Changes

Not Applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Card Prefix File (CPF)

NOTE: The CPF conversion program must be run to convert the CPF. No inline conversion is provided due to the increased size of some segments. Running the conversion program is required for all BASE24 customers.

The CPF conversion program on the BA60UC0D subvolume can be used to convert the CPF records from the BASE24 Release 6.0 Version 9 format to the BASE24 Release 6.0 Version 10 format. Follow the instructions in the BA60UC0D.CNVCPFR obey file for an automated conversion.

As some segment sizes have increased in BASE24 Release 6.0 Version 10, users must ensure that the new record length is correct when the new CPF is created. For further information on the new segment sizes, refer to the section on the Increase Size of Segments in the CPF enhancement.

If modifications to the conversion program are necessary, the source code may be modified by editing the CNVCPFS file.

To compile the conversion program, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#cpfcnv.t,name $cpfm,nowait/ cnvcpfm -  
force -list -listreplace
```

For BASE24 Release 5.3 customers, the CPF conversion program on the BA60UC04 subvolume can be used to convert the CPF records from the Release 5.3 format to the Release 6.0 Version 10 format. Before converting the CPF file, a new empty CPF file must be created using FUP and the BADDLFUP to correctly set the new record length.

BASE24 Release 5.3 users should update the CNVCPFR obey file on the BA60UC04 subvolume with the appropriate information. BASE24 Release 6.0 Version 9 users should update the CNVCPFR obey file on the BA60UC0D subvolume with the appropriate information. The CNVCPFR should then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREAD04 file on the BA60UC04 subvolume for Release 5.3 customers and to the AAREAD0D file on the BA60UC0D subvolume for Release 6.0 Version 9 customers.

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Increase Size of Segments in the CPF

This section describes the implementation requirements associated with the Increase Size of Segments in the CPF enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Increase Size of Segments in the CPF enhancement.

The Increase Size of Segments in the CPF enhancement expands the ATM, EMV, and POS segments in the Card Prefix File (CPF). The BASE24 Release 6.0 Version 10 project to support Network 2008 Mandates for POS Acquiring requires the expansion of the POS segment. Since both the ATM and EMV segments are also near capacity, this project expands all 3 segments at the same time.

While designing the changes required for the Network 2008 Mandates for POS Acquiring project, it was determined that additional space in the POS segment of the CPF would be needed. There should also be some room for growth in the event that a customer decides to modify this file in the future.

The ATM segment is very close to being at its capacity. A customer may need to enhance the BASE24 product to support any new features. To allow this to happen more easily, there should be some room for growth in the ATM segment of the CPF.

The EMV segment is very close to being at its capacity. A customer may need to enhance the BASE24 product to support any new features. To allow this to happen more easily, there should be some room for growth in the EMV segment of the CPF.

The Card Prefix File (CPF) is a segmented file. In Release 6.0 Version 9, the number of spare bytes available for ACI use in the Base, ATM, POS and EMV segments is shown in the table below.

Segment	Bytes available for ACI use	Bytes required for R6.0 Version 10 projects
<i>Base</i>	<i>1</i>	<i>0</i>
<i>ATM</i>	<i>1</i>	<i>0</i>
<i>POS</i>	<i>3</i>	<i>6</i>
<i>EMV</i>	<i>2</i>	<i>0</i>

In addition to increasing the length of the ATM, POS and EMV segments, part of the regional user field in the Base segment has become reserved for ACI use, but the length of the Base segment has not be increased.

The existing and new segment lengths (in bytes) in the CPF are shown in the table below.

Lengths	Base	ATM	POS	EMV	Card	SSBC	NCD	SV
<i>Current</i>	<i>340</i>	<i>98</i>	<i>164</i>	<i>86</i>	<i>478</i>	<i>72</i>	<i>110</i>	<i>96</i>
<i>New</i>	<i>340</i>	<i>110</i>	<i>220</i>	<i>116</i>	<i>478</i>	<i>72</i>	<i>110</i>	<i>96</i>

For the Base, ATM, POS and EMV segments, additional information about the changes is shown in the table below.

Segment	Bytes currently available for ACI use	Bytes added for ACI use in R6.0 Version 10	Bytes required for R6.0 Version 10 projects	Bytes available for future use
<i>Base</i>	<i>1</i>	<i>20</i>	<i>0</i>	<i>21</i>
<i>ATM</i>	<i>1</i>	<i>12</i>	<i>0</i>	<i>13</i>
<i>POS</i>	<i>3</i>	<i>56</i>	<i>6</i>	<i>53</i>
<i>EMV</i>	<i>2</i>	<i>30</i>	<i>0</i>	<i>32</i>

Software Impacts

All BASE24 customers:

- ☐ Inline changes to BA60DDL.DDLFCPF.
- ☐ Inline changes to BA60AFT.RQCPFS, SVCDFS.
- ☐ Inline changes to BA60EXT.OMFXG, OMFXS.
- ☐ Inline changes to BA60UC04.CNVCPFS.
- ☐ New conversion program (BA60UC0D.CNVCPFS) to convert the CPF to the required format.

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD0D.SCNCPI.

The following BASE24 processes use the CPF DDL, and must be recompiled and rebound to reference the extended segments:

- ☐ BASE24-atm Authorization;
- ☐ BASE24-pos Device Handler/Router/Authorization;
- ☐ BASE24-telebanking CPF Extended Memory Table (EMT).

Note that the BASE24-teller Authorization process should also be recompiled and rebound. However, this uses only the Base segment of the CPF (which is not being extended), and so it is not strictly necessary to rebuild this code.

CUSTMACS Changes

Not Applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

The CPF conversion program must be run to convert the Card Prefix File (CPF). The CPF conversion program converts the customer's current CPF to a new record format. Before converting the CPF file, a new empty CPF file must be created using FUP and BADDLFUP to correctly set the new record length. The BA60UC0D.CNVCPFR file automates the CPF conversion.

Details of this conversion program can be found in the **General Changes** earlier in Section 4.

Host/Co-Network Impacts

Super Extract Process

Host code changes are required for all BASE24-pos users that use OMF Extract to accept and process the new fields in records containing a CPF record image.

User/Operational Impacts

Disk Impacts

The CPF record length has increased as a result of this enhancement. However, this change may not affect the disk space required to hold the CPF, due to the way that ENSCRIBE manages disk blocks.

Where CPF records contain a Base segment and a POS segment only, and where CPF records contain a Base, ATM, POS, EMV and NCD segment, additional disk space for the CPF may be required. For any other valid combination of these five segments, additional disk space for the CPF is not required.

The CPF also supports Card, SSBC and Stored Value (SV). Where CPF records contain one or more of these segments, additional disk space for the CPF may be required.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Inhibit CVD Validation for EMV Transactions

This section describes the implementation requirements associated with the Inhibit CVD Validation for EMV Transactions enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Inhibit CVD Validation for EMV Transactions enhancement.

Section 5 contains BASE24-atm specific requirements associated with the Inhibit CVD Validation for EMV Transactions enhancement.

Section 6 contains BASE24-pos specific requirements associated with the Inhibit CVD Validation for EMV Transactions enhancement.

The Inhibit CVD Validation for EMV Transactions enhancement allows BASE24 to be configured to inhibit CVD validation on EMV transactions, where the expiration date of the card is prior to the card verification date on the EMV segment of the appropriate Card Prefix File (CPF) record.

When EMV cards were first issued, the Card Verification Digits (CVD) encoded on the Track-2 Equivalent Data on the EMV card were typically the same as the CVD encoded on the magnetic stripe Track-2 on the same card. As a fraud prevention measure, Visa and MasterCard subsequently decided that the CVD encoded on the Track-2 Equivalent Data should differ from the CVD encoded on the magnetic stripe Track-2. A different value for the CVD is created by replacing the 3-digit service code (used in the generation of the conventional CVD) with a fixed value (typically '999').

For Visa, the CVD encoded on the magnetic stripe Track-2 are known as the CVV, and the CVD encoded on the Track-2 Equivalent Data are known as the iCVV. For MasterCard, the CVD encoded on the magnetic stripe Track-2 are known as the CVC1, and the CVD encoded on the Track-2 Equivalent Data are known as the Chip CVC. For the purposes of this document, the CVD encoded on the magnetic stripe Track-2 is referred to generically as the CVD, and the CVD encoded on the Track-2 Equivalent Data is referred to generically as the iCVD.

The card verification date (EMV CV DATE) field on the EMV segment of the CPF record is designed to hold the expiration date of the first cards (for that prefix) encoded with the iCVD (as opposed to the conventional CVD) on the EMV Track-2 Equivalent Data. So, for example, if an issuer started encoding the iCVD on cards valid from January 2008, and the cards for that prefix have a 3-year validity period, then the EMV CV DATE would be set to December 2010. In this way, BASE24 would

know to perform conventional CVD validation on cards expiring before December 2010, and iCVD validation on cards expiring at the end of December 2010 or later.

Support for iCVD validation was added to BASE24 in Release 6.0 Version 4, and the Implementation Guide includes the following text:

It is assumed that all EMV cards with the same prefix have CVD encoded on the magnetic stripe and CVD encoded on the chip. Older cards may have the same CVD on both the magnetic stripe and the chip, whereas newer cards may have different CVDs.

This assumption is valid as long as there is a fixed date (for a specific prefix) where all EMV cards expiring up to that date contain the conventional CVD on the EMV Track-2 Equivalent Data, and all cards expiring after that date contain the iCVD on the EMV Track-2 Equivalent Data. However, case #485844 identifies a BASE24 customer that has EMV cards expiring at the same time, some of which have been encoded with the iCVD and some of which have been encoded with the conventional CVD (on the EMV Track-2 Equivalent Data). In an attempt to resolve this issue, the customer set the EMV CV DATE to the expiration date from which ALL cards have been encoded with the iCVD, but this configuration resulted in an issue with those cards expiring prior to the EMV CV DATE. Essentially, BASE24 assumes that all of these cards have been encoded with the conventional CVD, whereas some of these cards have actually been encoded with the iCVD.

This enhancement resolves the issue by providing the option of configuring BASE24 to inhibit CVD validation on EMV transactions, where the expiration date of the card is prior to the EMV CV DATE on the appropriate CPF record. For these cards, it is irrelevant whether the EMV Track-2 Equivalent Data is encoded with the iCVD or with the conventional CVD, because the enhanced logic ensures that the value is not checked during authorization processing.

New values are supported in the CHECK METHOD field in the EMV segment of the CPF record to allow this option to be configured. This field is checked during EMV card verification processing, and the values currently supported in the field are defined as follows:

CHECK METHOD — Specifies the algorithm and data to be used when card verification is performed on EMV transactions. Valid values are as follows:

0 = Use non-EMV card verification.

1 = Replace the service code with EMV-CV-DATA.

Implicitly, this parameter has applied only to cards expiring on or after the EMV CV DATE, because conventional card verification processing was performed on cards expiring before the EMV CV DATE. With the introduction of support for new values in

the CHECK METHOD field, the definition of each value explicitly includes its relationship with the EMV CV DATE:

CHECK METHOD — Specifies the algorithm and data to be used when card verification is performed on EMV transactions. Valid values are as follows:

- 0 = Use non-EMV card verification for cards expiring on or after the EMV CV DATE; also use non-EMV card verification for cards expiring before the EMV CV DATE.**
- 1 = Replace the service code with EMV CV DATA for cards expiring on or after the EMV CV DATE; use non-EMV card verification for cards expiring before the EMV CV DATE.**
- 2 = Use non-EMV card verification for cards expiring on or after the EMV CV DATE; do not perform card verification on EMV transactions for cards expiring before the EMV CV DATE.**
- 3 = Replace the service code with EMV CV DATA for cards expiring on or after the EMV CV DATE; do not perform card verification on EMV transactions for cards expiring before the EMV CV DATE.**

The new values ('2' and '3') result in CVD validation not being performed on EMV transactions, if the expiration date of the card is prior to the EMV CV DATE.

Software Impacts

All BASE24 customers:

- ☐ Inline changes to BA60DDL.DDLFCPF.
- ☐ Inline changes to BA60AFT.SCRNCPF, RQCPFS, SVCPFS.

For full details on the ATM software impacts for this enhancement see Software Impacts in Section 5 (ATM) of this document.

For full details on the POS software impacts for this enhancement see Software Impacts in Section 6 (POS) of this document.

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD0D.SCNCVDI.

CUSTMACS Changes

Not Applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

This section describes the enhanced field in the CPF used that can be configured to support this enhancement.

Card Prefix File (CPF)

The following existing field supports new values that allow customers to configure how EMV card verification should be performed (using either the standard method or the method that excludes the service code), and whether EMV card verification should be performed (using the standard method) on cards expiring before the EMV CV DATE:

Screen Name: CHECK METHOD

Screen: CPF Screen 11

DDL Name: EMV-CV-CHK-MTHD

Valid Values are: 0 (Standard/Standard),
1 (No Service Code/Standard),
2 (Standard/None),
3 (No Service Code/None).

Default: 0

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Optional Enhancements

This section identifies the Base modules enhancements that can be enabled or disabled with Release 6.0 Version 10, along with the changes, conversion requirements, and implementation requirements associated with each.

There are no new optional enhancements available for base modules.

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5: BASE24-atm

Section 5 discusses the changes and enhancements made to BASE24-atm in Release 6.0 Version 10, along with the resulting conversion and implementation requirements that existing BASE24-atm users must consider when upgrading from Release 6.0 Version 9.

This section is divided into two parts: General Enhancements and Optional Enhancements.

General enhancements describe the changes that affect all BASE24-atm users of a particular product or module, along with the associated conversion and implementation requirements that all BASE24-atm users must address.

Optional enhancements describe the changes that affect only those BASE24-atm users that choose a new separate licensable module, along with the associated conversion and implementation requirements that BASE24-atm users must address to implement the enhancements.

Section 5 must be used in conjunction with Section 4 to identify the changes that BASE24-atm users need to consider when upgrading to Release 6.0 Version 10.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24-atm users must address in order to upgrade to Release 6.0 Version 10.

Changes and requirements are identified using a standard format defined in the introduction to Section 4.

General Changes

ATM Bulk Check Infrastructure Support

Diebold Agilis 2.3 Upgrade

Diebold Agilis Enhanced Note Acceptor Upgrade

Diebold ENTIBOLD Entry Program Reorganization

Inhibit CVD validation for EMV Transactions

Issuer Fees for Non-Financial Transactions

No Default Account on Withdrawal and Balance Inquiry

Shared DH Maintenance

Shared DH Wincor EMV v1.2 Support

Triton MACing On/Off Support

TSS New AKDS Message Format for NCR and Diebold

Wincor SlimCash ATMs in NDC

These areas are summarized on the following pages.

Section 2 contains a functional overview of the general enhancements described here.

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General Changes

Changes for all BASE24 Release 6.0 Version 10 enhancements affecting the DDL files and the BASE24 AFT Subsystem files (i.e., screens, requesters, servers) have been merged and cataloged under one SCN for each affected file. Applying changes for individual enhancements to the screen files does not allow BASE24 users to identify which changes go with a particular enhancement, as screen files do not have sequence numbers at the end of each line. Since the screen file changes usually involve new fields being added, all associated requester, server and DDL changes are required. Even though requester, server and DDL files do contain sequence numbers, all SCNs would need to be applied in order to accept the changes to the screen files. Therefore, all requester, server and DDL file changes, for all enhancements, were done under one SCN per file as well.

The ATM Terminal Data File (ATD) has been modified to support a new additional offset/length area to store data required for the Shared DH 912 CPM Support enhancement. The ATD has also been modified to support an additional static field in the Diebold device-dependent data for the Shared DH 912 CPM Support enhancement. This field is displayed on ATD screen 10 for Diebold terminals, so there are also modifications to the Diebold specific screens. For further information, see Shared DH 912 CPM Support later in Section 5.

The Terminal Data File (TDF) has been modified to include two additional fields in the Triton device-dependent data for the Triton MACing On/Off Support enhancement. These fields are not displayed on the TDF screens, so there are no modifications to the Triton specific screens. For further information, see Triton MACing On/Off Support later in Section 5.

Software Impacts

All BASE24-atm Diebold 10XX/478X Device Handler customers:

- ☐ Inline changes to AT601000.C100DDLs, RQT10XXS, SCRN10XX, SV10XXTS

All BASE24-atm Triton DH customers:

- ☐ Inline changes to TRITDDLs

Note: The list of files impacted by the merged changes is also included in the **Software Impacts** section of the applicable enhancement(s).

A complete list of the merged files can be found in the SCN file BA60UD0D.SCNMERGE.

CUSTOMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

ATM Terminal Data (ATD)

The ATD conversion program must be run to convert the TDF records from the Release 5.3 format to the ATD/TDF Release 6.0 version 10 format or to convert the ATD records from the Release 6.0 version 9 format to the Release 6.0 version 10 format.

The ATD conversion program on the AT60UC04 subvolume can be used to convert the TDF records from the Release 5.3 format to the ATD/TDF Release 6.0 version 10 format depending on the device type in the record. If modifications to the conversion program are necessary, the source code may be modified by editing the CNVATDS file. This ATD conversion program is required for any BASE24 customer utilizing the ATD/TDF in transaction processing. Users must create the new ATD/TDF files using the BADDLFUP file

The ATD conversion program on the AT60UC0D subvolume can be used to convert the ATD records from the Release 6.0 version 9 format to the Release 6.0 version 10 format. If modifications to the conversion program are necessary, the source code may be modified by editing the CNVATDS file. Running this conversion program is required for any BASE24 customer utilizing the ATD in transaction processing. Users must create the new ATD file using the BADDLFUP file

To compile the conversion program, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#atdcnv.t,name $atdm,nowait/ cnvatdm –  
force –list –listreplace
```

Release 5.3 users should update the CNVATDR obey file on the AT60UC04 subvolume with the appropriate information. Release 6.0 version 9 users should update the CNVATDR obey file on the AT60UC0D subvolume with the appropriate information. The CNVATDR should then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREAD04 file on the AT60UC04 subvolume for Release 5.3 customers and to the AAREAD0D file on the AT60UC0D subvolume for Release 6.0 version 9 customers.

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

ATM Bulk Check Infrastructure Support

This section describes the implementation requirements associated with the ATM Bulk Check Infrastructure Support enhancement.

The implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the ATM Bulk Check Infrastructure Support enhancement.

The ATM Bulk Check Infrastructure Support enhancement allows the BASE24-atm Authorization and Host Interface (HISO) modules to support Bulk Check Deposits from ATM devices. **Please note:** At this time no Device Handler modifications have been made to support Bulk Check transactions. A Device Handler CSM would be required in order to process these transactions.

Business Requirements

Bulk Check is a new feature in ATMs, which allows depositing multiple checks without an envelope into the ATM. The Bulk Check feature is positioned as the next generation of check deposit at ATMs. In order to support this new feature, ACI has enhanced its existing check support, which is offered by its Check Processing Module.

There are a number of business drivers for BASE24 customers to enhance their ATMs with the Bulk check functionality:

- The US market requirements are driven by Check 21 regulations, requiring the imaging of checks, and truncation of the physical means of payment to improve check processing efficiency. US financial institutions will be looking to automate that process further from the first point of deposit, which is typically the ATM.

- The replacement of aging ATM fleets globally by newer models, especially the Self-service styled ATM models which offer self serving banking hardware modules and are highly marketed by the ATM vendors.

- The increase in demand from financial institutions for ATM deposit automation features resulting from consumer self service preference trends, where consumers seek independence and convenience, rather than dependence on bank tellers during limited banking hours. For Bulk Check, this is particularly applicable as it concerns small merchants looking to deposit stacks of checks and to obtain immediate confirmation of the deposit, outside of typical banking hours.

Functional Overview

The major ATM vendors, such as Diebold, NCR and Wincor Nixdorf, are introducing deposit devices in their ATMs that are capable of supporting Bulk Check deposit transactions. The Bulk Check deposit transactions allow the deposit of multiple checks without an envelope at the ATM, and allow the cardholders at the ATM to view a scanned image of their checks, and also to verify the ATM reading of the courtesy amount. Once the checks are reviewed and approved for deposit by the ATM, the ATM sends the information about each check (MICR, amount etc.) to BASE24 in a single transaction for authorization.

In BASE24-atm, new tokens have been defined to hold the additional data that must be passed through the system to support the Bulk Check functionality. The MICR data is stored in a new Bulk Check MICR token, and the amount data is stored in a new Bulk Check Amount token. Also, a new Bulk Check SSBC token is used to store SSBC data for all the checks that are part of this single transaction. A maximum of 20 checks are supported in a single Bulk Check transaction. BASE24 processes the transaction such that individual checks that do not meet authorization criteria are rejected and all checks are returned to the cardholder. If the cardholder decides to deposit the checks that meet the authorization criteria, a new transaction request is initiated using only the checks that meet the authorization criteria.

The ATM Authorization module interrogates the MICR data and amount from each check (stored in the new tokens) in order to process the transaction successfully. ATM Authorization adds another new token (the Bulk Check disposition token), with an indicator for each check that identifies whether the particular check is to be deposited (approved) or returned to the customer (denied). The Authorization processing for single check transactions and envelope deposits has not been modified.

Bulk Check transactions are logged to the Transaction Log File (TLF) as a single transaction. A value of '5' (indicating a bulk check depositor) is now supported in the AVAIL-DEP-TYP field and DEP-TYP field in the Standard Internal Message (STM). The DEP-TYP field is written to the DEP-TYP field in the TLF record, so customers can identify transactions as performed at a bulk check device. A Bulk Check deposit transaction is recognized as a regular deposit transaction (transaction code "20", "11" or "24") within BASE24. A new subtype "ABC0" is included in the Transaction Subtype Token (ID 'BM'), which can be used to identify Bulk Check transactions.

Bulk Check transactions are passed to the Host via the ISO Host Interface (HISO) as a single transaction. HISO has been enhanced to be able to process this transaction with large amounts of MICR and amount data. The four new tokens are not placed in the external message in the current token buffer (Secondary field 126), as they are too large. Instead, other secondary fields in the external message that are not currently used for ATM transactions have been redefined to carry this data.

Detailed information for the Bulk Check transaction is displayed on the TLF screens. Two new screens (18 and 19) have been added to the TLF perusal facility to show the information related to Bulk Check transactions. These screens include the index number (1-20) of each check. Screen 18 shows the check information for up to 12 checks; screen 19 shows the check information from check 13 to 20 (if the number of checks in the Bulk Check transaction is greater than 12). Each screen entry includes the first 58 bytes of MICR data, the check amount, and an indicator of whether the check was returned to the customer or was approved for deposit.

Authorization Processing

The processing of this Bulk Check transaction by the ATM Authorization module depends on whether the Self Service Banking Enhanced ATM Module (SSB) or the Self Service Banking Enhanced Check processing ATM Module (SSBC) is bound in.

If the Self Service Banking Enhanced Check processing ATM module is present, the ATM Authorization module retrieves the following new tokens:

Bulk Check MICR token (AV)

Bulk Check Amount token (AW)

Bulk Check SSB token (AX)

Using these tokens, each check in the transaction is validated against the Stop Payment File (SPF), Check Status File (CSF) and Corporate Check File (CCF). The Authorization module uses these files to perform the authorization aspect of the enhanced check processing features. Deposit credit is calculated on a check-by-check basis, based on the check type (pre-approved, registered or personal).

The Authorization process has also been enhanced to add the following new token:

Bulk Check Disposition token (AY)

This token is used to mark every check in the transaction as approved (forwarded for deposit) or denied (returned to the customer). The disposition token is populated with "Y" if the validation against the SPF, CSF and CCF is approved and with "N" if it is denied. The Amount field in the STM includes the sum of the amounts from all the checks, regardless of whether they are to be deposited (approved) or not.

If the Self Service Banking Enhanced Check processing ATM module is not present, the authorization is based on the total deposit amount only. So, customers can process Bulk Check deposit transactions using only the Self Service Banking Enhanced ATM Module, but in that case the checks are not processed individually.

HISO Processing

The ATM HISO module has been enhanced to populate the appropriate SEM fields in the outgoing request with the four new Bulk Check tokens (if the tokens exist in the STM):

Field Secondary Reserved2 National (S-113) holds the Bulk Check Amount token.

Field Secondary Reserved3 National (S-114) holds part of the Bulk Check MICR token. The check count and the first 10 occurrences of the MICR data are held in this field. If this Bulk Check transaction contains less than 10 checks, this field holds only the amount of data required for that smaller number of checks.

Field Secondary Reserved4 National (S-115) holds part of the Bulk Check MICR token. The next 10 occurrences of the MICR data are held in this field. If this Bulk Check transaction contains less than 10 checks, this field is not required. If this Bulk Check transaction contains more than 10 checks but less than 20 checks, this field holds only the amount of data required for that smaller number of checks.

Field Secondary Reserved5 National (S-116) holds the Bulk Check SSBC token.

Field Secondary Reserved6 National (S-117) holds the Bulk Check Disposition token.

The new tokens are not included in the standard token location, the Secondary Reserved7 Private (S-126).

The new tokens are present in the STM if the appropriate device handler has created them; this would currently only be the case if the device handler had been CSM'd. The tokens will be present regardless of whether the customer is configured to use the Self Service Banking Enhanced Check application. Customers can process Bulk Check deposit transactions without using the Self Service Banking Enhanced Check application, but in that case the authorization is based on the total deposit amount only. Host changes are required only if there is a device handler capable of placing the new Bulk Check tokens in the STM.

Assumptions and Restrictions

1. This enhancement requires Self-Service Banking (SSB).
2. BASE24 supports the authorization of a maximum of 20 checks in a single transaction, due to size constraints in various files and messages, although the ATMs are able to physically process from 30 – 50 checks at a time depending on the ATM vendor. The ATM must be configured to allow only a maximum of 20 checks in order for BASE24 to be able to process these transactions.

3. Features supported by the SSB Enhanced ATM Authorization extension (AT60BATH) for single check processing are also supported for bulk check processing.
4. Features supported by the SSB Enhanced Check ATM Authorization extension (AT60CATH) for single check processing are also supported for bulk check processing.
5. One Transaction Request message is sent for the entire check bundle, and a Bulk Check transaction is logged to the Transaction Log File (TLF) as a single item.
6. Check Deposit with cash back is supported for Bulk Check Transactions.
7. Cash Check is supported for Bulk Check Transactions.
8. If surcharging is enabled, there is a single surcharge for the Bulk Check Transactions.
9. Cash Check transactions are not surcharged.
10. Split deposit transactions are not supported. They will be denied with a response code "57" – transaction not allowed.
11. The BASE24-atm reports will not be modified for this enhancement. Additional changes to BASE24-atm reports would be required by customers if they need to distinguish between a check deposit and a regular deposit.
12. BASE24-atm Device Handlers have not been modified for this enhancement. Device handler changes will be required by customers to support Bulk Check transactions.

Software Impacts

All BASE24 customers:

- ☐ Inline changes to BA60AFT.COBTKN
- ☐ Inline changes to BA60DDL.DDLATTKN, DDLGISO, DDLGPTH, DDLGSTM
- ☐ Comment only changes to BA60DDL.DDLFCAF, DDLFCPF, DDLFUAF
- ☐ Inline changes to BA60SRC.ATTKNCVS, ATTKNID, BACOUTLS, SUBTPTBL

All BASE24-atm customers:

- ☐ Inline changes to AT60AFT.RQTLFS, SCRNTLF, SVTLFS
- ☐ Inline changes to AT60SRC.ATCOUTLS
- ☐ Inline changes to AT60AUTH.AUTHD, AUTHG, AUTHM, AUTHLIBS, AUTHS, AUTHMCUD

- ☐ Inline changes to AT60HISO.ATHISOG, ATHISOM, ATHISOS
- ☐ Updated event (log) message manual BA60LOGM.ATAUTH
- ☐ Updated event (log) message manual BA60LOGM.ATHISO

All BASE24-atm Self-Service Banking Check (SSBC) customers:

- ☐ Inline changes to AT60CATH.AUTHSSCG, AUTHSSCM, AUTHSSCS

All BASE24-atm Multiple Currency (MCU) customers:

- ☐ Inline changes to AT60UATH.AUTHMCUS

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD0D.SCNBCA.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Token file (TKN)

The following new BASE24-atm tokens are used by this enhancement:

- Bulk Check MICR token (AV)
- Bulk Check Amount token (AW)
- Bulk Check SSB token (AX)
- Bulk Check Disposition token (AY)

To ensure that one or more of these tokens are logged to the BASE24-atm Transaction Log File (TLF), the appropriate token records must be configured in the Token file.

To ensure that one or more of these tokens are extracted from the TLF, the appropriate records must be configured in the Token file.

The new tokens will be sent to the Host if they are present in the STM. This will be regardless of any settings in the Token File, as these tokens will not be placed in the standard token location, the Secondary Reserved7 Private (S-126).

Cardholder Authorization File (CAF)

Fields used to determine the number of times a card is used will now be used slightly differently with this enhancement. Each check that is included as part of a bulk check transaction will count as an individual usage. These fields are:

- Number of deposit credits (CAF screen 8)
- Times used this period limit (CAF Screen 8)
- Times used this period (CAF screen 8)
- CSF Checks allowed per period (CAF screen 16)
- CSF Checks accepted this period (CAF screen 16)

In each case, if there are 20 checks included in this bulk check transaction, these counts will be incremented by 20, and the limits verified against the 20 checks.

Card Prefix File (CPF)

Fields used to determine the number of times a card is used will now be used slightly differently with this enhancement. Each check that is included as part of a bulk check transaction will count as an individual usage. These fields are:

- Times used per period limit (CPF Screen 4)
- CSF Checks allowed per period (CPF screen 14)

In each case, if there are 20 checks included in this bulk check transaction, these counts will be incremented by 20, and the limits verified against the 20 checks.

Usage Accumulation File (UAF)

Fields used to determine the number of times a card is used will now be used slightly differently with this enhancement. Each check that is included as part of a bulk check transaction will count as an individual usage. These fields are:

- Times used this period (UAF Screen 4)
- CSF Checks accepted this period (UAF screen 10)

In each case, if there are 20 checks included in this bulk check transaction, these counts will be incremented by 20, and the limits verified against the 20 checks.

Customers may wish to review the values of any limits in the above files based on this change.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Devices must be able to provide BASE24 with the data required to authenticate the Bulk Check deposit transactions.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

ISO Host Interface Process

The Host may require modifications to process the new tokens added for this enhancement. These tokens will not be sent to the Host unless this enhancement has been configured, and a device handler is configured that is capable of creating the new tokens.

The tokens will not be sent to the Host in the standard Token area (Secondary field 126), but can be sent in the following Secondary fields of the Standard External Message:

The Bulk Check MICR token (AV) is split between two fields. The first 10 occurrences of the MICR data (Checks 1 – 10) are included along with the total number of checks in this transaction, in the S-114 field. The second 10 occurrences of the MICR data (Checks 11 – 20) are included in the S-115 field. Both of these fields have been increased to a maximum of 800 bytes in size.

The Bulk Check Amount token (AW) is sent in the S-113 field and that field has not increased in size.

The Bulk Check SSBC token (AX) is sent in the S-116 field and that field has not increased in size.

The Bulk Check Disposition token (AY) is sent in the S-117 field and that field has not increased in size.

The Host may also require modifications to process the new value “ABC0” in the Transaction Subtype token (identifying a Bulk Check transaction).

Super Extract Process

Host programs that process TLF extract and/or SAF Extract data may require modifications to support the new tokens added for this enhancement. These tokens will not be sent to the Host unless this enhancement has been configured. These programs may also require modifications to support the new value “ABC0” in the Transaction Subtype token (identifying a Bulk Check transaction).

Host programs that process TLF extract data may require modifications to support the new value “5” in the DEP-TYP field in the TLF record (identifying a bulk check depositor).

User/Operational Impacts

Disk Impacts

The size of the TLF record will be substantially bigger if a full 20 checks are processed in a single transaction. A full 1908 bytes of additional token data would be present. This means that only a single TLF record is likely to fit in a single disk block as opposed to three records per block with a single check transaction. Though these transactions are not likely to be performed in large numbers, this could result in a sizable increase in disk space requirements for a customer.

Similarly, if these tokens are extracted from the TLF, an increase in the size of the extracted TLF files should be expected.

The size of the SAF record written by HISO when the connection to the Host is down, will be substantially bigger if a full 20 checks are processed in a single transaction and the Self Service Banking enhanced check application is being used. A full 1908 bytes of additional message data would be present in each record. This means that only a single SAF record will fit in a single block as opposed to two records in a block with a single check transaction. Though these transactions are not likely to be performed in large numbers, this could result in a sizable increase in disk space requirements for the customers.

Similarly, if the SAF is extracted, an increase in the size of the extracted SAF file should be expected.

Operational Impacts

New screens have been added to the TLF perusal facility by this enhancement. Screens 18 and 19 display the data defined in the new Bulk Check MICR token (AV), the new Bulk Check Amount token (AW), and the new Bulk Check Disposition token (AY). Operators must be trained to recognize the new information on the screen.

System Limitations

BASE24 users installing this enhancement must be aware of their internal system limitations for messaging to the TLF. There are system limitations regarding the amount of data that can be passed through the BASE24 system and logged to a log file. BASE24 users should be aware of these limitations and be cognizant that support of new data in online messages and log files in the form of tokens can cause them to exceed these limitations. Appendix A of the **BASE24 Tokens Manual** provides information on assessing the impacts of tokens on log file disk space requirements.

Event Messages

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate remedy for a given message. Refer to the event message documentation files on the BA60LOGM subvolume.

Performance Impacts

If the Self Service Banking enhanced check application module is bound in, the ATM Authorization processing for this enhancement verifies the individual check amount and MICR data for each of the up to 20 checks in the transaction. This means that an I/O on the Stop Payment File (SPF) and on the Check Status File (CSF) is performed for each check in the transaction. This additional processing could cause ATM Authorization to take longer to process this transaction than a single check transaction. The Corporate Check File (CCF) is also accessed for each check in the transaction, but this file is held in an extended memory table, so the performance impact of this additional work is negligible.

Note: A customer may now choose to perform a single bulk check transaction instead of up to 20 individual check deposit transactions. In this case, the overall number of I/O's will not increase, in fact the number of I/Os may decrease as the CAF is not read for each of the individual transactions. However, there is an increase in the number of I/Os within the one transaction.

It could take longer to extract the TLF since the size of the TLF has increased.

It could take longer to extract the SAF since the size of the SAF has increased.

Customers should analyze the timer values for their ATMs, Device Handlers (Auth timeout value) and for the Host. These timer values may have to be increased due to the extra time Auth may need to process this transaction, or the extra time the Host may need to process this transaction.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Diebold Agilis 2.3 Upgrade

This section describes the implementation requirements associated with the Diebold Agilis 2.3 Upgrade enhancement.

The implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Diebold Agilis 2.3 Upgrade enhancement.

The Diebold Agilis 2.3 Upgrade enhancement provides support for a variety of new functions that Diebold has added to its Agilis ATM software in version 2.3. The enhancement also provides backwards compatibility for all ATM configurations currently supported by the Diebold 10XX/478X Device Handler.

The following changes are included:

Support for the Diebold ESC-ZV Screen Code to Allow Persistent Browser Windows

The Device Handler has been enhanced to support the Diebold ESC-ZV screen code to allow persistent XML-based browser windows to be opened on the ATM's terminal screen. The persistent browser form of the ESC ZV control sequence allows for the creation and management of one or more browser windows that persist across screen changes on the consumer display. The browser windows do not shift or resize any non-browser screen contents; they simply overlay the specified area of the non-browser screens.

The format of the command is:

ESC ZV C "ID" ZZ "URL spec" XXXX YYYY www hhhh ttt bbb

Where:

The C parameter is a one-character command identifier. This command identifier can have any one of the following values:

C - Create. Open a new browser window.

D - Destroy. Close an existing browser window.

N - Navigate. Change the web page of an existing browser window.

H - Hide. Make an existing browser window invisible

R - Restore. Redisplay an existing, hidden browser window.

S - Size/position. Change the size and/or position of an existing browser window.

The ID parameter is a brief text string (one or more characters) that identifies which browser window the command acts upon.

For the Create command, the ID text string becomes the identifier for the browser window created by the command. If an attempt is made to create a browser window where the ID matches one that already exists, the Create command is ignored and the existing browser window is used for any subsequent commands. All other commands must specify a valid ID that was previously defined by a Create command.

For the Destroy, Hide, and Restore commands, an ID of * (asterisk) can be used as a wildcard. If this wildcard is specified for one of these commands, the command is applied to all existing persistent browser windows.

The ZZ parameter specifies a z-order index (visibility priority) for the browser window. This parameter is used only by the Create command. The valid range is 00 through 99, where 00 is the lowest priority (least visible) and 99 is the highest priority (most visible).

The URL spec parameter specifies a web page or local document to be displayed in the browser window. This parameter is used by the Create and Navigate commands. The parameter must be a valid URL or, optionally when using a Create command, nothing at all. The parameter must be enclosed by double quotes.

The position parameters (xxxx and yyyy) specify the placement of the upper left corner of a browser window, measured in global coordinates. These parameters are used by the Create, Navigate, and Size/position commands.

The size parameters (www and hhhh) specify the width and height of a browser window, respectively, in global coordinates. These parameters are used by the Create, Navigate, and Size/position commands.

The ttt parameter specifies the number of seconds to wait for a URL to successfully load. This parameter is used by Create and Navigate commands that specify a valid URL. The valid range is 000 through 999. A value of 000 causes the system to use the Browser Timeout value set in the Agilis Configuration Utility.

The bbb parameter specifies a backup screen number for use in the event that the browser is unable to load the specified URL.

For a detailed description of each of the above parameters, refer to the Diebold Agilis version 2.3 Terminal Programming Manual.

Support for Envelope Rejects Count

The Device Handler has been enhanced to count the number of envelopes in the reject bin, to increase or reset the count when relevant envelope dispense status messages are received, and to display the count itself. The count is incremented only if the following status is received:

SD01:23:40:01 (envelope not taken by consumer)

The count is reset to zero only if one of the following statuses is received:

DP01:00:00:00 (depositor self-test OK)

SD01:00:00:00 (envelope dispenser self-test)

A display-only field, for the rejected envelope count in the bin, has been added to ATD screen 8.

Support for New Coin Dispenser Status Messages

Version 2.3 of the Agilis software has modified the format of the existing Coin Dispenser status messages (both solicited and unsolicited). The Device Handler has been enhanced to support the new formats. The new Coin Dispenser statuses are:

CN01:3D:31:37 (Feed failure in hopper 1)

CN01:3D:31:38 (Feed failure in hopper 2)

CN01:3D:31:39 (Feed failure in hopper 3)

CN01:3D:32:30 (Feed failure in hopper 4)

CN01:3E:32:35 (Jam failure in hopper 1)

CN01:3E:32:36 (Jam failure in hopper 2)

CN01:3E:32:37 (Jam failure in hopper 3)

CN01:3E:32:38 (Jam failure in hopper 4)

CN01:3E:33:33 (Jam between picker and chute sensor)

CN01:00:00:00 (No error)

CN01:3E:00:01 (Network fault. Coin dispenser is not present or is not enabled)

CN01:3D:33:35 (Network error (invalid coin amount field))

CN01:3E:36:30 (Jam at chute sensor before dispense. No coins dispensed)

CN01:3A:36:33 (Memory error)

CN01:3E:38:41 (Hopper 1 coin sensor is stuck clear/blocked)

CN01:3E:38:42 (Hopper 2 coin sensor is stuck clear/blocked)

CN01:3E:38:43 (Hopper 3 coin sensor is stuck clear/blocked)

CN01:3E:38:44 (Hopper 4 coin sensor is stuck clear/blocked)

CN01:38:00:00 (Internal communications problem)

CN01:3E:01:01 (Coin dispenser is not docked)

CN01:2D:0m:0n (Coin dispenser supply low)

The network action taken for each of these messages is documented in the Diebold Agilis 91x Status Reference Manual.

Support for New Alarm Status Messages

The Device Handler has been enhanced to support the logging of an Alarm Message to the EMS whenever it receives any Alarm Status message, including the status of each of the four security devices, i.e. Ambush detector (Ad), Burglary sensor (Bs), Safe door (Sd), Cabinet door (Cd). Sixteen statuses are now supported (four devices, two possible states for each), which are:

AL01:3F:00:00 (Ad: ON, Bs: OFF, Sd: OPEN, Cd: CLOSED)
AL01:3F:01:00 (Ad: ON, Bs: OFF, Sd: CLOSED, Cd: CLOSED)
AL01:3F:10:00 (Ad: ON, Bs: OFF, Sd: OPEN, Cd: OPEN)
AL01:3F:11:00 (Ad: ON, Bs: OFF, Sd: CLOSED, Cd: OPEN)
AL01:3F:40:00 (Ad: ON, Bs: ON, Sd: OPEN, Cd: CLOSED)
AL01:3F:41:00 (Ad: ON, Bs: ON, Sd: CLOSED, Cd: CLOSED)
AL01:3F:50:00 (Ad: ON, Bs: ON, Sd: OPEN, Cd: OPEN)
AL01:3F:51:00 (Ad: ON, Bs: ON, Sd: CLOSED, Cd: OPEN)
AL01:3F:80:00 (Ad: OFF, Bs: OFF, Sd: OPEN, Cd: CLOSED)
AL01:3F:81:00 (Ad: OFF, Bs: OFF, Sd: CLOSED, Cd: CLOSED)
AL01:3F:90:00 (Ad: OFF, Bs: OFF, Sd: OPEN, Cd: OPEN)
AL01:3F:91:00 (Ad: OFF, Bs: OFF, Sd: CLOSED, Cd: OPEN)
AL01:3F:C0:00 (Ad: OFF, Bs: ON, Sd: OPEN, Cd: CLOSED)
AL01:3F:C1:00 (Ad: OFF, Bs: ON, Sd: CLOSED, Cd: CLOSED)
AL01:3F:D0:00 (Ad: OFF, Bs: ON, Sd: OPEN, Cd: OPEN)
AL01:3F:D1:00 (Ad: OFF, Bs: ON, Sd: CLOSED, Cd: OPEN)

If one or more of these devices is found not to be in a secure state, then an Alarm Status message is generated, the status is logged to the EMS and the ATM is shut down.

Support for Skimmer Code "CR01" for Message Type "5"

The Device Handler has been enhanced to process Card Reader status messages (skimmer code "CR01") for message type "5". The new processing is in addition to the existing support for a message type of "J".

Software Impacts

All BASE24-atm customers:

- ☐ Inline changes to AT60DDL.DDLGATD

All BASE24-atm Diebold 10XX/478X Device Handler customers:

- ☐ Inline changes to AT601000.C1000G, C1000S, C100DDLS, SCRN10XX, SV10XXTS
- ☐ Inline changes to AT20IBLD.ENTCRTS, ENTIBLDG, ENTLOWS
- ☐ A new Diebold Entry program object (ENTIBOLD)
- ☐ Updated event (log) message manual BA60LOGM.AT1000

All BASE24-atm Diebold 10XX/478X Device Handler Self-Service Banking customers:

- ☐ Inline changes to AT60D100.C100SSDS

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD0D.SCND AU.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Diebold ATMs must be loaded with Diebold Agilis v2.3 ATM software.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

A new display-only field of 2 bytes has been added to ATD screen 8. The REJECTED ENVELOPE COUNT field displays the rejected envelope count for the terminal. If this field contains ASCII spaces, then the value is interpreted as “unknown”, for compatibility with ATMs not running Agilis version 2.3.

Event Messages

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate remedy for a given message. Refer to the event message documentation files on the BA60LOGM subvolume.

Vendor Impacts

Not applicable

Reference Documents

Diebold: Agilis version 2.3 Terminal Programming Manual, June 2008

Diebold: Agilis 91x Status Reference Manual, April 2008

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Diebold Agilis Enhanced Note Acceptor Upgrade

This section describes the implementation requirements associated with the Diebold Agilis Enhanced Note Acceptor Upgrade enhancement.

The implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Diebold Agilis Enhanced Note Acceptor Upgrade enhancement.

The Diebold Agilis Enhanced Note Acceptor Upgrade enhancement ensures that the Diebold 10XX/478X device handler can correctly process transactions and status messages for ATMs equipped with either Bunch Note Acceptor (BNA) hardware or Enhanced Note Acceptor (ENA) hardware.

Diebold is discontinuing the use of the Bunch Note Acceptor (BNA) hardware which was manufactured by a third party. It is instead introducing the Enhanced Note Acceptor (ENA), manufactured by Diebold itself. There are some differences between these hardware devices. The BNA is an explicit shutter control device whilst the ENA is an implicit shutter device:

An explicit shutter control device requires the cardholder to place the currency into the consumer input area, then press a key to indicate they are ready to continue. The application controls the shutter on the currency acceptor.

An implicit shutter control device requires the cardholder to hand a stack of currency to the device, which then automatically takes possession of the stack and closes the shutter. The application does not control the shutter on the currency acceptor.

The BASE24-atm Diebold 10XX/478X device handler currently supports cash deposits via a BNA device. This enhancement modifies the device handler to support Diebold's ENA devices as well.

Functional differences between the BNA and ENA devices are primarily related to the shutter control, as detailed above. Changes have been made to the Enhanced Currency Acceptor states (CI states) and screens to deal with these differences. The ENA can function with the CI states defined in the BNA load group 712 of the configuration file. However, a new load group (812) has been defined to support the ENA, and changes have been made to the CI states in order to document the differences in the way that timers are handled. This approach allows customers to use the new load group for ATMs supporting the ENA only.

In addition, new status values are possible in solicited and unsolicited messages for the ENA. The device handler also supports the logging of new messages to EMS

whenever it receives any of the new status messages issued by an ATM with an ENA device installed.

Assumptions and Restrictions

1. Support has not been included for operation 006 in the Enhanced Currency Acceptor state for the ENA.
2. Support for Suspect and Counterfeit information (amount and note counts) has not been included in this enhancement. These definitions within the Enhanced Currency Acceptor state are optional.
3. Multi-currency cash deposits are not supported for the BNA, and are also not supported for the ENA.
4. Support for the Currency Code Buffer ID 2 has not been included in this enhancement. If Multi-currency deposits are not supported, then the function of this buffer is identical to the function of Currency Code Buffer ID 1, which is already supported.

Software Impacts

All BASE24-atm Diebold 10XX/478X Device Handler customers:

- ☐ Changes to AT601000.CN100DOC, CNFGIX
- ☐ Changes to AT20IBLD.ENTSTATS
- ☐ A new Diebold Entry program object (ENTIBOLD)
- ☐ Updated Entibold documentation AT20IBLD.ENIXDOC
- ☐ Updated event (log) message manual BA60LOGM.AT1000

All BASE24-atm Diebold 10XX/478X Device Handler Self-Service Banking customers:

- ☐ Inline changes to AT60D100.C100SSDS

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD0D.SCND AEN.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Diebold 10XX Configuration File (CNFGIX)

The Diebold 10XX Configuration File (CNFGIX) has been amended to include a new load group 812 for the ENA. This load group is loaded to the ATM on top of groups 0, 12, and 712 for ATMs using the ENA only. ATMs still using the BNA load the existing 0, 12 and 712 load groups with no change.

New load group 812 includes the following small changes:

State 771 – type CI (Enhanced Currency Acceptor State):

Consumer Response Time	904
Amount Buffer ID	000
Currency Buffer ID	000
Count Buffer ID	000

State 773 – type CI (Enhanced Currency Acceptor State):

Consumer Response Time	904
Count Buffer ID	000

Screen 172:

This is the same as Diebold BNA Group 712 screen 172, but with text amended to read “GET READY TO INSERT CASH – PRESS OK TO CONTINUE”.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Event Messages

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate remedy for a given message. Refer to the event message documentation files on the BA60LOGM subvolume.

Vendor Impacts

Not applicable

Reference Documents

Diebold: Enhanced Currency Acceptor State/Function, January 9, 2009

Diebold: ENA Differences in Operation, February 10, 2009

Diebold: Agilis version 2.4 Terminal Programming Manual, July 2009

Diebold: Agilis 91x Status Reference Manual, July 2009

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Diebold ENTIBOLD Entry Program Reorganization

This section describes the implementation requirements associated with the Diebold ENTIBOLD Entry Program Reorganization enhancement.

The implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Diebold ENTIBOLD Entry Program Reorganization enhancement.

The Diebold ENTIBOLD Entry Program Reorganization enhancement splits the Diebold Entry program into three distinct objects. Going forward, this was necessary because of increasingly limited code space available for additional state support.

Prior to BASE24 Release 6.0 Version 10, the Diebold Entry program consisted of a single set of source files compiled into a single object (ENTIBOLD). This object provided support for all Diebold-specific states, and Wincor-specific states, as well as cardholder screens, FIT records, Miscellaneous records, etc. The large number of supported states left a limited amount of code space available for future support of additional states.

The Diebold Entry program has been restructured for BASE24 Release 6.0 Version 10. The new structure now supports three distinct objects resident on the AT20OBJ subvol.

NOTE: Customers generally receive only the AT20OBJ subvol, containing executable objects and documentation files. Customers do not generally receive the Diebold Entry program source files (AT20IBLD subvol), which are included in this section for clarification purposes only.

The three objects are as follows:

ENTIBOLD

ENTWBOLD

ENTIBLD2

These objects are resident on the AT20OBJ subvol. Each object is compiled from a distinct set of source files. Some files are common to all objects, while others are specific to that object.

There is a single set of Make files for all three objects:

IBOLDM: This Make file compiles all three objects.

IBOLDMFM: Make Macro file includes defines for all three objects.

IBOLDMM: Make Macro file includes defines for all three objects.

The three Diebold Entry Program objects are described in more detail below.

ENTIBOLD

This object supports all Diebold states, as well as cardholder screens, FIT records, Miscellaneous records, etc. Support for Diebold state 'SC' is limited to the NEW Operation codes only. This object does not support the old state 'SC' Operation codes, or any Wincor-specific states or tables. Any new/future Diebold-specific support will be added to this object.

State 'SC' Operation Codes supported:

[070, 071, 072, 073, 074, 075, 076, 077, 078]

The following files comprise this object. These source files (AT20IBLD subvol) are not normally distributed to customers.

ENTAPPLS	ENTCNFGS	ENTCRTS	ENTFITS	ENTFLDS	ENTFORMS	ENTIBLDG
ENTIBLDS	ENTLISTS	ENTLOWS	ENTMENUS	ENTMISCS	ENTPRNTS	ENTRBALS
ENTSCHMS	ENTSTATS	ENTSTMTS	ENTTERMS	SCRNFORM		

ENTWBOLD

This object supports all Wincor states, Wincor table(s), as well as cardholder screens. The object does not support Diebold-specific states, tables, etc. Any new/future Wincor-specific support will be added to this object.

The following files comprise this object. These source files (AT20IBLD subvol) are not normally distributed to customers.

ENTCNFGS	ENTCRTS	ENTFLDS	ENTLOWS	ENTPRNTS	ENTWAPPS	ENTWBLDG
ENTWBLDS	ENTWFRMS	ENTWMENS	ENTWSTAS	SCRNFORM		

ENTIBLD2

This object supports all Diebold states, as well as Wincor EMV state 'e', Wincor EMV AID table, cardholder screens, FIT records, Miscellaneous records, etc. Support for Diebold state 'SC' is limited to the old Operation codes only. This object does not support the new state 'SC' Operation codes, and will not receive further updates.

State 'SC' Operation Codes supported:

[010, 011, 095, 097, 099]

The following files comprise this object, new support will not be added to this object. These source files (AT20IBLD subvol) are not normally distributed to customers.

ENTAPPLS	ENTBLD2G	ENTBLD2S	ENTCNFGS	ENTCRTS	ENTFITS	ENTFLDS
ENTFRM2S	ENTLISTS	ENTLWS	ENTMEN2S	ENTMISCS	ENTPRNTS	ENTRBALS
ENTSCHMS	ENTSTA2S	ENTSTMTS	ENTTERMS	ENTWAPPS	SCRNFORM	

Software Impacts

NOTE: Software Impacts are included for completeness only, and for those customers who may have Diebold Entry program source files. Generally, the Diebold Entry program source files (AT20IBLD subvol) are not distributed to customers, who receive object files only (AT20OBJ subvol).

All BASE24-atm Diebold 10XX/478X Device Handler customers:

- ☐ Inline changes to AT20IBLD.ENTBLD2G, ENTBLD2S, ENTFORMS, ENTFRM2S, ENTIBLDG, ENTIBLDS, ENTMENUS, ENTSTA2S, ENTSTATS, SCRNFORM
- ☐ Changes to Make files AT20IBLD.IBOLDM, IBOLDMM
- ☐ Updated Entibold documentation AT20IBLD.AAREADME, ENIXDOC

All BASE24-atm Diebold 10XX/478X Shared Device Handler customers:

- ☐ Inline changes to AT20IBLD.ENTWAPPS
- ☐ New files AT20IBLD. ENTMEN2S, ENTWBLDG, ENTWBLDS, ENTWFRMS, ENTWMENS, ENTWSTAS
- ☐ New Diebold Entry program objects (ENTWBOLD)

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD0D.SCNENT.

CUSTOMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Inhibit CVD Validation for EMV Transactions

This section describes the implementation requirements associated with the Inhibit CVD Validation for EMV Transactions enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Inhibit CVD Validation for EMV Transactions enhancement.

For additional detailed information on the Inhibit CVD Validation for EMV Transactions enhancement see Section 4 (Base).

Software Impacts

For full details on the Base software impacts for this enhancement see Software Impacts in Section 4 (Base) of this document.

All BASE24-atm EMV customers:

- ☐ Inline changes to AT60IATH.AUTHEMVS

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD0D.SCNCVDI.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

For full details on the database configuration impacts for this enhancement see Database Configuration in Section 4 (Base) of this document.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Issuer Fees for Non-Financial Transactions

This section describes the implementation requirements associated with the Issuer Fees for Non-Financial Transactions enhancement.

The implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Issuer Fees for Non-Financial Transactions enhancement.

The Issuer Fees for Non-Financial Transactions enhancement allows BASE24-atm to support non-financial transactions that contain an access fee, i.e. a surcharge.

In Banknet Release 09.2 and MasterCard Debit Switch (MDS) Release 09.3, MasterCard introduced changes that allow acquirers to impose an access fee on certain non-financial transactions (balance inquiry, EMV PIN unblock, PIN change) performed at an ATM. MasterCard now requires issuers to support non-zero transaction amounts in these types of transaction.

Traditionally, BASE24-atm has supported Balance Inquiry, PIN Change and PIN Unblock transactions as non-financial transactions, i.e. no amount processing has been performed.

The Issuer Fees for Non-Financial Transactions enhancement ensures that BASE24-atm can process the following transactions, where the transaction amount is greater than zero.

- Balance Inquiry (BASE24-atm transaction code 30);

- PIN Unblock (BASE24-atm transaction code 82);

- PIN Change (BASE24-atm transaction code 81).

The BASE24-atm Authorization process has been enhanced to support Balance Inquiry, PIN Change and PIN Unblock transactions, where the transaction amount is non-zero, as follows:

- When processing Balance Inquiry, PIN Change and PIN Unblock transactions, the BASE24-atm Authorization process checks CPF/CAF limits, as it does for conventional financial transactions.

- When processing Balance Inquiry, PIN Change and PIN Unblock transactions, the BASE24-atm Authorization process checks PBF balances, as it does for conventional financial transactions.

- When processing Balance Inquiry, PIN Change and PIN Unblock transactions, the BASE24-atm Authorization process updates UAF/CAF accumulators, as it does for conventional financial transactions.

When processing Balance Inquiry, PIN Change and PIN Unblock transactions, the BASE24-atm Authorization process updates PBF balances, as it does for conventional financial transactions.

The BASE24-atm Authorization process has also been enhanced to support reversals of Balance Inquiry transactions, by updating UAF/CAF accumulators and PBF balances. Note that BASE24-atm already supports the reversal of PIN Change and PIN Unblock transactions.

These changes apply to all authorization types and levels, except for Level 2 (offline only) Type 1 (NEG only) and Type 4 (NEG/UAF), because BASE24-atm does not support the processing of Balance Inquiry, PIN Change and PIN Unblock transactions using these configurations. Balance Inquiry transactions are also not supported using Level 2 (offline only) Type 2 (CAF only) authorization (because a PBF is required if no Host authorization is available).

The BASE24-atm Authorization process has also been modified to ensure that an account type is specified in PIN Change and PIN Unblock transactions. This is required to allow a PBF record to be accessed (to check/update the balances) and to allow the appropriate CPF/CAF limit(s) to be checked and the corresponding UAF/CAF accumulators to be updated where the transaction amount is non-zero. The FAST CASH ACCOUNT TYPE on Screen 13 of the IDF is used as the default account type for PIN Change and PIN Unblock transactions.

The BASE24-atm Authorization process has also been modified to ensure that all appropriate authorization checks are performed on PIN Change and PIN Unblock transactions, regardless of whether they contain a fee. When Level 3 authorization is configured, and the HOST PIN CHANGE OPTION on IDF Screen 13 is set to '0' (approve, don't send to Host), PIN Change and PIN Unblock transactions are not sent to the Host for authorization. This resulted in certain authorization functions being bypassed, if these checks were usually performed by the Host. The processing flow has been modified to ensure that PIN Change and PIN Unblock transactions are routed through Level 2 (offline) authorization logic rather than Level 3 (online/offline) authorization logic, if the HOST PIN CHANGE OPTION is set to '0'.

The BASE24-atm Authorization process has also been modified to ensure that the CPF Screen 4 field STANDARD CASH ADV INCR is not used in transaction processing, when the original currency of the transaction differs from the currency used in authorization processing.

The BASE24-atm HISO process has been enhanced to move the transaction amount from the external message to the internal message when processing a balance inquiry from an acquiring Host system.

The BASE24-atm HISO process already supports the reversal of Balance Inquiry transactions, if configured to do so by the LCONF parameter REVERSE-BAL-INQ. The BASE24-atm HISO process has been enhanced to support reversals of Balance Inquiry transactions where the transaction amount is non-zero, regardless of the setting of the LCONF parameter REVERSE-BAL-INQ.

Assumptions and Restrictions

1. When the cardholder is performing a Balance Inquiry, PIN Change or PIN Unblock, the direct charging fee is included in the Surcharge Data token, and is the same as the transaction amount.
2. BASE24-atm has not been enhanced to support a direct charging fee when the cardholder is performing a Balance Inquiry, PIN Change or PIN Unblock at an ATM managed by BASE24. So, BASE24-atm surcharging logic has not been enhanced to allow a surcharge to be applied to these types of transaction.
3. BASE24-atm has not been enhanced to support a direct charging fee when the transaction is a generic Balance Inquiry, i.e. where the inquiry relates to more than one account.
4. Even if the transaction amount is non-zero, a Balance Inquiry is still permitted on an account with a status of “restricted to deposits” or “primary account restricted to deposits”.
5. For PIN Change and PIN Unblock transactions, cardholders have not been provided with the option to select an account (or account type) at the ATM, because these transaction types are card-based and have had no financial impact. If a PIN Change or PIN Unblock transaction contains a direct charging fee, then BASE24-atm may have to determine the account type (checking, savings, credit) to use in authorization processing. However, it is still not possible for the cardholder to select an account (or account type) at the ATM.
6. For PIN Change and PIN Unblock requests, customers have the option of authorizing the transaction on BASE24, even if the card is configured for Level 3 (online/offline) authorization. This feature may be used if the Host system cannot process PINs, and is enabled by setting the HOST PIN CHANGE OPTION field to ‘0’ on IDF Screen 13. There is no issue with BASE24 authorizing a non-financial PIN Change or PIN Unblock transaction, because the Host is not required to check accumulators/balances for the card/account. However, some issuers may require the Host system to authorize PIN Change and PIN Unblock requests where the transaction amount is non-zero. These issuers must ensure that the HOST PIN CHANGE OPTION field is not set to ‘0’ on IDF Screen 13.
7. Enhancements to the BIC ISO Interface are out of scope for this project.

Software Impacts

All BASE24-atm customers:

- ☐ Inline changes to AT60AUTH.AUTHG, AUTHTBL, AUTHLIBS, AUTHS
- ☐ Inline changes to AT60HISO.ATHISOG, ATHISOS

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD0D.SCNAFEE.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Institution Definition File (IDF)

If BASE24-atm is configured to authorize PIN Change and PIN Unblock transactions where the transaction amount is non-zero, then the FAST CASH ACCOUNT TYPE on Screen 13 of the IDF may be configured with the default account type to be used for PIN Change and PIN Unblock transactions.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

ISO Host Interface Process

Host programs that authorize ATM transactions or receive completion messages for ATM transactions, must be able to process Balance Inquiry, PIN Change and PIN Unblock transactions, where the transaction amount is greater than zero.

Host programs that authorize ATM transactions or receive completion messages for ATM transactions, must be able to process PIN Change and PIN Unblock transactions which include a specific account type (checking, savings, credit).

Host programs that authorize ATM transactions must be able to process Balance Inquiry reversals, and may have to support Balance Inquiry completion messages. Host programs that receive completion messages for ATM transactions must be able to process Balance Inquiry reversals and Balance Inquiry completion messages.

Super Extract Process

Host programs that process TLF extract data may require modifications to support TLF records for Balance Inquiry, PIN Change and PIN Unblock transactions, where the transaction amount is greater than zero.

Host programs that process TLF extract data may require modifications to support TLF records for PIN Change and PIN Unblock transactions which include a specific account type (checking, savings, credit).

Host programs that process TLF extract data may require modifications to support TLF records for Balance Inquiry reversals and Balance Inquiry completion messages.

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

MasterCard: Authorization and Clearing Release 09.2 Document, 24 June 2009.

MasterCard: MDS Release 09.3 Document, 22 July 2009.

No Default Account on Withdrawal and Balance Inquiry

This section describes the implementation requirements associated with the No Default Account on Withdrawal and Balance Inquiry enhancement.

The implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the No Default Account on Withdrawal and Balance Inquiry enhancement.

The No Default Account on Withdrawal and Balance Inquiry enhancement allows BASE24-atm to process cash withdrawal and balance inquiry transactions successfully when no account type is specified.

The BASE24 Banknet Interface has recently been enhanced to support ATM transactions, to allow European customers to communicate with MasterCard using Banknet instead of EPS-Net. The account type in Banknet transactions must be set to '00' (default), and this requirement has highlighted incompatibilities between the Banknet Interface and the ATM Authorization process.

In issuer processing, the Banknet Interface maps an external account type of '00' to an internal account type of '00'. For cash withdrawal transactions, the ATM Authorization process always defaults the account type to a non-zero value (using the FAST CASH ACCOUNT TYPE on IDF Screen 13) before authorizing the transaction. This logic is not performed for balance inquiries, so these transactions are declined as the ATM Authorization process expects a non-zero account type to be specified. This issue was raised by a customer in case #489305.

In acquirer processing, the Banknet Interface sends an external account type of '00' only if the internal account type is '00'. However, the CPF/IDF routing logic in the ATM Authorization process always defaults the account type in a cash withdrawal transaction to a non-zero value (using the FAST CASH ACCOUNT TYPE on IDF Screen 13) before sending the transaction message to an interface process. This logic is not performed for balance inquiries, but the ATM Authorization process expects a non-zero account type to have been set by the ATM Device Handler process. This issue was raised by a customer in case #504580.

To resolve these issues, BASE24-atm processing has been enhanced as described below.

Issuer Processing

The “fast cash” functionality in ATM authorization processing has been extended to balance inquiries. The FAST CASH ACCOUNT TYPE on IDF Screen 13 will be used as a “default” account type for balance inquiries as well as for cash withdrawals. This logic allows balance inquiries to be processed successfully, where no account type has been specified.

The enhancement affects transactions configured for Authorization Level 1 (online only), Authorization Level 3 (online/offline) and Authorization Level 2 (offline only) Type 3 (positive balance). The enhancement does not affect transactions configured for Authorization Level 2 (offline only) that are not Type 3 (positive balance), as balance inquiries cannot be authorized by BASE24 if balance information is not available.

Acquirer Processing

The BASE24-atm Diebold 10XX/478X and NCR 5XXX Device Handlers have been enhanced to support new operation codes which allow cash withdrawals and balance inquiries from “default” accounts to be supported. Note that Non-Currency Dispense (NCD) transactions from “default” accounts are not permitted.

A new value of ‘99’ (default) has been added to the FAST CASH ACCOUNT TYPE on IDF Screen 13. This value indicates that the CPF/IDF routing logic in the ATM Authorization process must not default an account type of ‘00’ to a non-zero value. This logic allows balance inquiries and cash withdrawals to be sent to an interface process with an account type of ‘00’. Note that the ATM Authorization process does not default an account type of ‘00’ to a non-zero value if the transaction is routed to the interface process using an IPF.

The enhancement to the ATM Authorization process affects transactions configured for Authorization Level 1 (online only), which is the typical configuration for transactions routed to an interchange interface using the CPF/IDF. For all other Authorization Levels and Types, a value of ‘99’ in the FAST CASH ACCOUNT TYPE on IDF Screen 13 is treated in the same way as the existing value of ‘00’.

Software Impacts

All BASE24 customers:

- ☐ Inline changes to BA60DDL.DDLFIDF
- ☐ Inline changes to BA60AFT.SCRNIDF, SVIDFS

All BASE24-atm customers:

- ☐ Inline changes to AT60AUTH.AUTHG, AUTHS

All BASE24-atm Diebold 10XX/478X Device Handler customers:

- ☐ Inline changes to AT601000.C100NAMS, C1000S
- ☐ Changes to AT601000.CN100DOC, CNFGIX

All BASE24-atm NCR 5XXX Device Handler customers:

- ☐ Inline changes to AT60N50.N50NAMS, N50S
- ☐ Changes to AT60N50.CNN50DOC, CNFG5070

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD0D.SCNDFLT.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Institution Definition File (IDF)

A new value of '99' is supported in the FAST CASH ACCOUNT TYPE field on IDF Screen 13, to indicate that an account type of '00' must not be defaulted to a non-zero value.

Receipt File (RCPT)

Entries for "withdrawal from default" and "inquiry from default" must be added as needed.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Diebold 10XX Configuration File (CNFGIX)

The Diebold 10XX Configuration File (CNFGIX) has been amended to include two new states in group 21 that overlay the group 0 state. This state presents the account selection screen for withdrawals and inquiries. Instead of prompting for an account, this state simply places the appropriate character in the op code buffer and points to the next state in the flow. The changes are also documented in the CN100DOC file on the AT60C1000 subvolume.

NCR 5XXX Configuration File (CNFG5070)

The NCR 5XXX Configuration File (CNFG5070) has been amended to include three new states in group 21 that overlay the group 0 state and the group 1 state. The group 0 state presents the account selection screen for withdrawals and inquiries, and the group 1 state is used for Fast Cash withdrawals. Instead of prompting for an account, these states simply place the appropriate character in the op code buffer and point to the next state in the flow. Because those states are also used by groups 17, 18, and 19 (card before cash, parallel dispense, FC with card before cash), the new group can be loaded on top of any of these as well. The changes are also documented in the CNN50DOC file on the AT60N50 subvolume.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

External Interfaces

Host and Co-Network programs that process BASE24 transaction messages must be able to receive an account type of '00' (default) from BASE24-atm, if the FAST CASH ACCOUNT TYPE field on IDF Screen 13 is set to '99'.

Super Extract Process

Host programs that process TLF extract files must be able to receive an account type of '00' (default) from BASE24-atm, if the FAST CASH ACCOUNT TYPE field on IDF Screen 13 is set to '99'.

User/Operational Impacts

BASE24-atm Diebold 10XX/478X Device Handler customers and BASE24-atm NCR 5XXX Device Handler customers should add appropriate entries into the APCF for the generic balance inquiry transaction code (300000) and rebuild the APCFEMT.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Shared DH Maintenance

This section describes the implementation requirements associated with the Shared DH Maintenance enhancement.

The implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Shared DH Maintenance enhancement.

The Shared DH Maintenance enhancement allows BASE24-atm Shared NDC DH customers to support the same size (in bytes) of Data Objects (in the EMV Data Object list accessed via the DCT) for Shared NDC ATMs as it does for NCR NDC ATMs. The NCR AID table allows for 68 bytes in the request Data Object list, while the Shared AID table allowed for only 54 bytes in the request Data Object list. This difference meant that it was not possible to configure as many EMV data elements for inclusion in request messages from Shared NDC ATMs, compared to request messages from NCR NDC ATMs. This enhancement increases the size of the Data Objects List for Shared NDC ATMs so that it is equal to that of NCR NDC ATMs, by increasing the size of the Data Objects List for Shared NDC.

This enhancement also supports deletion of EMV AID (Application Identifier) table records from both Shared NDC and Shared DDC ATMs. The BASE24-atm Shared NDC Device Handler and Shared DDC Device Handler did not provide support for deleting unwanted and/or obsolete Application ID (AID) records which had previously been downloaded to Shared NDC, or Shared DDC EMV ATMs. However, this support is provided in the standard NCR NDC Device Handler. AID records are deleted from the ATM by downloading blank AID records with the same record number as the existing AID to be deleted. The Shared EMV AID Table has been modified to allow blank AIDs, thus allowing the deletion of unwanted EMV AID records from both Shared NDC and Shared DDC ATMs.

Software Impacts

All BASE24-atm NCR 5XXX Device Handler customers:

☐ Inline changes to AT60N50.N50DDL

All BASE24-atm Shared NCR 5XXX EMV Device Handler customers:

☐ Inline changes to AT60IN50.SCNWEMAI, RQWEMAI

All BASE24-atm Shared Diebold 10XX/478X Device Handler customers:

Inline changes to AT20IBLD.ENTWAPPS

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD0D.SCNWMNT.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

EMV N5XXX DCT support has been enhanced with respect to the Shared EMV Application ID (AID) Table (accessed from DCT Screen 45):

- Support for a total of 68 characters (an increase of 14 characters) in the request Data Object list, which is used to configure the tags of EMV data elements that must be included in EMV request messages by the ATM.

- Support for deleting AID table entries at the ATM by a subsequent download of a blank AID (i.e., an AID table record with the same index, but with a blank AID).

Diebold Entry Program support has been enhanced with respect to the Shared EMV Application ID (AID) Table (accessed from the ENTWBOLD Program menu):

- Support for deleting AID table entries at the ATM by a subsequent download of a blank AID (i.e., an AID table record with the same index, but with a blank AID).

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Shared DH Wincor EMV v1.2 Support

This section describes the implementation requirements associated with the Shared DH Wincor EMV v1.2 Support enhancement.

The implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Shared DH Wincor EMV v1.2 Support enhancement.

The Shared DH Wincor EMV v1.2 Support enhancement allows the Shared NDC EMV Device Handler and the Shared DDC EMV Device Handler to support functional enhancements in version 1.2 of Wincor's EMV hardware specification.

The BASE24-atm Release 6.0 Shared NDC EMV Device Handler and Shared DDC EMV Device Handler have been modified to support the following new functionality:

- Support for the new Confirmed Candidates Checking Flag. The Confirmed Candidates Checking Flag is used to determine whether the confirmed candidates should be removed from the candidate list. This checking is done in the Contact Chip state (State 'e'), when the EMV card is presented to the terminal. The Candidate List Creation state (Operation '000') creates the candidate list using data provided in the AID Options Table. For NDC, Table Entry 5 of the Extended State (State 'z') of Operation '000' is used to send a Confirmed Candidates Checking Flag of either '000' (candidate should not be removed) or '001' (candidate should be removed) to the ATM. For DDC, Table Entry 11 provides the equivalent functionality.

- Support for inhibiting the sending of default tags in the Status Message, if the Data Object List for Solicited Status Messages is specified in the Application Identifier table. If the Data Object List sent in Solicited Status Messages contains a default list of tags that have been populated using the Application Identifier (AID) Table, then the default tags are no longer sent in the message. This includes EMV Fault Status messages with a Logical Error, i.e. an Error Type of '1'. The Device Handlers (both the Shared N50 and Shared Diebold EMV extension modules) have been enhanced so that messages can be processed correctly, where the default tags are not present. The default tags are '9F27' (Cryptogram Information Data), '9F06' (Selected Application ID), '95' (TVR) and '9B' (Transaction Status Info).

- Support for EMV Fault Status Messages containing a Field Separator (FS) instead of a Group Separator (GS), to distinguish between the Data Objects and Issuer Application Data. EMV Fault Status Messages (Solicited) for both NDC and DDC, now use a FS instead of a GS to mark the end of the Data Objects field. The Device Handlers (both the Shared N50 and Shared Diebold EMV extension

modules) have been be enhanced to support a Field Separator used for this purpose. The Device Handlers continue to support a Group Separator as well, to ensure backwards compatibility.

Support to extend the Initialize EMV Transaction Data state (Operation 003) to allow the setting of the “Amount/Other” data element and optionally not to set the transaction type and currency exponent. The Initialize EMV Transaction Data State (Operation 003) of the Contact Chip state (State ‘e’), now allows the setting of the “Amount/Other” field while initializing the EMV transaction data. Also, it will be possible not to set the Transaction Type (tag 9C) and Transaction Currency Exponent (tag 5F36). For this, the Transaction Type will support a value of ‘100’ (do not set Transaction Type) and the Transaction Currency Exponent will support a value of ‘010’ (do not set Currency Exponent). These data elements correspond to Table Entries 4 and 6 respectively.

Support for new EMV data tags. The Device Handlers (both the Shared N50 and Shared Diebold EMV extension modules) have been be enhanced to support data tags 9F21 (Transaction Time), 9F03 (Amount/Other Binary), 9F04 (Amount/Other Numeric) and 9F41 (Transaction Sequence Counter).

Support for new AID Options in the Application Identifier Table. The Device Handlers (both the Shared N50 and Shared Diebold EMV extension modules) have been be enhanced to support two new values in the AID Options field in the Application Identifier Table. These two AID Options values will allow the contents of the Track2 Equivalent Data (tag 57) to be compared against the PAN (tag 5A), as required for MasterCard testing. Hence, the existing one-byte AID Options field has been changed to a two-byte field, supporting values 00, 01, 10 and 11.

Software Impacts

All BASE24-atm NCR 5XXX Device Handler customers:

Inline changes to AT60N50.N50DDL5, N50S, RECIMGCS, RQ50STS, SVLIBS, SVN50S

All BASE24-atm NCR 5XXX EMV Device Handler customers:

☐ Inline changes to AT60IN50. RQWEMAIS, SCNWEMAI

All BASE24-atm Shared NCR 5XXX EMV Device Handler customers:

☐ Inline changes to AT60IWN5.N50WEMVG, N50WEMVS

All BASE24-atm Diebold 10XX/478X Device Handler customers:

Inline changes to AT20IBLD.CNFGDDL5, ENTWAPPS, ENTWBLDG, ENTWSTAS, SCRNF0RM

All BASE24-atm Shared Diebold 10XX/478X EMV Device Handler customers:

☐ Inline changes to AT60IW10.C10WEMVG, C10WEMVS

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD0D.SCNWNEM.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

This enhancement includes changes to screens (e.g. State screens) accessed through the DCT (for NDC) and ENTWBOLD (for DDC).

Vendor Impacts

Not applicable

Reference Documents

Wincor-Nixdorf NDC/Diebold D91x Message Format Extension for EMV (Release 1.2, 26th November 2007)

TSS New AKDS Message Format for NCR and Diebold

This section describes the implementation requirements associated with the TSS New AKDS Message Format for NCR and Diebold enhancement.

The implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the TSS New AKDS Message Format for NCR and Diebold enhancement.

The TSS New AKDS Message Format for NCR and Diebold enhancement changes Automated Key Distribution System (AKDS) processing to use newer TSS message calls, which include the Host field in all applicable cases.

TSS supports several message calls that provide the ability for the Device Handler to pass a host or key signer name to TSS. The field makes it possible for more than one set of public keys to be defined in a BASE24 system for each ATM type. The ability to utilize multiple keys becomes important in situations where multiple certificate authorities are used for a fleet of the same type of ATMs.

The reason for this change is that Diebold has announced it will be changing Certificate authority in 2010. By August 2010, all new Diebold ATMs will utilize a new certificate key, so BASE24 customers will need the ability to support multiple keys as part of the Diebold AKDS solution. Note that the Diebold 10XX/478X AKDS module already utilizes the new format, but only for signature-based messages. A similar enhancement to the NCR AKDS solution (which previously utilized the old format for all messages) ensures consistency across all Device Handlers within BASE24-atm.

Note that the Shared NDC, Shared DDC, Tidel and Triton AKDS modules already utilize the new TSS format for all messages.

The Diebold Device Handler AKDS extension and the NCR Device Handler AKDS extension have been enhanced to take the Host name from the ATD for each ATM, and to pass this information to TSS, using a new set of calls. This enhancement allows more than one set of keys to be available for each device type in a customer's BASE24-atm system.

Software Impacts

All BASE24-atm Shared Diebold 10XX/478X AKDS Device Handler customers:

- ☐ Inline changes to AT60A100.C100AKDS
- ☐ Updated event (log) message manual BA60LOGM.AT1000

All BASE24-atm Shared NCR 5XXX AKDS Device Handler customers:

- ☐ Inline changes to AT60AN50.N50AKDS and AT60AN50.N50AKDG
- ☐ Updated event (log) message manual BA60LOGM.AT5070

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD0D.SCNTSSM.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

The KEY SIGNER field on ATD screen 22 must contain the appropriate Host name for Diebold and NCR terminals supporting AKDS.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Event Messages

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate remedy for a given message. Refer to the event message documentation files on the BA60LOGM subvolume.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Triton MACing On/Off Support

This section describes the implementation requirements associated with Triton MACing On/Off Support enhancement.

The implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Triton MACing On/Off Support enhancement.

The Triton MACing On/Off Support enhancement allows MACing to be turned on or off at the Triton terminal level by issuing commands from NCPCOM.

When ACI originally introduced MACing in the Triton Device Handler, the function could be disabled via the Supervisor Menu at the ATM. Recently, Triton removed this facility for security reasons. This left no reasonable way to turn off MACing. Once MACing had been enabled in BASE24, the only way to disable MACing was to reset the ATM. This means that any issue encountered with MACing or MAC keys results in the ATM being out of service for some time, until the new configuration on BASE24 is implemented.

Some customers have also faced performance issues with the Triton MACing feature (see case #482878, for example). The obvious way to overcome these issues in the short term is to turn off MACing at Triton ATMs, so it is important that the function can be disabled easily.

This enhancement allows the Triton Device Handler to turn on/off MACing at the terminal level. This is accomplished by issuing a new NCPCOM command for a specific terminal:

To disable MACing, the command is "9500MACINGOFF <terminal-id>;

To enable MACing, the command is "9500MACINGON <terminal-id>".

These commands cause the Triton Device Handler to include the field ID code "\ (solidus) in the next response message sent to ATM. The value in this field indicates whether the ATM should enable or disable the MACing function. When enabling the MACing function, the Triton Device Handler also includes the new MAC key to be used by the ATM. This is contained in field ID code "m", and the second half of the key will be present in field code "}" if double length keys are configured.

Note that Triton ATMs support the field ID code "\ (solidus) in a transaction response message, but not in a response to a configuration download, a Host totals message, RKT message or a reversal message.

Also note that MACing can still be turned on for a particular ATM by setting its TDF MAC Support Type field to a value of "3" and warmbooting the Device Handler. This

existing functionality remains unchanged by this new enhancement. However, MACing cannot be turned off for a particular ATM by setting its TDF MAC Support Type field to a value of “0” and warmbooting the Device Handler.

Software Impacts

All BASE24-atm Triton DH customers:

- ☐ Inline changes to AT60TRIT.TRITDDL, TRITONS.
- ☐ Changes to AT60TRIT.TRITMAN.
- ☐ Updated event (log) message manual BA60LOGM.ATTRIT.

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD0D.SCNTRIT.

CUSTOMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Terminal Data File (TDF)

The MAC SUPPORT TYPE on Screen 9 of the TDF, for a Triton ATM, will reflect whether MACing is currently supported.

If a 9500MACINGON command has been issued for this ATM, from the NCPCOM, this field will contain a value of ‘3’ (Hardware MAC Support). If a 9500MACINGOFF command has been issued for this ATM, this field will contain a value of ‘0’ (No MAC Support).

A customer can still choose to turn MACing on for a particular ATM by manually setting the MAC Support Type field to a value of “3” and warmbooting the Device Handler. This existing functionality remains unchanged by this new enhancement. Note, however, that a customer cannot choose to turn MACing off for a particular ATM by manually setting the MAC Support Type field to a value of “0” and warmbooting the Device Handler.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

The following command disables MACing at a Triton ATM:

“9500MACINGOFF <terminal-id>”

The following command enables MACing at a Triton ATM:

“9500MACINGON <terminal-id>”

The <terminal-id> is a mandatory parameter for both commands, and must contain the identifier from the TDF for the terminal for which the MACing status is being changed. A single space character must immediately precede the <terminal-id>.

The new NCPCOM commands do not support wildcard characters for the terminal-id parameter, so the commands can be used to turn on/off the MACing for only one terminal at a time. It is possible to create a batch file to be used via NCPCOM if there is a requirement to turn MACing on/off at multiple terminals simultaneously.

Event Messages

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate remedy for a given message. Refer to the event message documentation files on the BA60LOGM subvolume.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

Wincor SlimCash ATMs in NDC

This section describes the implementation requirements associated with the Wincor SlimCash ATMs in NDC enhancement.

The implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Wincor SlimCash ATMs in NDC enhancement.

The Wincor SlimCash ATMs in NDC enhancement allows the NCR 5XXX Device Handler to support the Wincor SlimCash NDC device in addition to Wincor ProCash NDC and NCR NDC devices.

The BASE24-atm NCR 5XXX Device Handler has always supported NCR NDC ATMs, and also supports Wincor's ProCash ATMs emulating NCR NDC ATMs. There are several technical differences between Wincor's ProCash and SlimCash devices. The BASE24-atm NCR 5XXX Device Handler has now been enhanced to interface with a Wincor SlimCash device. Since the SlimCash device is operating essentially as an NDC device, the modifications have been made to the main NCR device handler module, rather than within any extension code. Therefore, the BASE24-atm NCR 5XXX Device Handler is now capable of supporting any combination of NCR NDC, Wincor ProCash NDC, and Wincor SlimCash NDC devices, using a single NCR device handler object.

The primary difference between SlimCash devices and standard NDC devices is that SlimCash devices have a maximum bill limitation of 20 bills per dispense, whereas standard NDC devices have a maximum of 40 bills per dispense. Consequently, the NCR 5XXX Device Handler has been modified to recognize a Wincor SlimCash ATM so that it will allow no more than 20 bills on any dispense. If it is not possible to complete the transaction by dispensing a maximum of 20 bills, the customer is properly notified of the transaction cancellation.

In addition to support for the 20-bill maximum withdrawal limit, the following standard function set is also supported for Wincor SlimCash ATMs:

- Partial/Short Dispense with properly updated customer screens.

- Partial/Short Dispense with updated customer screen information also printed on the customer receipt.

- Support for 60mm/40 char width receipt paper.

- Support for Note Jam/Short Dispense messages from the ATM.

- Support for increased time to complete the transaction (up to 30 seconds).

Software Impacts

All BASE24-atm NCR 5XXX Device Handler customers:

- ☐ Inline changes to AT60N50.N50G, N50S
- ☐ Inline changes to BA60LOGM.AT5070

An SCN file is not provided for this enhancement.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Wincor Nixdorf Wincor SlimCash 200 Adaptation for an NDC Host, 5 November
(version 1.1, 05th May, 2008)

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Optional Enhancements

This section identifies the BASE24-atm enhancements that can be enabled or disabled with Release 6.0 Version 10, along with the changes, conversion requirements, and implementation requirements associated with each. These enhancements include the following:

- Shared DH 912 BNA Support

- Shared DH 912 CPM Support

BASE24-atm users can choose whether or not to implement each of the above enhancements. Users who plan to implement a particular enhancement must address the corresponding Release 6.0 Version 10 changes, conversion requirements, and implementation requirements. Users who do not plan to implement a particular enhancement can simply review the corresponding Release 6.0 Version 10 changes, conversion requirements, and implementation requirements for future reference.

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Shared DH 912 BNA Support

This section describes the implementation requirements associated with the Shared DH 912 BNA Support enhancement.

The implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Shared DH 912 BNA Support enhancement.

Wincor Nixdorf has developed a new module for their ATMs to enhance their deposit capabilities. Wincor calls this new module the CashIn module. It is present on CRS and RM2 ATMs, and it is part of the Check Cash Deposit Module (CCDM) on the CCDM ATMs. This new module allows a bunch of notes (bills) and/or coins (if the CoinIn NOVOTEC Module is present) to be deposited into a secured part of the ATM. It can accept, count and verify a stack of bills and/or a number of coins, and can reject any that are invalid or unrecognizable (i.e., torn, dirty, etc.) before it sends the transaction request to the processing Host (BASE24). The bills and coins can be of different denominations as well as up to four different currencies that are supported by the ATM.

This enhancement is applicable to BASE24 systems running ATMs loaded with firmware that conforms to the Shared 912 CashIn specification created by Wincor Nixdorf. These ATMs will run off BASE24's Diebold 10XX/178X Device Handler.

The ATM is configured with signature files. Each of these files provides the ATM with the information needed to recognize a particular denomination of note. The ATM is also configured with a Note Id List, which determines which notes of which denominations it can accept. Valid signature files must exist for particular note denominations before they are configured in the Note Id List, and the Note Id List must contain details of all note denominations for which signature files exist, although certain note denominations can be set as not being accepted. BASE24 also maintains a Note ID File (NIDF), detailing the denominations of the notes that can be accepted at the ATM, and this file is used by BASE24 to determine the transaction amount deposited at the ATM. The customer must ensure that these two files are kept in sync. Failure to do so will result in the cardholder being credited for an amount different from the amount of money that was deposited at the ATM.

At present, while the Wincor ATMs can be configured to support recycling of notes, BASE24 does not support recycling of notes at the ATM.

Commands available at the Device Control Terminal (DCT) will allow the customer to inquire as to the ATM's detailed configuration for accepting notes, change whether a particular note denomination is accepted or rejected, warmboot processes using the NIDFEMT and collect count information from the CashIn device.

This enhancement is provided in the form of two bindable extensions to the BASE24 Diebold 10XX/178X Device Handler. The first extension, the Shared Bunch Note Acceptor (BNA) extension, is required to support the CashIn deposits. It will accept deposits made in any of the denominations of the terminal's currency, configured in the Note ID file as being accepted at the particular ATM. The second extension, the BNA Multiple Currency extension, is required if any ATM is to support the acceptance of more than just the terminal's currency. In this case the Note ID file will contain details of denominations in a maximum of four different currencies that will be accepted by the ATM.

Prerequisite: Wincor Nixdorf ATMs loaded with firmware conforming to Wincor Nixdorf's specification for CashIn are required to support the Shared DH 912 BNA Support enhancement.

Software Impacts

All BASE24 customers:

- ☐ Changes to CUSTMACS and FLGSDOC

All BASE24-atm customers:

- ☐ Minor inline changes to Make files to include the new CUSTMACS flags, AT60SRC.ATMFM, and ATMMM
- ☐ Inline changes to AT60DCT.SVDCTS
- ☐ Minor inline changes to AT60AFT.COBRCP

All BASE24-atm Diebold 10XX/178X Device Handler customers:

- ☐ Changes to AT601000.C1000S and C1000D
- ☐ Minor inline changes to AT601000.C1000G, C100NAMS and C1000M
- ☐ Inline changes to AT601000.C100DDL and C100DDL
- ☐ Changes to AT601000.CNFGIX and CN100DOC
- ☐ Inline changes to AT601000.RQD10XXS, SCND10XX and SVD10XXS
- ☐ Inline changes to AT601000.RQT10XXS, SCRN10XX and SV10XXTS
- ☐ Inline changes to AT20IBLD.CNFGDDL, ENTWBLDG, ENTWBLDS, ENTWFRMS, ENTWSTAS and SCRNFORM
- ☐ A new Diebold Wincor Entry program object (ENTWBOLD)
- ☐ Updated Entibold documentation AT20IBLD.ENIXDOC
- ☐ Updated event (log) message manual BA60LOGM.AT1000

All BASE24-atm Diebold 10XX/178X SSB device handler customers:

- ☐ Inline changes to AT60D100.C100SSDS
- ☐ Minor inline changes to AT60D100.C100SSDM.

All BASE24-atm Diebold 10XX/178X Device Handler Multiple Currency Dispense customers:

- ☐ Minor inline changes to AT60U100.C100MCUS

All BASE24-atm Diebold 10XX/178X Device Handler customers requiring the new Shared 912 BNA Support functionality:

- ☐ A new extension subvolume, AT60DW10, with all of the files required to support the Shared 912 BNA Support functionality.

In addition to the above, customers must make sure they install the following, already existing code, available since Release 6.0 Version 8:

- ☐ Inline changes to BA60DDL.DDLGSTM, DDLATTKN, DDLFOMF
- ☐ Minor inline changes to AFT files for the NIDF, BA60AFT.MEGATBL, SECTBL, CCAFMLM and SVMHLPS
- ☐ Changes to Make files to include the new CUSTMACS flags, BA60SRC.BASEMM, BASEFM and BASEM
- ☐ Minor inline changes to BA60AFT.COBTKN, BA60SRC.ATTKNCVS and ATTKNID
- ☐ Minor inline changes to BA60DCT.RQDECTS and SVDPCTS
- ☐ Minor inline changes to BA60EMTB.BEMTBLM and EMTBRUN
- ☐ Minor inline changes to BA60RPT.OMFTBL
- ☐ Minor Inline changes to BA60MISC.BAMISCTD
- ☐ Inline changes to AT60DDL.DDLFTLF and DDLGTLFX
- ☐ Minor inline changes to AT60DDL.DDLFATD and DDLGATD
- ☐ New DDL for the NIDF, AT60DDL.DDLFNIDF
- ☐ Minor inline changes to AT60DDL.ATDDLMM and ATDDLMM
- ☐ Minor inline changes to AT60AFT.SCRNATD1, SVATDTBL and SVATDTS
- ☐ New Requester and Server code for the NIDF, AT60AFT.RQNIDFSS, RQNIDFXM, RQNIDFXS, SVNIDFTG, SVNIDFTS and SVNIDFTG
- ☐ Minor inline changes to Make files to include the NIDF, AT60AFT.ATAFTM and ATAFTMM
- ☐ Minor inline changes to AT60EXT.ATOMFXG, ATOMFXM and ATOMFXS
- ☐ New Extended Memory table build (EMTBUILD) code AT60EMTB.NIDEMTTG, NIDEMTTS and NIDEMTTM
- ☐ Minor inline changes to AT60EMTB.AEMTBM, AEMTBMM and AEMTDS
- ☐ Inline changes to AT60DCT.SCNDCOMM and RQDCOMMS
- ☐ Minor Inline changes to BA60MISC.LCONFAT

In addition to the above, any customer wishing to perform the Multiple Currency cash deposits must make sure they install the following, already existing, code:

- ☐ The ERF Requester/Server code(BA60AFT.RQERFS, RQERFM, SCRNERF, SVERFM and SVERFS)
- ☐ The ERFEMT Build code (BA60EMTB.ERFEMTTG, ERFEMTTS and ERFEMTTM)
- ☐ Inline changes to AT60UMCD.ATMCDHLG and ATMCDHLS
- ☐ Updated event (log) message manual BA60LOGM.ATATMCDH

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0D.SCNWDBN.

CUSTMACS Changes

The following Make flags have been added to the CUSTMACS file.

Set the following flag to true to bind the Shared 912 BNA module to the Diebold 10XX/178X device handler.

```
atm_dh_c1000_shrd_bna_on
```

If Multiple Currency Cash Deposits are to be accepted, then set the following flag to true to bind the Shared 912 BNA MCU module to the Diebold 10XX/178X device handler.

```
atm_dh_c1000_shrd_bna_mcu_on
```

Then set the following flag to true to bind the Multiple Currency Utilities into the Diebold 10XX/178X device handler.

```
atm_bna_mcu_on
```

LCONF Assign Changes

This section describes the existing assign records that are required to be present in the LCONF to support this enhancement. These following assigns therefore already exist in the LCONFAT file located on the BA60MISC subvolume.

ATM-NIDF

ATM-NIDFEMT

In addition, if BNA Multiple Currency is being used, several assigns that already exist in a multiple currency environment will also be used. These following assigns therefore already exist in the LCONFAT file located on the BA60MISC subvolume.

ERFEMT

ERF

LCONF Param Changes

This section describes the existing param records that are required to be present in the LCONF to support this enhancement. These following params therefore already exist in the LCONFAT file located on the BA60MISC subvolume.

ATM-NIDFEMT-MAX-TBL-SIZE

ATM-NIDFEMT-WARMIdentifier

In addition, if BNA Multiple Currency is being used, several params that already exist in a multiple currency environment will also be used. These following params therefore already exist in the LCONFAT file located on the BA60MISC subvolume.

CURRENCY-CODE

BASE-CURRENCY-CODE

Database Configuration

Token file (TKN)

The following existing BASE24-atm tokens are used by this enhancement:

The Shared BNA Counts Token (token ID "AS") contains the note counts for the Bulk Note Acceptor (BNA) device. These counts can only be obtained from the ATM. The data in this token gets written to the BASE24-atm Transaction Log File (TLF) whenever the ATM is balanced. This token is created only for ATMs supporting either the Shared NDC+ BNA enhancement or the Shared DH 912 BNA Support enhancement.

The BNA Multiple Currency Token (token ID "AT") carries details of the separate deposit amounts for each of the currencies that were deposited at the ATM, if BNA Multiple Currency Cash deposits are supported. It is added by the Device Handler and used during the response processing to format the receipt.

If the tokens for this enhancement are to be extracted with the TLF, sent to the host, or passed through the BIC ISO Interchange Interface, the appropriate records must be configured in the Token file.

ATM Terminal Definition Files (ATD)

This section describes the new and existing fields in the ATD file that are required to support this enhancement.

The NOTE ID Profile is an existing field on screen 2 of the ATD. It is only used by the Shared NDC+ BNA enhancement and the Shared DH 912 BNA Support enhancement.

This field should be set up for each ATM that is to support the Shared DH 912 BNA Support enhancement. It must contain the key value to be used in reading up the profile from the NIDF.

The Bunch Note Acceptor is a new field on screen 10 of the ATD. It is only used by the Shared DH 912 BNA Support enhancement.

Screen Field Name:	BUNCH NOTE ACCEPTOR
Screen Field Length:	1 alphabetic character
Range of Values:	'Y' indicates that the device supports Bunch Note Acceptor (BNA) deposits 'N' indicates that the device does not support BNA module deposits
Default Value:	'N'
DDL Name:	ATDS1.DEV-INFO.SHRD-BNA

Coin is a new field on screen 10 of the ATD. It is only used by the Shared DH 912 BNA Support enhancement.

Screen Field Name:	COIN
Screen Field Length:	1 alphabetic character
Range of Values:	'Y' indicates that the device supports Coin BNA deposits 'N' indicates that the device does not support Coin BNA module deposits
Default Value:	'N'
DDL Name:	ATDS1.DEV-INFO.SHRD-BNA-COIN

These fields should be set up for each ATM that is to support the Shared DH 912 BNA Support enhancement.

A full description of these fields can be found in the **BASE24 Files Maintenance Manual**.

Diebold Entibold Module

Support was added for a new state type, the CashIn state (r). Customers wanting to take advantage of this state will need to make their own configuration changes. Example states and screens for the Shared DH 912 BNA Support enhancement are available in the CNFGIX file on the AT601000 subvolume, in group 800.

Receipt file (RCPT)

New field type codes were added to the RCPT file for use when processing Multiple Currency Cash Deposits. The customer can configure these codes if wished, on the deposit receipt, to allow for exchange rate information to be produced.

The following new codes were defined:

- CA Converted amount – produces the amount for the particular currency deposited, once converted to the terminal's currency.
- OA Original amount – produces the amount in the currency that was deposited
- EA Exchange rate – produces the exchange rate used to convert the Original amount to the converted amount
- DC Produces the text “CONVERTED AMOUNT” on the receipt
- DO Produces the text “ORIGINAL AMOUNT” on the receipt
- DE Produces the text “EXCHANGE RATE” on the receipt

Each ATM can accept up to a maximum of four currencies, if Shared BNA Multiple currency support is implemented. Therefore, a suggested layout for the receipt is as follows:

```
#E(@RECORD NO.        #SQ>
#DO#OACC$> #DE#EA> #DC#CACC$>
#DO#OACC$> #DE#EA> #DC#CACC$>
#DO#OACC$> #DE#EA> #DC#CACC$>
#DO#OACC$> #DE#EA> #DC#CACC$>
#TT   #A1$>
@TO   #TO>
#XL#LB$>
#XA#AB$>
```

Each of the 4 lines of new field codes above will print details for one of the currencies deposited. If less than four currencies are deposited, then the lines will only be printed for those currencies that were present.

See below for an example of a receipt for a BNA deposit of four different currencies, produced using the suggested template above.

1	A	C	I	D	E	M	O
2							
3	DATE	TIME	STATION ID				
4	MM/DD/YY	HH:MM	S1AWIN10				
5							
6							
7	CARD NUMBER						
8	1234567000000003-003						
9							
10	LOCATION						
11	330 SOUTH 108TH AVE						
12	RECORD NO.		570				
13	ORIG AMT		\$210.00				
14	EXCH RATE		1.000000				
15	CNVRT AMT		\$210.00				
16	ORIG AMT		CAD20.00				
17	EXCH RATE		1.063500				
18	CNVRT AMT		\$21.27				
19	ORIG AMT		MXN50.00				
20	EXCH RATE		4.596220				
21	CNVRT AMT		\$4.60				
22	ORIG AMT		GBP100.00				
23	EXCH RATE		2.091200				
24	CNVRT AMT		\$209.12				
25							
26	DEPOSIT		\$444.99				
27	TO 013333						
28	LEDGER BAL		\$3412.00				

1	2	3
123456789012345678901234567890		

Exchange Rate File (ERF)

The ERF (and the corresponding ERF Extended Memory Table (ERFEMT)) are used by the Shared 912 BNA Multiple Currency extension if Multiple Currency cash deposits are to be accepted at any of the customer's ATMs. This file must contain the exchange rates to be used, and the ERFEMT must be successfully built from the ERF.

Note ID File (NIDF)

The BASE24-atm Note ID File (NIDF) contains in each record a set of note ids that will be accepted by the ATM. It is anticipated that several ATMs will share a similar setup of note ids, and so the records in the NIDF have a Note ID Profile as their key, allowing many ATD records to point to the same NIDF record.

Each NIDF record has details of the note id, currency code, and denomination of the note that is accepted. The NIDF File Maintenance sorts the note ids into ascending order in the NIDF. Note ids of 01 – 99 are valid, although BASE24 only supports a maximum of 45 note ids, so two screens are available to allow a maximum of 45 note ids to be configured. Note ids of 80 – 99 must be used to configure coins. Consecutive note ids do not have to be used. The denomination field is in whole units of the major currency unit if the note id is a note and not coin. For example, for USD, 10 would indicate \$10.00. It is whole units of the minor currency unit if the note id is for a coin. For example, for USD, 10 would indicate 10 cents.

The NIDF record also contains a modifier for each of the note ids. This indicates whether the note id should be accepted at the ATM or not. The LoadNoteIDs command from the DCT will load the appropriate NIDF record down to the ATM and allows the customer to remotely turn on or off acceptance of a particular note id that has a valid signature file at the ATM, and is in the ATM's Note ID List. Coin note ids (values of 80 – 99) will not be sent to the ATM as part of this command, as they cannot be updated remotely.

Note ID File Extended Memory Table (NIDFEMT)

The NIDFEMT is an Extended Memory Table built by the EMTBuild process from the NIDF. This allows the device handler to access the data in the NIDF in an efficient manner.

New Data Files

Not applicable

Template Changes

The customer must configure the AT60C100.C100NAMS file. This table contains the rcpt_money_sym_tbl_g which provides a location to configure the currency symbol that should appear on the receipt, for any currency that the customer wishes to accept at an ATM supporting the Shared 912 BNA enhancement.

Device Configuration

Diebold 10XX/178X Configuration File (CNFGIX)

The Diebold 10XX/178X Configuration File (CNFGIX) requires new states and screens to support the new Shared 912 BNA enhancement. For more detailed information, see the CN100DOC file on the AT60C1000 subvolume. A default set of screens is provided, but these may require specific modifications depending upon the institution's requirements.

Load group 800 is the new SSB BNA load group, and is required to support the Shared 912 BNA Support enhancement. It must be loaded on top of base load group 0, SSB DH load group 12 and Diebold BNA load group 712.

ATM Registry settings

The following registry setting is required to ensure that the ATM can correctly respond to a COUNTS command within the correct timeframe:

[HKEY_LOCAL_MACHINE\SOFTWARE\Wincor Nixdorf\ProTopas\CurrentVersion\CCOPEN\PROTOCOL\DDC\H2TSendCinSupplyCounters]

"VAL_FIELD_001"="VAL_WRONG_ATM_MODE,DC_MODE_TRAN_ACTIVE:20"

The 20 indicates the number of seconds that the ATM will wait to determine if the internal processing of the COUNTS command is complete. If the command has not finished being processed at that time, the command will be rejected. This time limit can be changed by the customer as required.

The following registry setting is required to ensure that a single cashin retract does not result in the ATM hitting the maximum allowed retracts:

[HKEY_LOCAL_MACHINE\SOFTWARE\Wincor Nixdorf\ProTopas\CurrentVersion\CCOPEN\CCCashInFW]

"MaxRetract"="99"

This value can be set to the number of retracts that the customer determines is their maximum. If left to the default value of "-1" the first retract will be the maximum.

The customer may need to request a CCERRMAP.ina or 6446.reg file from Wincor Nixdorf, to run against the registry, to ensure that the error mappings are set up correctly to match the Wincor Nixdorf specification.

CashIn device

A CashIn device must be installed on every ATM implementing Shared 912 BNA.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

ATM Terminal Definition Files (ATD)

For users wishing to take the Shared DH 912 BNA Support enhancement, either the ATD conversion program must be run to convert the ATD, or the customer may modify the fields manually via PATHWAY for affected ATMs. If the conversion program is to be used, the customer must create the new ATD file using the ATDDLUP file.

The ATD conversion program must be run to convert the ATD. Details of this conversion program can be found in the **General Changes** earlier in Section 5.

Host/Co-Network Impacts

Host impacts are incurred by those BASE24 users implementing this enhancement. Modifications to existing files and processing impact the External Interfaces and Super Extract processes.

External Interfaces

The host or co-network may require modification to process the BNA Multiple Currency token or Shared BNA Counts token. If the tokens for this enhancement are

to be sent to the host, or passed through the BIC ISO Interchange Interface, the appropriate records must be configured in the Token file.

Super Extract Process

If the tokens for this enhancement are to be included in the TLF Extracts, BASE24-atm users must consider this additional data in host programs that use the data from the log files.

Host code changes are required for all BASE24-atm users that want to use OMF Extract to accept and process information from the NIDF record.

User/Operational Impacts

BASE24 users must understand the concept of EMT Builds and the procedure to create an Extended Memory Table file (EMT). BASE24 users must understand the EMT Build Utility report that is generated by the EMT Build Utility. BASE24 users must also understand the concept of EMT warmboots and the procedure to warmboot an EMT. The NIDFEMT is used in this enhancement, along with the ERFEMT, if multiple currency cash deposits are being supported.

If changes are made to the NIDF, the NIDFEMT must be rebuilt, and any processes with the NIDFEMT open must be warmbooted in order to pick up these changes.

Disk Impacts

The TLF requires additional disk space as the tokens for this enhancement can be included in the transaction records logged.

The NIDF and NIDFEMT also require additional disk space. It is anticipated that most ATMs will be configured to share a particular profile in the NIDF; therefore the number of records to be stored in this file is expected to be small.

Diebold 10XX/178X Device Handler – DCT Commands

The following command is available on the Diebold 10XX Commands Screen (screen 40) of the Device Control Terminal (DCT).

LOAD NOTE IDS

When the operator places the cursor on this command and presses function key F1-ENTER, the DCT sends a command to the device handler to initiate a Load command to the ATM, to update its Note ID List to turn acceptance on or off for the note ids from the relevant record in the BASE24 NIDF.

The following command is available on the ATM Terminal Data Screen (screen 7) of the Device Control Terminal (DCT), as a SF3:

BNA STATUS

When the operator performs a SF3, the DCT sends a command to the device handler to request configuration information from the ATM. Details of note ids configured at the ATM, and the status of the drums and cashin and coinin boxes is returned. The information is displayed via EMS, using many EMS events. Please note, as a large number of EMS events are generated by this command, it is suggested that it is used sparingly in a production environment.

The Balance to Current Cash Screen (screen 8) and the Balance to Standard Cash Screen (screen 9) now support the use of the field titled COUNTS, for the Shared DH 912 BNA Support enhancement.

COUNTS

For an ATM equipped with a Shared BNA CashIn device, the BNA currency counts must be obtained from the device, before the ATM can be balanced. To obtain the BNA currency counts, the operator places the cursor in the COUNTS field and presses the F1 Function key. After obtaining the BNA currency counts, the operator can then balance the terminal as usual.

Diebold 10XX/178X Device Handler – NCS/NCPCOM Commands

The following 9500 command is supported in the Diebold 10XX/178X Device Handler. It is entered at the BASE24 AFT NCS screen or through NCPCOM.

LOADNOTEIDS

This 9500 command is an alternative to the DCT's LOAD NOTE IDS command. It causes the device handler to initiate a Load command to the ATM, to update its Note ID List to turn acceptance on or off for the note ids from the relevant record in the BASE24 NIDF.

For more information refer to the LOAD NOTEIDS command in the preceding subsection (Diebold 10XX/178X Device Handler – DCT Commands).

The syntax of this command is as follows.

DELIVER PROCESS <process name>, "9500LOADNOTEIDS <terminal ID>"

<process name>: The symbolic name assigned to the Diebold 10XX/178X Device Handler process in the XPNET node.

<terminal ID>: The terminal ID defined for the ATM in the ATM Terminal Data file (ATD).

COUNTS

This 9500 command is an alternative to the DCT's COUNTS command. It causes the device handler to obtain the BNA currency counts before the ATM can be balanced. For more information refer to the COUNTS command in the preceding subsection (Diebold 10XX/178X Device Handler – DCT Commands).

The syntax of this command is as follows.

DELIVER PROCESS <process name>, "9500COUNTS <terminal ID>"

<process name>: The symbolic name assigned to the Diebold 10XX/178X Device Handler process in the XPNET node.

<terminal ID>: The terminal ID defined for the ATM in the ATM Terminal Data file (ATD).

The following command is also supported in the Diebold 10XX/178X Device Handler. It is entered at the BASE24 AFT NCS screen or through NCPCOM. Please note that this is not a 9500 command.

BNASTATUS

This command is an alternative to the DCT's BNA STATUS command. It causes the device handler to request configuration information from the ATM. For more information refer to the BNA STATUS command in the preceding subsection (Diebold 10XX/178X Device Handler – DCT Commands).

The syntax of this command is as follows.

DELIVER PROCESS <process name>, "BNASTATUS <terminal ID>"

<process name>: The symbolic name assigned to the Diebold 10XX/178X Device Handler process in the XPNET node.

<terminal ID>: The terminal ID defined for the ATM in the ATM Terminal Data file (ATD).

System Limitations

BASE24-atm users installing this enhancement must be aware of their internal system limitations for messaging to the TLF. There are system limitations regarding the amount of data that can be passed through the BASE24 system and logged to a log file. BASE24-atm users should be aware of these limitations and be cognizant that support of new data in online messages and log files in the form of tokens can cause them to exceed these limitations. Appendix A of the **BASE24 Tokens Manual** provides information on assessing the impacts of tokens on log file disk space requirements.

Event Messages

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate remedy for a given message. Refer to the event message documentation files on the BA60LOGM subvolume.

Warmboot ATD Command

A 9529ATD command must be performed from the NCS/NCPCOM after the ATD is first configured to support the Shared DH 912 BNA Support enhancement, to initialize the ATD additional offset for the CashIn device.

Load Procedures

Load procedures may have to be modified to load the new terminal groups described above under Database Configuration.

Vendor Impacts

Compatible Devices and Software

The following device platforms and software versions are compatible with BASE24's Shared DH 912 BNA Support enhancement.

ATMs running firmware compliant with Wincor Nixdorf's CashIn Specification – Wincor Nixdorf NCD/Diebold D91x Message Format Extension for CashIn release 1.44.

Wincor hardware is already compatible with the newly supported format. Since Wincor has made its standards available to other vendors, it is anticipated that other vendors will support it as well.

Reference Documents

Wincor Nixdorf NDC/Diebold D91x Message Format Extension for CashIn (Release 1.44. 23rd June 2009)

ACI Worldwide Inc

Shared DH 912 CPM Support

This section describes the implementation requirements associated with the Shared 912 CPM Support enhancement.

The implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Shared 912 CPM Support enhancement.

Wincor Nixdorf has developed a new module for their ATMs to enhance their deposit capabilities. Wincor calls this new module the Check Cash Deposit Module (CCDM). This module allows a bundle of up to 50 checks to be deposited into a secured part of the ATM. It will then scan each check's MICR data, attempt to read the amount (via the optional OCR functionality), store an image, or reject any checks that are invalid, unrecognizable (i.e., torn, dirty, etc.), or improperly inserted, all before it sends the first transaction request, for the first check in the bundle, to the processing Host (BASE24). Each check in the bundle, although collectively deposited as a single stack of checks, will be processed as single check transactions. This means that a distinct transaction request will be sent for each check in the bundle, and a corresponding response received and processed, cash dispensed (if applicable), and receipt dispensed. This cycle will repeat until all checks in the bundle are processed.

This enhancement is applicable to BASE24 systems running ATMs loaded with firmware that conforms to the Shared 912 Cheque In specification created by Wincor Nixdorf. These ATMs will run off BASE24's Diebold 10XX/178X Device Handler.

This enhancement requires the C1000 SSB Device Handler extension code (AT60D100 subvolume) to be present. Additionally, optional extensions that can be purchased to provide added check processing functionality are the SSB Enhanced ATM Application (AT60B100 subvolume), and SSB Enhanced Check Processing Application (AT60C100 subvolume).

At present, while the Wincor ATMs can be configured with an optional sidecar to dispense coins, this enhancement does not support the dispensing of coins from non-whole dollar amount checks.

Prerequisite: Wincor Nixdorf ATMs loaded with firmware conforming to Wincor Nixdorf's specification for Cheque In are required to support the Shared 912 CPM enhancement.

Software Impacts

All BASE24 customers:

- ☐ Inline changes to BA60MAKE.CUSTMACS, FLGDOCS

All BASE24-atm customers:

- ☐ Inline changes to Make files AT60SRC.ATMFM, and ATMMM to include the new CUSTMACS Make flag.
- ☐ Inline changes to AT60UC04.CNVATDS
- ☐ New conversion program (AT60UC0D.CNVATDS) to convert the current ATD to the required format

All BASE24-atm Diebold 10XX/478X Device Handler customers:

- ☐ Inline changes to AT601000.C1000D, C1000G, C1000M, C1000S, C100DDLM, C100DDLS, RQT10XXS, SCRNX10XX, SV10XXTS, SVD10XXS
- ☐ Changes to AT601000.CN100DOC, CNFGIX
- ☐ Inline changes to AT20IBLD.CNFGDDL, ENTWBLDG, ENTWBLDS, ENTWFRMS, ENTWSTAS, and SCRNFOM
- ☐ A new Diebold Entry program object (ENTWBOLD)
- ☐ Updated Diebold Entry program documentation file (AT20IBLD.ENIXDOC)
- ☐ Updated event (log) message manual BA60LOGM.AT1000

All BASE24-atm Diebold 10XX/478X SSB Device Handler customers:

- ☐ Inline changes to AT60B100.C100SSBM, C100SSBS
- ☐ Inline changes to AT60D100.C100SSDM, C100SSDS

All BASE24-atm Shared Diebold 10XX/478X Device Handler customers requiring the new Shared 912 CPM Support functionality.:

- ☐ A new extension subvolume, AT60CW10, with all of the files required to support the Shared 912 CPM Support functionality

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD0D.SCNWDCI.

CUSTMACS Changes

The following Make flags have been added to the CUSTMACS file.

Set the following flag to true to bind the Shared 912 CPM module to the Diebold 10xx/478x Device Handler:

atm_dh_c1000_shrd_cpm_on

LCONF Assign Changes

Not applicable

LCONF Param Changes

This section describes the existing param records that are required to be added to the LCONF to support this enhancement. Details of the new parameters are provided in the LCONFAT file located on BA60MISC subvolume.

ATM-DH-MAX-TERMS-IN-CPLF-TBL

Identifies the maximum number of unique terminals that can exist at one time in the Check Proof List File (CPLF) extended memory table.

The default value is 1/5 of the value of the existing param, ATM-DH-MAX-TERMS.

ATM-DH-TERM-IN-CPLF-RETN-PRD

Identifies the length of time to retain a terminal in the Check Proof List File (CPLF) extended memory table before deleting. Unit of time is minutes.

The default value is 5 minutes.

Database Configuration

Token file (TKN)

The following existing BASE24-atm token is used by this enhancement:

- ☐ The Check Bundle Token (token ID "AU") contains the total number of checks in the bundle that was deposited in the Check Processing Module (CPM) and the index of the check currently being processed. The data in this token gets written to the BASE24-atm Transaction Log File (TLF) by the Authorization process for each check processed in the bundle.

If the token for this enhancement is to be extracted with the TLF, sent to the host, or passed through the BIC ISO Interchange Interface, the appropriate records must be configured in the Token file.

CPLF Extended Memory Table

This table is built by the device handler process. It provides the device handler with a temporary, internal storage location for CPLF records for checks that are marked to be returned to the cardholder after the last check in the bundle is processed. If the cardholder does not take the returned checks, these extended memory CPLF records will be written to the CPLF file on disk.

ATM Terminal Definition Files (ATD)

This section describes the new and existing fields in the ATD file that are required to support this enhancement.

The Check Cash Deposit Module is a new field on screen 10 of the ATD. It is only used by the Shared DH 912 CPM Support enhancement.

Screen Field Name:	CHECK CASH DEPOSIT MODULE
Screen Field Length:	1 alphabetic character
Range of Values:	'Y' indicates that the device supports Check Processing Module (CPM) deposits 'N' indicates that the device does not support CPM deposits
Default Value:	'N'
DDL Name:	ATDS1.DEV-INFO.SHRD-CCDM

This field should be set up for each ATM that is to support the Shared DH 912 CPM Support enhancement.

A full description of this field can be found in the **BASE24 Files Maintenance Manual**.

Diebold Entry Program (ENTIBOLD / ENTWBOLD)

Support has been added for the new Cheque In state '@q'. This state is required for Shared DH 912 CPM support. Load group 801 has been added to the CNFGIX configuration file for support of this new state, which includes a default set of states and screens. Customers may use this load group as an example to build on for their own customized device load. Refer to the AT601000.CN100DOC documentation file for more information regarding this new load group.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Diebold 10XX/178X Configuration File (CNFGIX)

The Diebold 10XX/178X Configuration File (CNFGIX) requires new states and screens to support the new Shared 912 CPM enhancement. For more detailed information, see the AT601000.CN100DOC. A default set of states and screens is provided, but these may require specific modifications depending upon the institution's requirements.

Load group 801 is the new SSB DH CPM (Shared) load group, and is required for the Shared 912 CPM Support enhancement. It must be loaded on top of base load group 0 and SSB DH load group 12.

ATM Registry settings

The following ATM registry settings are required to ensure the Device Handler will work as designed with the Cheque In device. Specific registry values are implementation dependent.

Current check ID screen:

Parameter name: CHQ_CURRENT_SCRN

Section in registry: [.\LYNXCI\SCREEN]

Total number of checks screen:

Parameter name: CHQ_TOTAL_SCRN

Section in registry: [.\LYNXCI\SCREEN]

Check image size on screen:

Parameter name: CHQ_SIZE

Section in registry: [.\LYNXCI\SCREEN]

Cheque In device

A Cheque In device must be installed on every ATM implementing Shared 912 CPM Support. ATM devices must be running software that conforms to Wincor Nixdorf's Message Format Extension for Cheque In specification, version 1.34.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Customers implementing this enhancement must verify security access to screen 10 of the ATD for Diebold terminals. Anyone responsible for configuring a Shared CCDM device on an ATM must be given access to this screen.

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

ATM Terminal Data File (ATD)

The ATD conversion program must be run to convert the ATD for the newly added SHRD-CCDM field (screen 10). Details of this conversion program can be found in the **General Changes** earlier in Section 5.

Host/Co-Network Impacts

External Interfaces

The host or co-network may require modifications to process the CHK-BNDL-TKN (AU) token and/or the MICR-DATA-TKN (12) token.

Super Extract Process

Host programs that process TLF extract data may require modifications to support the CHK-BNDL-TKN (AU) token and/or the MICR-DATA-TKN (12) token.

Host code changes are required for all BASE24-atm users that want to use OMF Extract to accept and process the new field from the ATD record.

User/Operational Impacts

Disk Impacts

The TLF requires additional disk space as the tokens for this enhancement can be included in the transaction records logged.

The CPLF requires additional disk space as CPLF records will be written to the CPLF file for all transactions involving checks retained by the ATM.

System Limitations

BASE24-atm users installing this enhancement must be aware of their internal system limitations for messaging to the TLF. There are system limitations regarding the amount of data that can be passed through the BASE24 system and logged to a log file. BASE24-atm users should be aware of these limitations and be cognizant that support of new data in online messages and log files in the form of tokens can cause them to exceed these limitations. Appendix A of the ***BASE24 Tokens Manual*** provides information on assessing the impacts of tokens on log file disk space requirements.

Event Messages

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate remedy for a given message. Refer to the event message documentation files on the BA60LOGM subvolume.

Warmboot ATD Command

A 9529ATD command must be performed from the NCS/NCPCOM after the ATD is first configured to support the Shared DH 912 CPM Support enhancement, to initialize the ATD additional offset for the Cheque In device.

Load Procedures

Load procedures may have to be modified to load the new terminal groups described above under Database Configuration.

Vendor Impacts

Compatible Devices and Software

The following device platforms and software versions are compatible with BASE24's Shared 912 CPM enhancement.

ATMs running firmware compliant with Wincor Nixdorf's Cheque In Specification – Wincor Nixdorf NCD/Diebold D91x Message Format Extension for Cheque In release 1.34.

Wincor hardware is already compatible with the newly supported format. Since Wincor has made its standards available to other vendors, it is anticipated that other vendors will support it as well.

Reference Documents

Wincor Nixdorf: NDC/Diebold D91x Message Format Extension for Cheque In specification, version 1.34, 13 June 2007.

6: BASE24-pos

Section 6 discusses the changes and enhancements made to BASE24-pos in Release 6.0 Version 10, along with the resulting conversion and implementation requirements that existing BASE24-pos users must consider when upgrading from Release 6.0 Version 9.

This section is divided into two parts: General Enhancements and Optional Enhancements.

- ☐ General enhancements describe the changes that affect all BASE24-pos users of a particular product or module, along with the associated conversion and implementation requirements that all BASE24-pos users must address.
- ☐ Optional enhancements describe the changes that affect only those BASE24-pos users that choose a new separate licensable module, along with the associated conversion and implementation requirements that BASE24-pos users must address to implement the enhancements.

Section 6 must be used in conjunction with Section 4 to identify the changes that BASE24-pos users need to consider when upgrading to Release 6.0 Version 10.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24-pos users must address in order to upgrade to Release 6.0 Version 10.

Changes and requirements are identified using a standard format defined in the introduction to Section 4.

- ☐ General Changes
- ☐ Address Verification Support for No ZIP Code
- ☐ Contactless from Unattended PIN Required Terminals
- ☐ Inhibit CVD Validation for EMV Transactions
- ☐ Network 2008 Mandates for POS Acquiring
- ☐ PTD Terminal Table Full Warning

These areas are summarized on the following pages.

Section 2 contains a functional overview of the general enhancements described here.

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General Changes

- ☐ Changes for all BASE24 Release 6.0 Version 10 enhancements affecting the DDL files and the BASE24 AFT Subsystem files (i.e., screens, requesters, servers) have been merged and cataloged under one SCN for each affected file. Applying changes for individual enhancements to the screen files does not allow BASE24 users to identify which changes go with a particular enhancement, as screen files do not have sequence numbers at the end of each line. Since the screen file changes usually involve new fields being added, all associated requester, server and DDL changes are required. Even though requester, server and DDL files do contain sequence numbers, all SCNs would need to be applied in order to accept the changes to the screen files. Therefore, all requester, server and DDL file changes, for all enhancements, were done under one SCN per file as well.
- ☐ The POS Retailer Definition File (PRDF) has been modified to include an additional field on screen 4 for the Network 2008 Mandates for POS Acquiring enhancement. The POS Retailer Definition File screen, requester and server have been modified to process this new field. For further information, see Network 2008 Mandates for POS Acquiring, where the impacts of this enhancement are described, later in Section 6.

Software Impacts

All BASE24-pos customers:

- ☐ Inline changes to PS60DDL.DDLFPRDF
- ☐ Inline changes to PS60AFT.SCRNPRDF, RQPRDFS

Note: The list of files impacted by the merged changes is also included in the **Software Impacts** section of the applicable enhancement(s).

A complete list of the merged files can be found in the SCN file BA60UD0D.SCNMERGE.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

POS Retailer Definition File (PRDF)

The PRDF conversion program may be run to convert the PRDF. Users must create the new PRDF file using the BADDLFUP file.

The PRDF conversion program on the PS60UC04 subvolume can be used to convert the PRDF records from the BASE24 Release 5.3 format to the BASE24 Release 6.0 Version 10 format. The PRDF conversion program on the PS60UC0D subvolume can be used to convert the PRDF records from the Release 6.0 Version 9 format to the Release 6.0 Version 10 format. If modifications to the conversion program are necessary, the source code may be modified by editing the CNVPRDFS file.

To compile the conversion program, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#prdfcnv.t,name $prdfm,nowait/ cnvprdfm -  
force -list -listreplace
```

Release 5.3 users should update the CNVPRDFR obey file on the PS60UC04 subvolume with the appropriate information. Release 6.0 Version 9 users should update the CNVPRDFR obey file on the PS60UC0D subvolume with the appropriate information. The CNVPRDFR should then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREAD04 file on the BA60UC04 subvolume for Release 5.3 customers and to the AAREAD0D file on the BA60UC0D subvolume for Release 6.0 Version 9 customers.

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Address Verification Support for No ZIP Code

This section describes the implementation requirements associated with the Address Verification Support for No ZIP Code enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Address Verification Support for No ZIP Code enhancement.

The Address Verification Support for No ZIP Code enhancement allows BASE24-pos to perform address verification on blank as well as non-blank ZIP codes.

The Address Verification Service (AVS) is an optional function which allows the BASE24-pos Router/Authorization process to validate the two components (address and ZIP code) that comprise a cardholder's billing address. When BASE24-pos authorizes a transaction containing AVS data, the add-on Address Verification module compares the address and ZIP code provided in the transaction data to corresponding fields stored in the Address Verification segment in the appropriate Cardholder Authorization File (CAF) record. The module returns a status code that indicates the outcome of the address verification process. For the purpose of this comparison, the address field contains only the numeric data in the cardholder's address. Because some addresses do not contain any numeric data, it is valid for the address component to contain no data (i.e. blanks), and this does not prevent BASE24-pos from performing the verification. However, the Router/Authorization process has previously not performed address verification if the transaction contains a blank ZIP code.

Support for this function is enabled by configuring a new LCONF parameter called POS-ISS-AVS-BLANK-ZIP-CDE-ALWD. If a blank ZIP code is sent in the transaction along with a valid address (and the LCONF parameter is set to "Y"), then BASE24-pos provides a status code of "A" (Addresses matched but ZIP codes did not match) after performing the address verification.

Software Impacts

All BASE24-pos customers:

- ☐ Inline changes to PS60RTAU.RTAUG, ROUTERS, AUTHLIBS
- ☐ Updated event (log) message manual BA60LOGM.PSROUTER

All BASE24-pos AVS customers:

- ☐ Inline changes to PS60ADDR.ADDRS

A complete list of files affected by this enhancement can be found in the SCN file BA60UD0D.SCNAVSZ.

CUSTMACS Changes

Not Applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

This section describes the param record that must be added to the LCONF to support these changes. Details of the new parameter are provided in the LCONFPS file located on the BA60MISC subvolume.

POS-ISS-AVE-BLANK-ZIP-CDE-ALWD

This param, used by the BASE24-pos Router/Authorization process, specifies whether the add-on Address Verification module should perform address verification if the ZIP code in the AVS data is blank.

Y = Perform address verification if the ZIP code in the AVS data is blank.

N = Do not perform address verification if the ZIP code in the AVS data is blank.

The default value is "N".

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Users should be aware that additional event messages will be generated as a result of implementing this enhancement.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Contactless from Unattended PIN Required Terminals

This section describes the implementation requirements associated with the Contactless from Unattended PIN Required Terminals enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Contactless from Unattended PIN Required Terminals enhancement.

The Contactless from Unattended PIN Required Terminals enhancement allows BASE24 to be configured to accept transactions from an unattended terminal that does not support PIN entry. This option can be used to prevent contactless transactions from being declined because a PIN is not entered.

The enhancement also allows BASE24 to be configured to accept contact transactions from an unattended terminal that does not support PIN entry, and to accept contactless transactions without a PIN, even when the terminal does support PIN entry.

Issuers can control the conditions under which cardholders are required to enter a PIN in POS transactions by configuring the PIN PROCESSING FLAG on screen 7 of the CPF. Traditionally, BASE24 has supported three options for setting this field.

PIN PROCESSING FLAG — A code indicating whether a PIN entry is required to perform POS transactions with this card prefix. Valid values are as follows:

0 = PIN entry is not required to complete any transaction.

1 = PIN entry is required for every transaction; transactions from terminals without PIN capabilities are declined.

2 = PIN entry is required with transactions from terminals with PIN capabilities; however, transactions from attended terminals without PIN capabilities are processed without the PIN. Transactions from unattended terminals without PIN capabilities are declined.

If an issuer requires a PIN with transactions from terminals with PIN capabilities, then these definitions mean that an issuer cannot configure the following options:

- ☐ A PIN is not required for contact transactions from unattended terminals **without** PIN capabilities.
- ☐ A PIN is not required for contactless transactions from unattended terminals **without** PIN capabilities.

- ☐ A PIN is not required for contactless transactions from **attended** terminals with PIN capabilities.
- ☐ A PIN is not required in contactless transactions from **unattended** terminals with PIN capabilities.

To allow these options to be configured, this enhancement adds support for the following new PIN PROCESSING FLAG values:

3 = PIN entry is required with transactions from terminals with PIN capabilities; however, transactions from terminals without PIN capabilities are processed without the PIN.

4 = PIN entry is required with transactions from terminals with PIN capabilities, except for contactless transactions; however, transactions from terminals without PIN capabilities are processed without the PIN.

Note that BASE24 uses the following logic to determine whether a terminal is attended and supports PIN entry, and whether the transaction is contactless:

- ☐ An unattended terminal is identified by a value of "02" or "27" in the POS Condition Code;
- ☐ A PIN-capable terminal is identified by a value of "2" in the third byte of the POS Entry Mode;
- ☐ A contactless transaction is identified by a value of "07" or "91" in the first two bytes of the POS Entry Mode.

The new PIN PROCESSING FLAG values ('3' and '4') provide issuers with greater flexibility in controlling the conditions under which cardholders are required to enter a PIN in POS transactions.

Software Impacts

All BASE24 customers:

- ☐ Inline changes to BA60DDL.DDLFCPF
- ☐ Inline changes to BA60AFT.SCRNCPF, RQCPFS, SVCDFS

All BASE24-pos customers:

- ☐ Inline changes to PS60RTAU.AUTHLIBS

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD0D.SCNCUT.

CUSTOMACS Changes

Not Applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

This section describes the enhanced field in the CPF used that can be configured to support this enhancement.

Card Prefix File (CPF)

The following existing field supports new values to allow customers to configure the conditions under which a PIN is required in POS transactions:

Screen Name: PIN PROCESSING FLAG

Screen: CPF Screen 7

DDL Name: PIN-REQ

Valid Values are: 0 (PIN Not Required),

1 (PIN Required),

2 (PIN Not Required from Attended Terminal),

3 (PIN Required from PIN-Capable Terminal),

4 (PIN Not Required in Contactless Transactions).

Default: 0

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

Inhibit CVD Validation for EMV Transactions

This section describes the implementation requirements associated with the Inhibit CVD Validation for EMV Transactions enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Inhibit CVD Validation for EMV Transactions enhancement.

For additional detailed information on the Inhibit CVD Validation for EMV Transactions enhancement see Section 4 (Base).

Software Impacts

For full details on the Base software impacts for this enhancement see Software Impacts in Section 4 (Base) of this document.

All BASE24-pos EMV customers:

- ☐ Inline changes to PS60IRTA.RTAUEMVS

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD0D.SCNCVDI.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

For full details on the database configuration impacts for this enhancement see Database Configuration in Section 4 (Base) of this document.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Network 2008 Mandates for POS Acquiring

This section describes the implementation requirements associated with the Network 2008 Mandates for POS Acquiring enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Network 2008 Mandates for POS Acquiring enhancement.

The Network 2008 Mandates for POS Acquiring enhancement provides support for network mandates introduced by MasterCard and Visa.

Partial Approval Support

Functional Overview:

MasterCard requires that all acquirers in the U.S. region support partial approval of authorization transactions. This mandate applies to specific acquirers or merchants based on Merchant Category Code (MCC). In the February 2008 Global Operations Bulletin, implementation was scheduled to begin in May 2009, but this was revised to May 2010 in the July 2008 Global Operations Bulletin. For Debit MasterCard card issuers (including prepaid), the mandate is November 1, 2008, and is effective with the Banknet 08.2 release.

BASE24-pos Release 6.0 Version 6 and Version 8 included enhancements to support Partial Approvals (i.e., Authorizations for Lesser Amount / Partial Authorizations) for Purchase, Pre-Authorization Purchase, Pre-Authorization Purchase with Cash Back, Mail Phone Order, Cash Advance, and Purchase with Cash Back transactions. This enhancement extends this functionality to provide full compliance with the MasterCard requirements.

Current System:

Support for partial approvals already exists in BASE24-pos, configured using the PTD field `pre_auth_less_amt_flg`. This field is mapped to the PSTM `pre^auth^opt` field, and is used by the Router/Authorization process to allow transactions to be partially-approved, if necessary. Partial approvals are now fully supported in the SPDH, as a small error in the processing of Purchase with Cash Back (PWCB) transactions was resolved with the application of a fix (described in case #429277) as part of a Release 6.0 Version 9 project (SPDH Support for Network Mandates).

If a transaction is authorized for a lesser amount on a back-end Host, then a fix applied to HISO for case #430755 in 2007 ensures that the requested amount is

replaced by the authorized amount in the internal message. However, an issuer Host (using the HISO Interface) should never authorize a transaction for a lesser amount, because there is no field in the external message that indicates whether this function is supported by the acquirer and/or the terminal. Also, transactions acquired via HISO from a front-end Host cannot be partially-approved, because there is no mechanism for setting the PSTM pre^auth^opt field, although HISO does provide the capability for updating the Transaction Amount field (P-4) in a response message sent to the Host.

Like HISO, transactions acquired via CRT Auth cannot be partially-approved, because there is no mechanism for setting the PSTM pre^auth^opt field, although the CRT DH does provide the capability for updating the Transaction Amount field in a response message sent to the OLT Screen.

Finally, there were a couple of restrictions in the BASE24-pos implementation of partial authorizations, which needed to be addressed:

- ☐ The PTD field pre_auth_less_amt_flg applies to all transactions performed at the terminal, but the MasterCard mandate applies to acquired transactions only. Therefore, a customer may wish to comply with the mandate, but may not want to allow partial approvals for on-us cards.
- ☐ The BASE24-pos logic applies to a number of different transaction types, including Purchase with Cash Back. MasterCard does not permit partial approvals on PWCB transactions (unless the whole of the cash amount is excluded), so the processing needs to cater for this.

Enhanced System:

The existing BASE24-pos support of partial approvals has been enhanced to allow the function to be configured by card type, e.g. prepaid debit, and to ensure that the function is supported by all system endpoints that are required to do so.

A new value is supported in the PTD field pre_auth_less_amt_flg, to specify that the terminal supports partial authorizations, but only on transactions not involving cash. This option allows the PTD to be configured to ensure that transactions such as PWCB from an SPDH terminal are not partially-approved.

A new field (called PARTIAL AMOUNT SUPPORT on the AFT screen) has been added to the Card Prefix File (CPF) to specify whether partial authorizations for a lesser amount are supported by the card type. This field is checked in conjunction with the existing PSTM pre^auth^opt field that indicates whether partial authorizations are supported by the terminal. The transaction can be partially-approved only if both fields indicate that this is allowed. This approach provides greater flexibility in controlling partial approvals based on card type, as opposed to being applicable for all transactions from a specific terminal.

A new field has been added to the POS Data1 token to indicate whether partial authorizations for a lesser amount are supported by the terminal. Tokens can be configured for inclusion in the BASE24 external message, so this enhancement allows BASE24 to notify an issuer Host whether authorizations for a lesser amount are supported by the terminal. This enhancement also allows an acquirer Host to notify BASE24 whether partial authorizations are supported by the terminal. A field for the same purpose (called AUTH PARTIAL AMOUNT on the AFT screen) has been added to the POS Retailer Definition File (PRDF), in case the token is not present (or the new token field is not set) in the incoming message from a Host.

Status Check Transaction Support

Functional Overview:

Visa originally came out with the partial approval and \$1 status check requirement as part of their *April 2008 Business Enhancements*. The Visa mandate was vague and affected all MCC types with the exception of 6011 (ATMs). The crux of the Visa requirement was to add support for partial authorization of \$1 status check messages, where the partial approved amount is greater than the original transaction amount (\$1).

MasterCard has released more definitive requirements regarding their support for \$1 Status Check transactions, which are identified in their Banknet Release 08.2 and Banknet Release 09.1 documents (partial approval amount edit changes). The MasterCard requirement is to allow issuers to provide a maximum transaction amount in responses to Status Check transactions performed at Automated Fuel Dispensers (AFDs). The merchant receives chargeback protection up a specific limit.

The enhancement allows the issuer (BASE24-pos Router/Authorization) to respond to specific transactions with a partial approval amount that is greater than the initial transaction amount (e.g. \$1 status check), which also does not exceed the issuer's chargeback liability. BASE24 is required to accept the status check message, and then set the transaction amount to a configurable value. U.S. customers today may set this value to \$75, which is typically the chargeback liability supported by several issuers. If the account does not have \$75, then the POS Router/Authorization process performs a partial approval, and (if the acquirer supports partial authorizations) sets the approved amount to the available balance + draft protection – holds, and returns a partial approval response code to the acquirer. Where a chargeback liability needs to be defined, the acquiring endpoints must have the capability to set the transaction amount to a default to limit chargeback liability.

Current System:

BASE24-pos does not currently support partial authorization for a greater amount.

A fix for case #413866 was applied to the VisaNet Interface in 2007 to allow a response message sent to VisaNet to contain an authorized amount greater than the requested amount. This fix is redundant in terms of BASE24 product code, because BASE24-pos never authorizes a transaction for an amount greater than the requested amount, and an issuer Host (using the HISO Interface) should never authorize a transaction for an amount greater than the requested amount, because there is no field in the external message that indicates whether this function is supported by the acquirer and/or the terminal.

Enhanced System:

The best method of supporting this requirement is to make use of the existing (and enhanced) BASE24-pos support of partial approvals.

The acquiring endpoint (device handler or interface) must set an indicator to specify that 'partial approval for a greater amount' is supported. The indicator is set if partial approvals in general are supported, and the transaction amount is one (major) unit of currency. The indicator is the existing PSTM pre^auth^opt field, but a new value is supported to specify that the transaction may be authorized for a greater amount. The new value does not affect modules not being enhanced by this project, e.g. BIC ISO, as these modules simply treat the value as indicating that partial approvals are not supported.

Because MasterCard is restricting the card types for which this function is supported (debit or prepaid card only), a flag should be added to the CPF to specify whether partial approvals for a greater amount are supported for the card type. This approach also allows partial approvals to be supported for not-on-us cards (as required for MasterCard), but for issuers to inhibit the functionality for on-us cards. So, an additional value is supported by the PARTIAL AMOUNT SUPPORT field in the CPF (added for partial approval support) to specify whether partial authorizations for a lesser amount only are supported by the card type, or whether partial authorizations for both a lesser amount and a greater amount are supported by the card type.

This enhancement requires a mechanism which allows BASE24 to notify an issuer Host whether authorizations for a greater amount are supported by the terminal. So, the new field in the POS Data1 token (added for partial approval support) is also used to indicate whether partial authorizations for a greater amount are supported by the terminal. This approach has the benefit of allowing transactions acquired via HISO from a front-end Host to be authorized for a greater amount, because the new token field can be used to set the PSTM pre^auth^opt field. For Hosts that do not include the new field in the POS Data1 token, the PSTM pre^auth^opt field is set from the new AUTH PARTIAL AMOUNT field in the PRDF (added for partial approval support).

The SPDH has been enhanced to set the new value in the PSTM pre^auth^opt field, based on the value of the existing PTD field pre_auth_less_amt_flg, and a

transaction amount of one (major) unit of currency. To allow authorizations for a greater amount at an SPDH terminal, the PTD field `pre_auth_less_amt_flg` should be set to the new value (added for partial approval support), because transactions involving cash should not be authorized for a greater amount than requested.

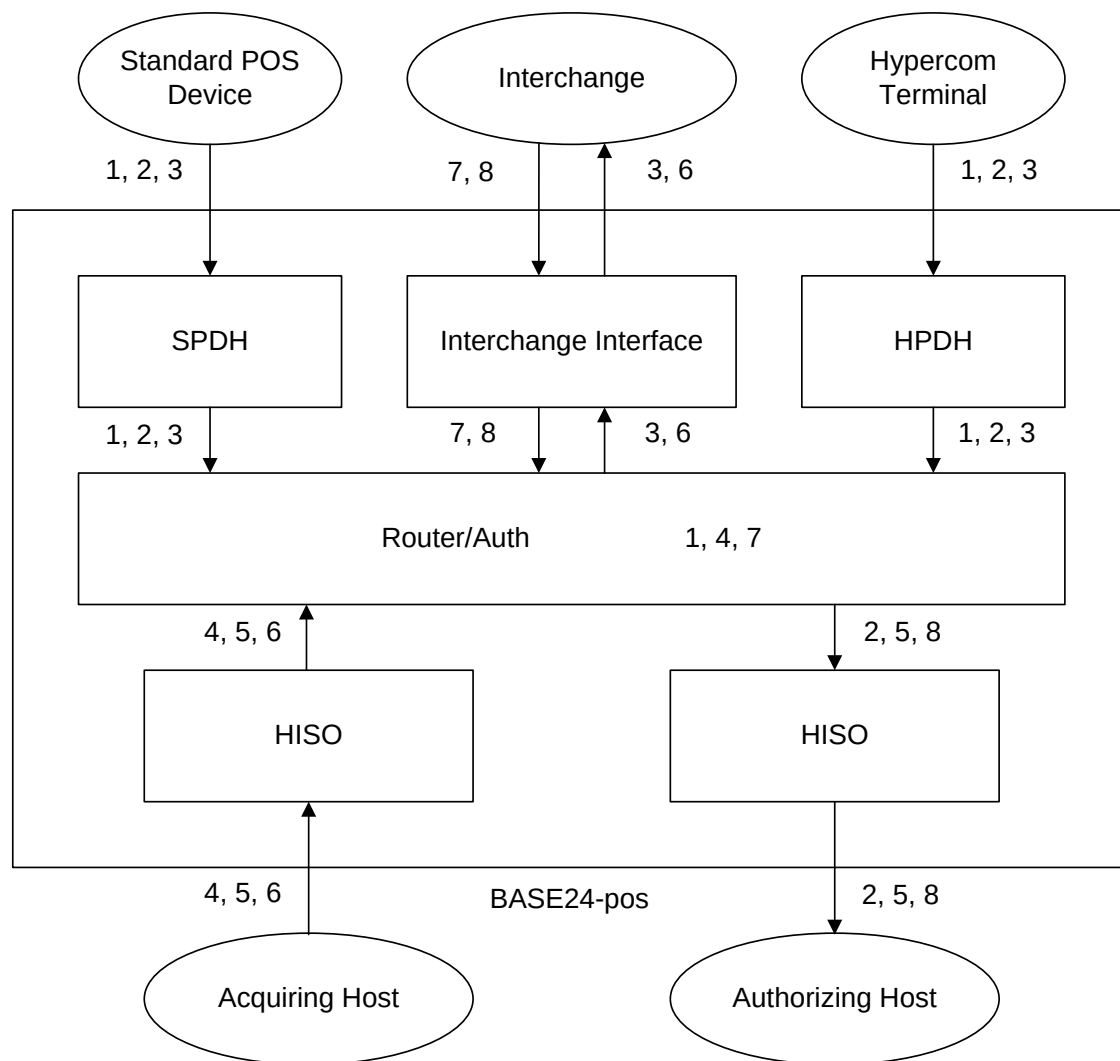
The POS Router/Authorization process has been enhanced to use the new indicator in the token and the new flag in the CPF to determine whether authorization for a greater amount is appropriate for the transaction. If it is, then the transaction amount can be set to the maximum permitted value for the transaction, and existing partial authorization processing reduces the amount authorized, if necessary. The maximum transaction amount is configured in another new field (called PARTIAL AUTH LIMIT on the AFT screen) on the CPF, as the value is likely to match the issuer's maximum chargeback protection, which typically differs by card type. If the new maximum transaction amount is set to zero on the CPF record, then the transaction is authorized for the original requested amount. Note that the maximum transaction amount configured on the CPF is also used when checking the Authorization Selection Table (AST) online limit, as this provides a more accurate estimate of the "transaction value" than the requested amount.

If the transaction is configured for authorization on a back-end Host, then it is recommended that the requested amount be sent to the back-end Host for authorization, and so the Host system is responsible for determining the maximum amount that can be authorized. If the BASE24 system is configured to perform limit checking before the transaction is sent to a back-end Host system for authorization, then it may be necessary to overwrite the transaction amount with the maximum transaction amount configured in the PARTIAL AUTH LIMIT field on the CPF (to allow the maximum transaction amount to be checked against the limits). For this configuration, it is important that the Host system also supports authorizations for a lesser amount, because the transaction could be declined incorrectly if the Host is unable to reduce the authorized amount. For example, the requested amount could be \$1, and the maximum transaction amount configured on the CPF record could be \$50, which is the transaction amount sent to the Host. If the cardholder's account has a balance of \$20, then the Host should authorize the transaction for \$20, because it would be incorrect to decline the transaction, as the original requested amount was only \$1.

If the Host system does not support partial authorizations, and the card prefix is configured for online/offline authorization, then another new field (called PARTIAL AUTH ROUTING on the AFT screen) on the CPF can be configured to allow a transaction to be processed on BASE24, without being sent to the back-end Host. If the Host system does not support authorizations for a greater amount, but does support authorizations for a lesser amount, then the same CPF field can be configured to allow a transaction to be sent to the back-end Host with the transaction amount set to the maximum transaction amount configured on the CPF.

Processing Diagram

This enhancement is designed to support the processing flows shown in the diagram below.



Processing Flow

The numbers below equate to those shown on the diagram above.

On-us card at a directly-connected terminal, authorized by BASE24-pos. The Device Handler may set the PRE-AUTH-OPT field in the PSTM to either “P” (transaction may be authorized for a lesser amount) or “G” (transaction may be authorized for a greater amount), if the appropriate field in the PTD record indicates that the transaction can be partially-authorized. The field is set to “G” only if the transaction amount is one major unit of currency. The Authorization module uses the new PARTIAL AMOUNT SUPPORT field on the CPF to

determine whether partial authorizations are permitted for the prefix, and the value of PSTM.PRE-AUTH-OPT may be modified accordingly. If PSTM.PRE-AUTH-OPT is set to "G" and the new PARTIAL AUTH LIMIT field on the CPF is non-zero, then the Authorization module sets the AMOUNT-1 field in the PSTM to the value of PARTIAL AUTH LIMIT, and sets PSTM.PRE-AUTH-OPT to "P". Existing partial authorization processing may then result in the value of the AMOUNT-1 field in the PSTM being reduced. This processing is also performed when BASE24-pos authorizes a transaction that could not be sent to an issuer Host for authorization.

On-us card at a directly-connected terminal, sent to an issuer Host for authorization. The Device Handler may set the PRE-AUTH-OPT field in the PSTM to either "P" (transaction may be authorized for a lesser amount) or "G" (transaction may be authorized for a greater amount), if the appropriate field in the PTD record indicates that the transaction can be partially-authorized. The field is set to "G" only if the transaction amount is one major unit of currency. The Authorization module uses the new PARTIAL AMOUNT SUPPORT field on the CPF to determine whether partial authorizations are permitted for the prefix, and the value of PSTM.PRE-AUTH-OPT may be modified accordingly. If PSTM.PRE-AUTH-OPT is set to "G", then the Authorization module uses the new PARTIAL AUTH ROUTING field on the CPF to determine whether the AMOUNT-1 field in the PSTM should be set to the new PARTIAL AUTH LIMIT field on the CPF, and the value of PSTM.PRE-AUTH-OPT may be modified accordingly. The POS Data1 token can be included in the message sent by the issuing HISO process, with the PARTIAL-AUTH-OPT field informing the Host that the transaction can be partially-authorized.

Not-on-us card at a directly-connected terminal, sent to an interchange for authorization. The Device Handler may set the PRE-AUTH-OPT field in the PSTM to either "P" (transaction may be authorized for a lesser amount) or "G" (transaction may be authorized for a greater amount), if the appropriate field in the PTD record indicates that the transaction can be partially-authorized. The field is set to "G" only if the transaction amount is one major unit of currency. In the message to the card scheme, the interchange interface specifies that the transaction can be partially-authorized, as long the transaction complies with the rules of the card scheme. For example, MasterCard supports partial authorizations for a greater amount only on transactions initiated at a fuel filling station.

On-us card received from an acquiring Host, authorized by BASE24-pos. The acquiring HISO process sets the PRE-AUTH-OPT field in the PSTM to the value of the PARTIAL-AUTH-OPT field in the POS-Data1 token in the message received from the Host. If the token is not present, the HISO process sets the PRE-AUTH-OPT field in the PSTM to "D", which results in the Router module using the new AUTH PARTIAL AMOUNT field on the PRDF to determine whether the transaction can be partially-authorized. The Authorization module uses the new PARTIAL AMOUNT SUPPORT field on

the CPF to determine whether partial authorizations are permitted for the prefix, and the value of PSTM.PRE-AUTH-OPT may be modified accordingly. If PSTM.PRE-AUTH-OPT is set to "G" and the new PARTIAL AUTH LIMIT field on the CPF is non-zero, then the Authorization module sets the AMOUNT-1 field in the PSTM to the value of PARTIAL AUTH LIMIT, and sets PSTM.PRE-AUTH-OPT to "P". Existing partial authorization processing may then result in the value of the AMOUNT-1 field in the PSTM being reduced. This processing is also performed when BASE24-pos authorizes a transaction that could not be sent to an issuer Host for authorization.

On-us card received from an acquiring Host, authorized by a Host system. The acquiring HISO process sets the PRE-AUTH-OPT field in the PSTM to the value of the PARTIAL-AUTH-OPT field in the POS-Data1 token in the message received from the Host. If the token is not present, the HISO process sets the PRE-AUTH-OPT field in the PSTM to "D", which results in the Router module using the new AUTH PARTIAL AMOUNT field on the PRDF to determine whether the transaction can be partially-authorized. The Authorization module uses the new PARTIAL AMOUNT SUPPORT field on the CPF to determine whether partial authorizations are permitted for the prefix, and the value of PSTM.PRE-AUTH-OPT may be modified accordingly. If PSTM.PRE-AUTH-OPT is set to "G", then the Authorization module uses the new PARTIAL AUTH ROUTING field on the CPF to determine whether the AMOUNT-1 field in the PSTM should be set to the new PARTIAL AUTH LIMIT field on the CPF, and the value of PSTM.PRE-AUTH-OPT may be modified accordingly. The POS Data1 token can be included in the message sent by the issuing HISO process, with the PARTIAL-AUTH-OPT field informing the Host that the transaction can be partially-authorized.

Not-on-us card received from an acquiring Host, sent to an interchange for authorization. The acquiring HISO process sets the PRE-AUTH-OPT field in the PSTM to the value of the PARTIAL-AUTH-OPT field in the POS-Data1 token in the message received from the Host. If the token is not present, the HISO process sets the PRE-AUTH-OPT field in the PSTM to "D", which results in the Router module using the new AUTH PARTIAL AMOUNT field on the PRDF to determine whether the transaction can be partially-authorized. In the message to the card scheme, the interchange interface specifies that the transaction can be partially-authorized, as long the transaction complies with the rules of the card scheme. For example, MasterCard supports partial authorizations for a greater amount only on transactions initiated at a fuel filling station.

On-us card received from an interchange, authorized by BASE24-pos. The interchange interface process may set the PRE-AUTH-OPT field in the PSTM to "P" (transaction may be authorized for a lesser amount) or "G" (transaction may be authorized for a greater amount) based on the information received from the interchange and the rules of the card scheme. For example, Visa supports authorizations for a greater amount only if the transaction amount

contains one major unit of currency. The Authorization module uses the new PARTIAL AMOUNT SUPPORT field on the CPF to determine whether partial authorizations are permitted for the prefix, and the value of PSTM.PRE-AUTH-OPT may be modified accordingly. If PSTM.PRE-AUTH-OPT is set to "G" and the new PARTIAL AUTH LIMIT field on the CPF is non-zero, then the Authorization module sets the AMOUNT-1 field in the PSTM to the value of PARTIAL AUTH LIMIT, and sets PSTM.PRE-AUTH-OPT to "P". Existing partial authorization processing may then result in the value of the AMOUNT-1 field in the PSTM being reduced. This processing is also performed when BASE24-pos authorizes a transaction that could not be sent to an issuer Host for authorization.

On-us card received from an interchange, authorized by a Host system. The interchange interface process may set the PRE-AUTH-OPT field in the PSTM to "P" (transaction may be authorized for a lesser amount) or "G" (transaction may be authorized for a greater amount) based on the information received from the interchange and the rules of the card scheme. For example, Visa supports authorizations for a greater amount only if the transaction amount contains one major unit of currency. The Authorization module uses the new PARTIAL AMOUNT SUPPORT field on the CPF to determine whether partial authorizations are permitted for the prefix, and the value of PSTM.PRE-AUTH-OPT may be modified accordingly. If PSTM.PRE-AUTH-OPT is set to "G", then the Authorization module uses the new PARTIAL AUTH ROUTING field on the CPF to determine whether the AMOUNT-1 field in the PSTM should be set to the new PARTIAL AUTH LIMIT field on the CPF, and the value of PSTM.PRE-AUTH-OPT may be modified accordingly. The POS Data1 token can be included in the message sent by the issuing HISO process, with the PARTIAL-AUTH-OPT field informing the Host that the transaction can be partially-authorized.

Standard message routing has been modified for transactions that may be authorized for a greater amount than requested. The new PARTIAL AUTH ROUTING field on the CPF is checked by the Router module when a CPF record has been located for the transaction. This field may be configured to indicate that "under floor" authorization (typically by BASE24) should be performed. This option may be useful for transactions where online/offline authorization is configured for a card prefix, but the Host system has not been enhanced to perform partial authorizations. If the Authorization Selection Table (AST) and Routing Table (RTBL) has been configured to send "over floor" transactions to a Host ("AUTHH") but to authorize "under floor" transactions on BASE24 ("AUTHF"), then this option allows any transaction that may be authorized for a greater amount to be processed on BASE24 only. If this field is not configured to indicate that "under floor" authorization should be performed, then the PARTIAL AUTH LIMIT field on the CPF is used when checking the Authorization Selection Table (AST) online limit. This value is used because it provides a more accurate estimate of the "transaction value" than the requested amount.

Assumptions and Restrictions

BASE24-pos will authorize a transaction for a greater amount only if the requested amount is one major unit of the transaction currency and the transaction does not involve cash, unless the incoming authorization message, e.g. from an acquiring Host, states explicitly that the transaction may be authorized for a greater amount.

It is possible for a transaction that can be authorized for a greater amount to end up being authorized for a lesser amount. This can occur when the cardholder has a tiny amount of funds available, which is less than the one major unit of currency included in the request message.

When a transaction is authorized on a back-end Host via HISO, BASE24 may adjust the transaction amount (if appropriate) before sending the authorization request to the Host. This means that the transaction amount received from the terminal may not be the same value sent to the Host.

BASE24-pos has not been enhanced to support an internal response code specific to partial approvals, but an ISO response code of '10' (approved for partial amount) is supported in the BASE24 external message, to allow a Host system to identify a partially-approved transaction.

The BASE24-pos Hypercom Device Handler (HPDH) has not been enhanced, because the implication from Version 3.44 of the Hypercom message specification is that partial approvals are supported only in Prepaid Card Redemption transactions (section 7.33). For this transaction type, the remaining amount that must be collected from the cardholder is sent in Field 54 (Additional Amounts) in the response message, but this field is not supported in other Authorization and Sale response messages.

Enhancements to the Visa II POS device handler are out of scope for this project.

Enhancements to the CRT Authorization sub-system are out of scope for this project.

Enhancements to the BIC ISO Interface are out of scope for this project.

Software Impacts

All BASE24 customers:

- ☐ Inline changes to BA60DDL.DDLFCPF, DDLGPSTM, DDLPSTKN
- ☐ Inline changes to BA60AFT.SCRNCPF, RQCPFS, SVCPFS
- ☐ Inline changes to BA60SRC.PSTKNCVS
- ☐ Inline changes to BA60UC04.CNVCPFS
- ☐ New conversion program (BA60UC0D.CNVCPFS) to convert the CPF to the required format

All BASE24-pos customers:

- ☐ Inline changes to PS60DDL.DDLFPRDF, DDLGPTD
- ☐ Inline changes to PS60AFT.SCRNPRDF, RQPRDFS
- ☐ Inline changes to PS60AFT.SCRNPTD, RQPTDS, SVPTDS, SVPTDTBL
- ☐ Inline changes to PS60RTAU.RTAUG, ROUTERS, AUTHLIBS
- ☐ Inline changes to PS60SPDH.ASPDHS
- ☐ Inline changes to PS60HISO.HISOG, HISOS
- ☐ Inline changes to PS60EXT.PSOMFXS
- ☐ Inline changes to PS60UC04.CNVPRDFS
- ☐ New conversion program (PS60UC0D.CNVPRDFS) to convert the PRDF to the required format
- ☐ Updated event (log) message manual BA60LOGM.PSHISO

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD0D.SCNETM.

Prerequisite: The Increase Size of Segments in the CPF enhancement must be installed prior to implementation of this enhancement.

CUSTMACS Changes

Not Applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Card Prefix File (CPF)

The following new field allows customers to configure whether transactions may be authorized for a lesser amount or may be authorized for either a lesser amount or a greater amount:

Screen Name: PARTIAL AMOUNT SUPPORT

Screen: CPF Screen 8

DDL Name: PARTIAL-AUTH-SPPT

Valid Values are: 0 (Not Allowed), 1 (Allowed), 2 (Lesser Allowed)

Default: 2

The following new field allows customers to configure that “under floor” authorization should be performed when the transaction can be authorized for a greater amount, because the authorizing Host system does not support partial authorizations. This field also allows customers to configure that the transaction amount should be set to the PARTIAL AUTH LIMIT (see below) when the transaction can be authorized for a greater amount, because the authorizing Host system does not support authorizations for greater amount:

Screen Name: PARTIAL AUTH ROUTING
Screen: CPF Screen 8
DDL Name: PARTIAL-AUTH-RTE
Valid Values are: 0 (Standard), 1 (Auth Limit), 2 (Under Floor)
Default: 0

The following new field allows customers to configure the maximum amount (in major currency units) that may be authorized when a transaction is authorized for an amount greater than requested. It is recommended that this field is set to zero if limit checking for the prefix is not performed on BASE24:

Screen Name: PARTIAL AUTH LIMIT
Screen: CPF Screen 8
DDL Name: PARTIAL-AUTH-LMT
Valid Values are: 0 – 99999999
Default: 0

POS Retailer Definition File (PRDF)

The following new field allows customers to configure whether the retailer supports partial authorizations for all transaction types, supports partial authorizations only for transactions not involving cash, or doesn’t support partial authorizations at all:

Screen Name: AUTH PARTIAL AMOUNT
Screen: PRDF Screen 4
DDL Name: AUTH-PARTIAL-AMT
Valid Values are: 0 (Not Allowed), 1 (Lesser Allowed), 2 (Non-Cash Only)
Default: 0

POS Terminal Data File (PTD)

A new value of ‘2’ is supported in the AUTH LESSER AMT field on PTD Screen 6, to indicate that the terminal supports partial authorizations, but only for transactions not involving cash.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Devices must be configured to support partial authorization.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Card Prefix File (CPF)

The CPF conversion program must be run to convert the Card Prefix File (CPF). The CPF conversion program converts the customer's current CPF to a new record format. Details of this conversion program can be found in the **General Changes** in Section 4.

POS Retailer Definition File (PRDF)

The PRDF conversion program must be run to convert the POS Retailer Definition File (PRDF). The PRDF conversion program converts the customer's current PRDF to a new record format. Details of this conversion program can be found in the **General Changes** earlier in Section 6.

Host/Co-Network Impacts

ISO Host Interface Process

Host programs that support partial approvals must be able to receive a response code of '10' (partial approval) from BASE24-pos, and may send a response code of '10' (partial approval) to BASE24-pos.

Host programs that support partial authorizations must be able to process the new field added to the POS Data1 Token (CH) in transactions received from BASE24-pos, and may set a value in the new field added to the POS Data1 Token (CH) in transactions sent to BASE24-pos.

Super Extract Process

Host programs that process PTLF extract data may require modifications to support the new field added to the POS Data1 Token (CH).

Host code changes are required for all BASE24-pos users that use OMF Extract to accept and process the new fields in records containing a CPF record image.

Host code changes are required for all BASE24-pos users that use OMF Extract to accept and process the new field in records containing a PRDF record image.

User/Operational Impacts

Screen Impacts

There are changes on the BASE24 AFT screens for the CPF, PRDF and PTD.

Disk Impacts

The POS Data1 Token (CH) is added to the PSTM for transactions eligible for partial approval for a greater amount. The token may subsequently be logged to the PTLF or ILF, and so additional disk space for the PTLF or ILF may be required.

Event Messages

Users should be aware that additional event messages will be generated as a result of implementing this enhancement.

Vendor Impacts

SPDH devices may require modifications to support partial approvals for a lesser amount.

SPDH devices may require modifications to support partial approvals for a greater amount.

Reference Documents

Visa: April 2008 VisaNet Business Enhancements, Technical Letter and Implementation Guide, Version 2.0.

MasterCard: Global Operations Bulletin No.9, September 2007

MasterCard: Global Operations Bulletin No.2, February 2008

MasterCard: Global Operations Bulletin No.7, July 2008

MasterCard: Banknet Release 08.2 Document

MasterCard: Banknet Release 09.1 Document

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PTD Terminal Table Full Warning

This section describes the implementation requirements associated with the PTD Terminal Table Full Warning enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the PTD Terminal Table Full Warning enhancement.

The PTD Terminal Table Full Warning enhancement provides timely notification when the number of entries in any of the POS Terminal Data (PTD) extended memory tables approaches the table's capacity limit, enabling operations staff to proactively increase the size of the table and avoid disruptions to transaction processing. With this enhancement, the ACI Standard POS Device Handler (SPDH) and Hypercom Device Handler (HPDH) monitor the growth of the PTD dynamic table and PTD terminal table. When the size of a table meets or exceeds a threshold parameter specified in the LCONF, a warning message is logged to the Event Management Service (EMS).

To perform this monitoring, the device handler determines the number of entries in each PTD extended memory table every time a PTD record is read. It uses this value, along with the maximum number of entries that each table can contain, to calculate each table's current percentage full value. The current percentage full value of each table is compared with the value of the new LCONF parameters POS-DH-PTD-NTFY-PCNT and POS-DH-PTD-NTFY-PCNT-DYN. When the current percentage full value of the PTD terminal table or PTD dynamic table meets or exceeds the value of the POS-DH-PTD-NTFY-PCNT and POS-DH-PTD-NTFY-PCNT-DYN LCONF parameters, respectively, the device handler logs an event message warning of the condition. Each time the current percentage full value of the table increases or decreases by 1% and still remains at or above the notification value, an event message is logged.

The event messages are as follows:

```
PTD dynamic extended memory table is <n>% full
PTD terminal extended memory table is <n>% full
```

When one of these messages is logged, operations staff should take corrective action by increasing the value of the POS-DH-MAX-TERMS-IN-DYN-TBL, POS-DH-MAX-TERMS-IN-STATIC-TBL, or POS-DH-MAX-TERMS LCONF parameter (respectively), followed by a warmboot of the device handler.

The default value for both the POS-DH-PTD-NTFY-PCNT and POS-DH-PTD-NTFY-PCNT-DYN LCONF parameters is 0 (turned off/do not monitor). The recommended

setting is 90 percent.

Software Impacts

All BASE24 customers:

- ☐ Changes to BA60MISC.BAMISCTD

All BASE24-pos customers:

- ☐ Changes to BA60MISC.LCONFPS

All BASE24-pos ACI Standard POS Device Handler (SPDH) customers:

- ☐ Inline changes to PS60SPDH.ASPDHG and ASPDHS
- ☐ Updated event (log) message manual BA60LOGM.PSSPDH

All BASE24-pos Hypercom Device Handler (HPDH) customers:

- ☐ Inline changes to PS60HPDH.HPDHG and HPDHS
- ☐ Updated event (log) message manual BA60LOGM.PSHYPRCM

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0D.SCNTTBL.

CUSTMACS Changes

Not Applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

This section describes the parameter records that are required to be added to the LCONF to support this enhancement. Details of the params are provided in the *BASE24 Logical Network Configuration File Manual*. Samples of the params are included in the LCONFPS file located on the BA60MISC subvolume.

POS-DH-PTD-NTFY-PCNT

This parameter is used by the SPDH and HPDH. When the current percentage full value of the PTD terminal table meets or exceeds the value of this parameter, the device handler logs a warning event message. Each time the current percentage full value of the table increases or decreases by 1% and still remains at or above this parameter's value, a warning message is logged.

Valid values for this parameter are 0 through 99. The default value is 0. The recommended value is 90.

POS-DH-PTD-NTFY-PCNT-DYN

This parameter is used by the SPDH and HPDH. When the current percentage full value of the PTD dynamic table meets or exceeds the value of this parameter, the device handler logs a warning event message. Each time the current percentage full value of the table increases or decreases by 1% and still remains at or above this parameter's value, a warning message is logged.

Valid values for this parameter are 0 through 99. The default value is 0. Recommend 90 percent.

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Users should be aware that additional event messages will be generated as a result of implementing this enhancement.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

Optional Enhancements

This section identifies the BASE24-pos enhancements that can be enabled or disabled with BASE24 Release 6.0 Version 10, along with the changes, conversion requirements, and implementation requirements associated with each.

There are no new optional enhancements available for BASE24-pos.

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7: BASE24-teller

Section 7 discusses the changes and enhancements made to BASE24-teller in Release 6.0 Version 10, along with the resulting conversion and implementation requirements that existing BASE24-teller users must consider when upgrading from Release 6.0 Version 9.

This section is divided into two parts: General Enhancements and Optional Enhancements.

- ☐ General enhancements describe the changes that affect all BASE24-teller users of a particular product or module, along with the associated conversion and implementation requirements that all BASE24-teller users must address.
- ☐ Optional enhancements describe the changes that affect only those BASE24-teller users that choose a new separate licensable module, along with the associated conversion and implementation requirements that BASE24-teller users must address to implement the enhancements.

Section 7 must be used in conjunction with Section 4 to identify the changes that BASE24-teller users need to consider when upgrading to Release 6.0 Version 10.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24-teller users must address in order to upgrade to Release 6.0 Version 10.

There are no general enhancements inherent to BASE24-teller.

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Optional Enhancements

This section identifies the BASE24-teller enhancements that can be enabled or disabled with Release 6.0 Version 10, along with the changes, conversion requirements, and implementation requirements associated with each.

There are no new optional enhancements available for BASE24-teller.

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8: BASE24-remote banking

Section 8 discusses the changes and enhancements made to BASE24-remote banking in Release 6.0 Version 10, along with the resulting conversion and implementation requirements that existing BASE24-remote banking users must consider when upgrading from Release 6.0 Version 9.

This section covers changes made to the following products within the BASE24 family:

- ☐ BASE24-telebanking
- ☐ BASE24-billpay
- ☐ BASE24 Kernel/IEP

This section is divided into two parts: General Enhancements and Optional Enhancements.

- ☐ General enhancements describe the changes that affect all BASE24-remote banking users of a particular product or module, along with the associated conversion and implementation requirements that all BASE24- remote banking users must address.
- ☐ Optional enhancements describe the changes that affect only those BASE24-remote banking users that choose a new separate licensable module, along with the associated conversion and implementation requirements that BASE24-remote banking users must address to implement the enhancements.

Section 8 must be used in conjunction with Section 4 to identify the changes that BASE24-remote banking users need to consider when upgrading to Release 6.0 Version 10.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24-remote banking users must address in order to upgrade to Release 6.0 Version 10.

There are no general enhancements inherent to BASE24-remote banking.

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Optional Enhancements

This section identifies the BASE24-remote banking enhancements that can be enabled or disabled with Release 6.0 Version 10, along with the changes, conversion requirements, and implementation requirements associated with each.

There are no new optional enhancements available for BASE24-remote banking.

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9: BASE24 Interchange Interfaces

Section 9 discusses the changes and enhancements made to BASE24 Interchange Interfaces in Release 6.0 Version 10, along with the resulting conversion and implementation requirements that existing BASE24 Interchange Interface users must consider when upgrading from Release 6.0 Version 9.

This section is divided into two parts: General Enhancements and Optional Enhancements.

- ☐ General enhancements describe the changes that affect all BASE24 Interchange Interface users, along with the associated conversion and implementation requirements that all BASE24 Interchange Interface users must address.
- ☐ Optional enhancements describe the changes that affect only those BASE24 Interchange Interface users that choose to implement the enhancements, along with the associated conversion and implementation requirements that BASE24 Interchange Interface users must address to implement the enhancements.

Section 9 must be used in conjunction with Section 4 to identify the changes that BASE24 Interchange Interface users need to consider when upgrading to Release 6.0 Version 10.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24 Interchange Interface users must address in order to upgrade to Release 6.0 Version 10.

Changes and requirements are identified using a standard format defined in the introduction to Section 4.

☐ General Changes

These areas are summarized on the following pages.

Section 2 contains a functional overview of the general enhancements described here.

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General Changes

☐ None

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Optional Enhancements

This section identifies the BASE24 Interchange Interface enhancements that can be enabled or disabled with Release 6.0 Version 10, along with the changes, conversion requirements, and implementation requirements associated with each. These enhancements include the following:

- ☐ There are no optional enhancements for the BASE24 Interchange Interface modules.

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10: BASE24 Value-Added Products

Section 10 discusses the changes and enhancements made to the BASE24 value-added products in Release 6.0 Version 10, along with the resulting conversion and implementation requirements that existing users of these products must consider when upgrading from Release 6.0 Version 9.

The value-added products discussed in this section include the following:

- ☐ BASE24-card
- ☐ BASE24-customer service
- ☐ BASE24-infoBase
- ☐ BASE24-inventory Voucher Manager

This section is divided into two parts: General Enhancements and Optional Enhancements.

- ☐ General enhancements describe the changes that affect all BASE24 value-added product users, along with the associated conversion and implementation requirements that all BASE24 value-added product users must address.
- ☐ Optional enhancements describe the changes that affect only those BASE24 value-added product users that choose to implement the enhancements, along with the associated conversion and implementation requirements that BASE24 value-added product users must address to implement the enhancements.

Section 10 must be used in conjunction with Section 4 to identify the changes that BASE24 value-added product users need to consider when upgrading to Release 6.0 Version 10.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all BASE24 Value-added product users must address in order to upgrade to Release 6.0 Version 10.

There are no general enhancements affecting the following BASE24 value-added product users:

- ☐ BASE24-card
- ☐ BASE24-customer service
- ☐ BASE24-infoBase
- ☐ BASE24-inventory Voucher Manager

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Optional Enhancements

This section identifies the BASE24 enhancements that can be enabled or disabled with Release 6.0 Version 10 that have an impact on the following BASE24 value-added products, along with the changes, conversion requirements, and implementation requirements associated with each.

There are no optional enhancements affecting the following BASE24 value-added product users:

- ☐ BASE24-card
- ☐ BASE24-customer service
- ☐ BASE24-infoBase
- ☐ BASE24-inventory Voucher Manager

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A: Logical Network Configuration File Changes

Appendix A contains the Logical Network Configuration File (LCONF) changes.

This information includes assigns and params that have been added, deleted or modified with BASE24 Release 6.0 Version 10 along with their associated page numbers. Refer to the specified page numbers for explanations of how these assigns and params can or must be set.

Refer to the ***BASE24 Logical Network Configuration File (LCONF) Manual*** for instructions on adding them to the LCONF.

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ASSIGNS

The following assigns were added or modified for BASE24 Release 6.0 Version 10. Refer to the specified page numbers for explanations of how these assigns can or must be set.

- ☐ There were no new LCONF assigns added as part of BASE24 Release 6.0 Version 10.

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PARAMS

The following params were added or modified for BASE24 Release 6.0 Version 10. Refer to the specified page numbers for explanations of how these params can or must be set.

<u>Address Verification Support for No ZIP Code</u>	
POS-ISS-AVS-BLANK-ZIP-CDE-ALWD	6-9
<u>PTD Terminal Table Full Warning</u>	
POS-DH-PTD-NTFY-PCNT	6-37

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B: Host/CO-Network Impacts

Appendix B contains a list of BASE24 Release 6.0 Version 10 enhancements that have host/co-network impacts, along with their associated page numbers. Refer to the specified page numbers for more information about these enhancements.

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Enhancements

The following enhancements have host/co-network impacts. Refer to the specified page numbers for more information about these enhancements.

Issuer Fees for Non-Financial Transactions	BASE24-atm
Network 2008 Mandates for POS Acquiring	BASE24-pos

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C: CUSTMACS File Changes

Appendix C contains a list of BASE24 Release 6.0 Version 10 enhancements that have CUSTMACS changes, along with their associated page numbers. Refer to the specified page numbers for more information about these enhancements.

BASE24 users should review these changes to determine how the CUSTMACS file should be set for their BASE24 environment before running MAKE to compile their system. More information on all MAKE flags used within BASE24 can be found in the FLGSDOC file located on the BA60MAKE subvolume delivered with BASE24 Release 6.0 Version 10 code.

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Enhancements

The following enhancements have CUSTMACS impacts:

Shared 912 BNA Support

atm_dh_c1000_shrd_bna_on.....	5-79
atm_dh_c1000_shrd_bna_mcu_on.....	5-79

Shared 912 CPM Support

atm_dh_c1000_shrd_cpm_on.....	5-95
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D: Re-Sequenced Files and New Files

Appendix D consists of the following information:

- ☐ A table illustrating the new files added to BASE24 for the Release 6.0 Version 10 enhancements.

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New Files Added to BASE24 with Version 10

The following table lists the files added by the BASE24 Release 6.0 Version 10 enhancements:

Subvolume	Filename	Language	Enhancement Name
AT20IBLD	ENTFRM2S	TAL	Diebold ENTIBOLD Entry Program Reorg
	ENTWBLDG	TAL	Diebold ENTIBOLD Entry Program Reorg
			Shared 912 BNA Support
			Shared 912 CPM Support
			Shared DH Wincor EMV v1.2 Support
	ENTWBLDS	TAL	Diebold ENTIBOLD Entry Program Reorg
			Shared 912 BNA Support
			Shared 912 CPM Support
	ENTWFRMS	TAL	Diebold ENTIBOLD Entry Program Reorg
			Shared 912 BNA Support
			Shared 912 CPM Support
	ENTWMENS	TAL	Diebold ENTIBOLD Entry Program Reorg
AT60DDLD	C1000B	N/A	Shared 912 BNA Support
			Shared 912 CPM Support
	C1000D	N/A	Shared 912 BNA Support
			Shared 912 CPM Support
AT60CW10	ATCW10FM	TACL	Shared 912 CPM Support
	ATCW10M	TACL	Shared 912 CPM Support
	ATCW10MM	TACL	Shared 912 CPM Support
	C10WCPMG	TAL	Shared 912 CPM Support
	C10WCPMM	TACL	Shared 912 CPM Support
	C10WCPMS	TAL	Shared 912 CPM Support
AT60DW10	ATDW10FM	TACL	Shared 912 BNA Support
	ATDW10M	TACL	Shared 912 BNA Support
	ATDW10MM	TACL	Shared 912 BNA Support
	C10WBNAG	TAL	Shared 912 BNA Support
	C10WBNAM	TAL	Shared 912 BNA Support
	C10WBNAS	TAL	Shared 912 BNA Support
	C10WBNMG	TAL	Shared 912 BNA Support
	C10WBNMM	TACL	Shared 912 BNA Support
	C10WBNMS	TAL	Shared 912 BNA Support
AT60UC0D	AAREAD0D	TACL	Shared 912 BNA Support
			Shared 912 CPM Support
	ATUC0DFM	TACL	Shared 912 BNA Support

Re-Sequenced Files and New Files
New Files Added to BASE24 with Version 10

Subvolume	Filename	Language	Enhancement Name
	ATUC0DM	TACL	Shared 912 CPM Support
			Shared 912 BNA Support
			Shared 912 CPM Support
	ATUC0DMM	TACL	Shared 912 BNA Support
			Shared 912 CPM Support
	CNVATDM	TACL	Shared 912 BNA Support
			Shared 912 CPM Support
	CNVATDR	TACL	Shared 912 BNA Support
			Shared 912 CPM Support
	CNVATDS	TAL	Shared 912 BNA Support
			Shared 912 CPM Support
BA60DDL	CPF0D	N/A	Increase Size of Segments in the CPF
BA60UC0D	AAREAD0D	N/A	Increase Size of Segments in the CPF
			Network 2008 Mandates for POS Acquiring
			Shared 912 BNA Support
			Shared 912 CPM Support
	BAUC0DFM	TACL	Increase Size of Segments in the CPF
			Network 2008 Mandates for POS Acquiring
	BAUC0DM	TACL	Increase Size of Segments in the CPF
			Network 2008 Mandates for POS Acquiring
	BAUC0DMM	TACL	Increase Size of Segments in the CPF
			Network 2008 Mandates for POS Acquiring
PS60DDL	PRDF0D	N/A	Increase Size of Segments in the CPF
			Network 2008 Mandates for POS Acquiring
PS60UC0D	CNVCPFM	TACL	Network 2008 Mandates for POS Acquiring
			Network 2008 Mandates for POS Acquiring
	CNVCPFR	TACL	Increase Size of Segments in the CPF
			Network 2008 Mandates for POS Acquiring
	CNVCPFS	TAL	Increase Size of Segments in the CPF
			Network 2008 Mandates for POS Acquiring
	PSUC0DFM	TACL	Network 2008 Mandates for POS Acquiring
	PSUC0DM	TACL	Network 2008 Mandates for POS Acquiring
	PSUC0DMM	TACL	Network 2008 Mandates for POS Acquiring

The following table lists the same files but by version 10 enhancement name:

Enhancement Name	Subvolume	Filename	Language
Diebold ENTIBOLD Entry Program Reorg	AT20IBLD	ENTFRM2S	TAL
		ENTWBLDG	TAL
		ENTWBLDS	TAL
		ENTWFRMS	TAL
		ENTWMENS	TAL
		ENTWSTAS	TAL
		ENTMEN2S	TAL
Increase Size of Segments in the CPF	BA60DDL	CPF0D	N/A
	BA60UC0D	AAREAD0D	N/A
		BAUC0DFM	TACL
		BAUC0DM	TACL
		BAUC0DMM	TACL
		CNVCPFM	TACL
		CNVCPFR	TACL
		CNVCPFM	TACL
		CNVCPFS	TAL
Network 2008 Mandates for POS Acquiring	BA60UC0D	AAREAD0D	N/A
		BAUC0DFM	TACL
		BAUC0DM	TACL
		BAUC0DMM	TACL
		CNVCPFM	TACL
		CNVCPFR	TACL
		CNVCPFS	TAL
	PS60DDL	PRDF0D	N/A
	PS60UC0D	AAREAD0D	N/A
		CNVPRDFM	TACL
		CNVPRDFR	TACL
		CNVPRDFS	TAL
		PSUC0DFM	TACL
		PSUC0DM	TACL
		PSUC0DMM	TACL
Shared 912 BNA Support	AT20IBLD	ENTWBLDG	TAL
		ENTWBLDS	TAL
		ENTWFRMS	TAL
		ENTWSTAS	TAL
	AT60DDL	C1000B	N/A
		C1000D	N/A
	AT60DW10	ATDW10FM	TACL
		ATDW10M	TACL
		ATDW10MM	TACL
		C10WBNAG	TACL
		C10WBNAM	TACL
		C10WBNAS	TAL
		C10WBNMG	TACL
		C10WBNMM	TACL

Re-Sequenced Files and New Files
New Files Added to BASE24 with Version 10

Enhancement Name	Subvolume	Filename	Language
	AT60UC0D	C10WBNMS	TAL
		AAREAD0D	N/A
		ATUC0DFM	TACL
		ATUC0DM	TACL
		ATUCODMM	TACL
		CNVATDM	TACL
		CNVATDR	TACL
		CNVATDS	TAL
	BA60UC0D	AAREAD0D	N/A
Shared 912 CPM Support	AT20IBLD	ENTWBLDG	TAL
		ENTWBLDS	TAL
		ENTWFRMS	TAL
		ENTWSTAS	TAL
	AT60DDL	C1000B	N/A
		C1000D	N/A
	AT60CW10	ATCW10FM	TACL
		ATCW10M	TACL
		ATCW10MM	TACL
		C10WCPMG	TACL
		C10WCPMM	TACL
		C10WCPMS	TAL
	AT60UC0D	AAREAD0D	N/A
		ATUC0DFM	TACL
		ATUC0DM	TACL
		ATUCODMM	TACL
		CNVATDM	TACL
		CNVATDR	TACL
		CNVATDS	TAL
	BA60UC0D	AAREAD0D	N/A
Shared DH Wincor EMV v1.2 Support	AT20IBLD	ENTWBLDG	TAL
		ENTWSTAS	TAL

E: Moved, Renamed, Deleted Files

Appendix E contains a list of files that were moved, renamed, or deleted in BASE24 Release 6.0 Version 10.

No files were moved, renamed or deleted in BASE24 Release 6.0 Version 10.

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F: Regional Specific Information

Appendix F contains the information that is required to implement the changes for BASE24 Release 6.0 Version 10 that are specific to each region.

There were no Regional Specific changes made as part of BASE24 Release 6.0 Version 10.

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